User's Guide

2D VOG

Version 4

SensoMotoric Instruments

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Instrument: 2D VOG - Video-Oculography® Version 4
Manual version: E1.0

Documentation: klipp und klar, Bremen
www.klippundklar.com

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FACSIMILE FORM XI

A

Introduction



1. Preface

In the past, a variety of techniques have been developed for measuring horizontal and vertical eye movements (2D) and have found widespread use, such as the Electro-Oculography (EOG) and the wide spectrum of photo-electric techniques.

However, EOG has its limitation in the achieved accuracy (around 2 degrees) due to its sensitivity to other electro-physiological artifacts and electromagnetic sources in the environment. Signal drift impairs measurement of absolute eye position and vertical eye position is highly non-linear. Photo-electric techniques provide higher accuracy than EOG (down to 0.25 degrees). However, these techniques are often limited only to the measurement of the horizontal eye movement component.

Around fifty years ago, first reports of the employment of image based techniques for oculometric measurement were published. These involved first manual photographic techniques to measure horizontal and vertical eye movement. Later, cinematographic techniques were employed to improve temporal resolution of the recordings.

Some 25 years later, with the establishment of electronic techniques for image scanning and reproduction, the image processing approach to oculometric measurement was introduced. First analog, then later replaced by more sophisticated digital analysis techniques of the video signal from a camera were employed. With the development of the video market in the past, small, high resolution, inexpensive CCD sensor cameras have become available and a variety of video-based eye movement systems for horizontal and vertical eye movement were developed (2D).

These systems enable a contact free recording of eye movements with a good accuracy (0.1 to 0.5 degrees), provide absolute eye position, and are easy to handle.

This provided the impetus for the development of a non-invasive three dimensional video-based eye movement technology, Video-Oculography (VOG). From 1989 to 1991 a prototype of this system was developed as a research tool in the framework of ground based space studies in the vestibular research laboratory of the Medical Center Steglitz, Free University of Berlin. It has been extensively used in clinical diagnosis and in various research environments to record eye movement responses from rotating chairs, linear sleds, centrifuges, parabolic aeroplanes and during space flights.

In 1991, SensoMotoric Instruments was founded to make the Video-Oculography technique available to other clinicians and researchers and to develop this technique further. The Video-Oculography was the first commercial video-based 3D eye movement recording system available on the market and is still unsurpassed. A space qualified version of VOG was developed, a binocular VOG recording option was integrated and further enhancements on algorithms were performed.

Thus the decision was made to develop the 2D VOG application with the goal to create a powerful but easy-to-use system for eye movement measurement and analysis. The system should be based on an integrated eye-tracker hardware that has the processing power to calculate the eye position in real time (50Hz for PAL and 60Hz for NTSC video signals).

The 2D VOG – Video-Oculography® system was the first system to measure and analyze eye movements which was completely integrated in the MS Windows operating system. Until now its online analysis capabilities are unsurpassed.

Wer are pleased that you are a new user of the 2D VOG – Video-Oculography® system and hope you will be working efficiently with the system. When you have any question concerning the system or encounter a technical problem, do not hesitate to contact your distributor or SensoMotoric Instruments GmbH.

2. Safety Notes

You 2D VOG system complies with the legall requirements of the safety standards for medical diagnosis devices (EU: Class IIa according Medical Device Directive 93/42/EEC)).



WARNING:

- **Proper use of this device depends on careful reading of this manual and all additional instructions and labels.**
- **Before connecting or disconnecting components or additional devices switch off the system.**
- **All devices connected to your system (PC, monitor, printer etc.) must comply with the IEC 950 safety directives.**
- **The system must not be used in humid environments as it is not protected against moisture (IPXo according to IEC 529).**
- **The system must not be used in presence of explosive or flammable gases.**
- **Patient examinations may be done on behalf of the doctor by trained madical staff only.**
- **The VOG mask should not be worn by patients with strong defective vision and abnormal rare blink. Please consult a specialist in such circumstances before using the mask on these types of patients. (This recommendation is made as there is infra red light that is given out in the VOG mask to illuminate the eye).**
- **The patients or test persons may not carry any optical utilities expect contact lenses.**
- **In case your 2D VOG – Video-Oculography® system or a component of the system is damaged, do not use the system until the system or its component has been repaired by a certified distributor or SensoMotoric Instruments.**
- **Do not repair the system by yourself. Electrical shock hazard.**
- **Switch off the system before cleaning.**



CAUTION:

– **Federal law in the U.S.A. and Canada restricts the sales, distribution, or use of this device, to, by, or on the order of a licensed medical practitioner.**

3. Feature List

Within the field of clinical vestibularic diagnostics, the 2D VOG – Video-Oculography® system is a computer based system for measuring and analyzing eye movements. Depending on the equipment of the system it has the following features:

- Video oculographic tests for: Spontaneous Nystagmus, Gaze, Smooth Pursuit, Saccade, Position and Positional and Caloric Irrigation.
- Monocular and binocular recording (Caloric Irrigation and Multi-Condition test monocular only)
- VOG combi mask for light-occluding and free-field-of-view testing
- Online display of eye images
- Automatic detection of nystagmus beats with two different algorithms (continuous and discrete)
- Automatic calculation of all essential statistical data
- Automatic detection of artefacts
- User defineable examination templates
- Digital generator for optocinetic stimulation (VisualLab)
- Detailled one-test-per-page and summarizing for-test-per-page hardcopy reports
- Letter wizard
- Foot switch
- Patient database compatible with Microsoft® Access®
- User-guided backup system of the database

4. The Documentation

The 2D VOG documentation consists of

– the present user's guide and
– the online-documentation (pdf)

4.1 Who Should Use the Documentation?

The documentation has been designed for persons with medical skills that are used to perform vestibular examination and that have a basic knowledge of how to use a Windows application.

4.2 The User's Guide

4.2.1 Structure and Contents

Due to its process-orientated structure this manual reflects your daily work which enables you to work efficiently with your 2D VOG system and the manual.

To give a better overview, we have divided the manual into sections:

A – Introduction
This part contains some introducing words about the basic idea of the program, safety notes, and how to use the documentation.

B – System Setup
In this section, you will find a description of the system requirements and how to connect the periphery devices. In addition, the VOG masks and the foolt switch will be described.

C – Installing and Using the Software
In this section, you will learn how to install and use 2D VOG. Moreover, you will get information about the license manager. Please read carefully the "Backup" chapter.

D – Working with 2D-VOG
This part of the manual contains the proper description of the software. Step by step, you will learn how to perform an examiniation - it starts with a description of how to select a patient and at the end, you will learn how to print test results and how to use the letter wizard.

E – Templates
This section describes how to work with examination and letter templates.

F – Tests
Here, you will find a detailed description of each test that can be performed with 2D VOG.

4.2.2 Spellings and Symbols Used

❍ Dialog Box Titles

The text shown in the title bar of a dialog box, in this manual is marked by double quotation mark and an italic font:

In the *"Patient Manager"* dialog box, you can ...

❍ Field Names

Field names are capitalized:

In the SEX field, select the ...

❍ Key Combinations (Shortcuts)

If you are prompted to press a particular key or to enter a key combination, the keys will be shown in brackets. Always make sure whether the keys are separated by the plus sign or the comma:

<ALT>, <S>, <K>	Press the <ALT> key followed by <S> and <K>.
<ALT> + <H>	Press the <ALT> and <H> keys at the same time.

❍ Pushbuttons

Pushbuttons are usually shown in angle brackets. The underlined letter indicates how you can activate the pushbutton using the <ALT> key without using the mouse:

[Add]

This corresponds to the <ALT> + <A> shortcut.

❍ Menus and Commands

A menu and its corresponding commands are shown in single quotation mark. The underlined letter also shows which is the shortest way to select a command:

Fromt the 'Test' menu, select the 'Print' command.

or

Select 'Test | Print'.

By using the key combination <ALT>, <T>, <D> you will achive the same result.

❍ Procedures

A procedure is a description of the steps a user takes to complete a specific task. For a better orientation, each step is numbered so you can always follow. An arrow (→) shows the result of a specific step, e.g.:

To select a patient, proceed as follows:

1. Open the Patient Manager.
   → All patients currently saved in the database will be listed in alphabetical order.
2. Select the patient by double clicking the name.
3. ...

❍ Cross References

When you can find additional information about a certain topic in the manual, a cross reference that contains the corresponding page numer will lead you to the desired page, e.g.:

☞ For more information on this topic, see page D-xx .

❍ Notes

In paragraphs which are highlighted as notes, you can find additional ideas, clues and tricks which you might find interesting:

Note

To acquire reliable data, you always have to adjust the VOG mask properly on the patient's head before performing the test.

❍ Warnings

Hints or very important information that affect the safety of a human being or a machine are marked as follows

Attention!

To clean the camera modules, they have to be switched off. When cleaning them, take care that the camera modules don't come into contact with liquids.

❍ Examples

Examples appear always in a colored box to highlight them from the normal text. Thus, if you are not interested in examples, you will see immediately which passages can be left out.

Example

In the *Graphs* tab of the configuration dialog box of a caloric test, you select the graphs

- Butterfly,
- Freyss,
- Claussen,
- Haid/Stoll,

saving them as default.

4.3 The PDF Online Documentation

The present manual has additionally been saved as pdf file (*manual.pdf*) on the installation CD. Furthermore, once the 2D VOG installation is completed, the pdf file will be copied to your system and a link is saved on your desktop.



Note

You need the Adobe® Acrobat® Reader to view pdf files. If you have not installed the Reader, you can easily get a free copy using the 2D VOG installation CD.

☞ For more detailed information, refer to page C-6.

4.3.1 Starting the Online Documentation

There are three different ways to access the online documentation:

... using the Start Menu

1. In the Start Menu, first select the *Programs* folder and then click the *2D VOG* folder.

   → The contents of the folder will be shown.

2. Click *Online-Help*.

   → The online documentation will be displayed and the Adobe® Acrobat® Reader will be started automatically.

... on your Windows desktop



On the desktop, click the icon shown to your left.

→ The online documentation will be displayed and the Adobe® Acrobat® Reader will be started automatically.

... using the program

In 2D VOG, open the *'?'* menu and select the *'Help'* command.

→ The online documentation will be displayed and the Adobe® Acrobat® Reader will be started automatically.

4.3.2 Navigation

Using the bookmarks and the hyper links, you can browse in the documentation which allows you to access the text you are interested in.

❍ Bookmarks

Nach dem Aufrufen der Datei wird links neben dem Deckblatt ein Lesezeichen angezeigt. Das Lesezeichen ist vergleichbar mit einem Inhaltsverzeichnis und bleibt ständig sichtbar, egal, wo Sie sich gerade im Text befinden.

After opening the pdf file, on the left side of the first page you will see bookmarks (vgl. Fig. A-1). Bookmakrs are similar to a table of contents, being shown all the time. It doesn't matter which page you have actually selected.

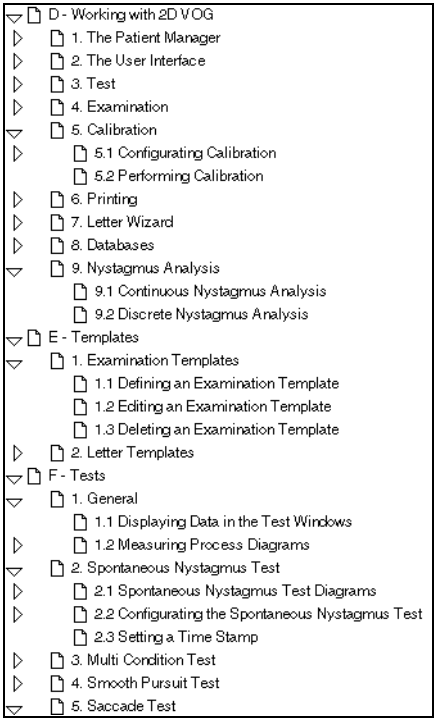


Fig. A-1: PDF documentation bookmarks

Chapters and Subchapters:
Bookmarks have a tree structure. You may find the following symbols in front of the listed headings:


This chapter contains several subchapters. Click the plus to show the subchapters.


This branch of the table of contents is shown. Click the minus to close the branch.

Showing a Chapter:
To view a specific chapter of the documentation, proceed as follows:

1. Open the corresponding contents branch until the heading of the chapter will be displayed you wish to read (refer to *"Chapters and Subchapters"*).

2. Using the mouse, point to the bookmark.

   → The mouse pointer becomes a hand.

3. Click the bookmark.

   → On the right part of the screen, the selected chapter of the documentation will be displayed.

❍ Hyperlinks

The text contains several hyperlinks (cross references) that are shown in blue color. The mouse pointer becomes a hand if you point to a hyperlink using the mouse.

If you click the hyperlink, you will directly access the text the hyperlinks refers to.

4.3.3 Printing

Using the 'Filei | Print' command, you can print the online help.



Note

For more detailed information on the Adobe® Acrobat® Reader features, please refer to the Adobe online guide.

B

System Setup



1. System Requirements

1.1 Hardware

- Processor pentium II 400Mhz
- 128 MB of RAM
- Empty PCI slot
- Another available slot of any type (option analog / digital in)
- Graphic board with at least 4 MB video RAM
- Hard disk with at least 4 GB of free capacity
- CD-ROM drive
- 3,5" disk drive
- PS/2 mouse adapter
- Available serial port (option VisualLab)
- Available parallel port (printer port)
- 17" monitor with 1024 x 768 resolution

1.2 Software

- Microsoft® Windows98®
- Monitor resolution 1024x768, 16 bit color depth

2. VOG Workstation Setup

The required type of cabling depends on

– the mask type that is delivered with your system
– and the selected system configuration.

Attention!

Before connecting the cables always make sure your 2D VOG system, the VisualLab, and all connected devices have been switched off!

2.1 Connecting the Mask Cameras

The SMI SoftEyes board behaves like a video switch for up to four video cameras.

The connectors of the adapter can be found at the back of your computer.

– **Monocular Systems**
  Connect the camera of your VOG mask to the CAM1 port.

– **Binocular Systems**
  In binocular systems, two cameras are used. Connect the camera that records the left eye to the CAM1 port. The camera that is used to record the right eye has to be connected to the CAM2 port.

It is not possible to connect an older camera model that is used for light occluding, combi and free-field-of view masks directly to the SMI SoftEyes board.

To purchase the proper adapter, contact your distributor.

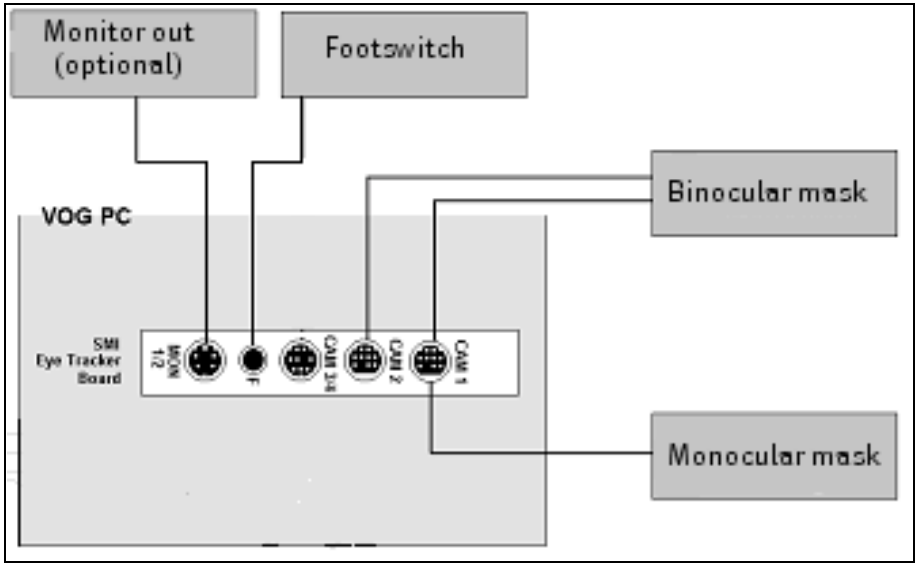


Fig. B-1: Overview of connecting VOG masks and foot switch to the VOG computer

If your system is equipped with a SMI framegrabber board, it should be connected to the SMI SoftEyes board, as described in Fig. B-2.

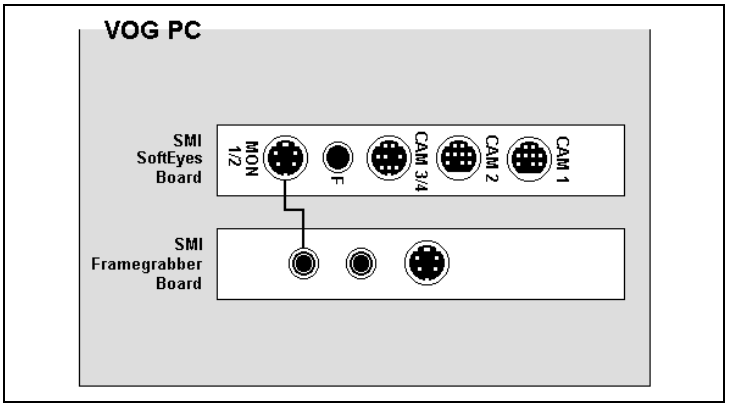


Fig. B-2: Connecting an additional SMI framegrabber board in 2D VOG systems

2.2 Connecting the Foot Switch

Insert the connector of the foot switch into the female connector of the SMI SoftEyes board labeled F (see Fig. B-1).

2.3 Monitor-Out

The SoftEyes board "MON 1/2" connector allows you to show the eye images on external video-monitors (no computer monitors!) or to record the images with a video recorder. To do so, use the provided monitor-out cable.

The image recorded with the camera connected to the "CAM1" input will then be available at the cable labelled "Monitor Out 1", whereas the "CAM2" image will be sent to the "Monitor-Out 2" cable.

Attention!

To avoid a short circuit, please cover the unused ends of the monitor-out cable with the provided blind connectors.

Please make sure the connected monitors and video recorders comply with the IEC 950 standard.

2.4 First Check of the VOG Workstation

To perform a first checking of the 2D VOG workstation, proceed as follows:

1. After booting the VOG computer, start 2D VOG.
2. From the *Patient Manager* dialog box, select a patient (for more information, see page D-3).



3. Start a spontaneous nystagmus test by clicking this button in the toolbar..

4. Put on the VOG mask and check whether the eye lid will appear at the upper right corner of the monitor.

5. Adjust the height of the VOG mask until the pupil is centered in the video image (for more information, see page B-24).

Note

Pupil detection will be perfect, if the pupil shown in the video image is colored completely white and a crosshair can be seen in the pupil center. (Due to its small size, the crosshair may not always be visible, however this doesn't affect the measurement.)



6. Start a measurement by clicking this button.

7. Make sure the system is actually recording data. The graph should be very clear, when the threshold is set properly. If necessary, you may readjust the threshold during measurement (refer to page D-15).

   → The system will work correctly once you can see the eye image and data is being recorded.

❍ Foot Switch

To test the foot switch, proceed as follows:

1. Perform a spontaneous nystagmus test

2. Use the foot switch instead of the return key.

   → If the foot switch doesn't work properly, you should check the cable connection.

3. System Setup VisualLab

3.1 Cabling

The VisualLab is an optimal extension to the basic VOG system. It is used to provide visual stimulation for calibration and for the following tests:

- Smooth pursuit test
- Optokinetic test
- Saccade test
- Gaze test.

The VisualLab module is controlled by the 2D VOG computer via a serial connection. Therefore, a serial cable is connected from the COM 1 (or COM 2) port of the VOG computer to the COM 1 port of the VisualLab module (see Fig. B-3).

The calculated image produced by the VisualLab module is provided in two different modes:

- as a VGA signal (640 x 480 pixels, frequency of 60 or 72Hz) and
- as a NTSC video signal (frequency of 60Hz).

Thus, you may either connect a computer monitor to the VGA output or connect a video monitor or projector to the NTSC output. The NTSC signal may be taken from the chinch connector as well as from the s-video connection.

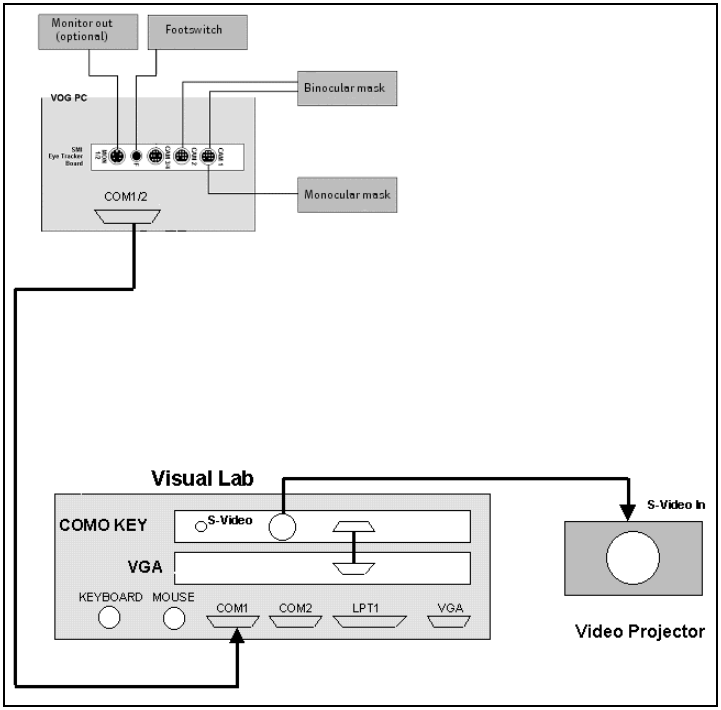


Fig. B-3: VisualLab cabling

3.2 Installing the Projector / Monitor

The projector and the patient should be positioned in the same distance to the projection screen. When looking straight the patient's eyes should be directed towards the center of the image.

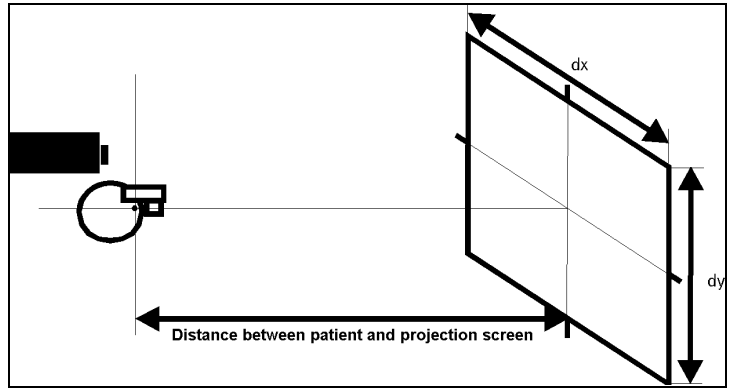


Fig. B-4: Positioning the projector

Great distortions of the image should be avoided, as they affect the results of smooth pursuit and saccade tests.

Note

If a projector is used, it should be placed at least 1.5m/70" from the projection screen - preferably over or slightly beside the patient's head.

As the patient position is part of the VisualLab geometry (see page B-15), is should not be changed. We recommend to mark the patient position on the floor.

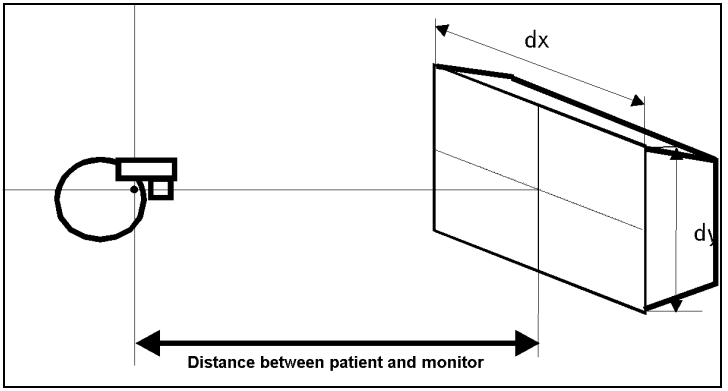


Fig. B-5: Positioning the monitor

3.3 VisualLab - a first Check

To connect the VisualLab, proceed as follows:

1. Switch on the VisualLab module.
2. Switch on the projector/monitor.

   → The VisualLab is ready, once the SMI can be seen on the projection screen or the monitor.

3.4 Checking serial Connection to VOG Workstation

Checking the serial connection to the VOG workstation can be proceeded as follows:

1. Start 2D VOG and select a patient from the *Patient Manager* (refer to page D-3).



2. Start an optokinetic test by clicking this button in the toolbar.

3. In the examination tree, click the test using the right mouse button.

   → A context menu will appear (for more information, see page D-22).

4. In the context menu, select the 'Settings' command.

5. Open the *Stimulus* tab and perform the following settings:

STIMULUS:	*Stripes*
DIRECTION OF PATTERN MOVEMENT:	right / left
WIDTH:	3°

6. In the lower area of the dialog box, click the [Test] button.

   → On the projector image, a stripe pattern should become visible.

If connection has failed...

1. If the VOG computer and the VisualLab are not connected (the SMI logo is still moving), check the cable connections and the ports.

2. If there is still no pattern on the projector image, close the 2D VOG application and restart it.

3. In case, connecting still has not been successful, repeat closing down 2D VOG and restart the VisualLab.

   Wait until the moving SMI logo on the video projector appears.

   Now, restart 2D VOG and repeat the steps 1. to 6. described on page B-13.

4. If again no connection can be established, first reboot your computer and then start the Visual Lab check unit. Wait until the SMI logo appears on the projector image before starting the 2D VOG software again.

5. If you haven't been successful at all, close the 2D VOG software. Open the 2D VOG program group in the Windows start menu. Start the program *"System Configuration"* and, in the *VisualLab* tab, select another COM port.

If the connection has failed in spite of the suggested solutions described above, please contact your distributor.

3.5 Configurating the VisualLab Geometry

To produce stimulation patterns that are correct in size, position and angle velocity, the VisualLab requires information on the geometry of the stimulation projection in your laboratory. These values will have to be updated, if the projection geometry is changed or if the patient's position in relation to the projection screen / monitor is changed.

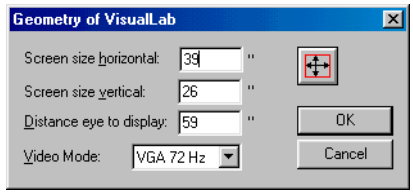


Fig. B-6: Setting VisualLab geometry

To configure the VisualLab geometry, proceed as follows:



1. Create an optokinetic test by clicking this button in the toolbar.

2. Double click into the test window (see Fig. D-1 on page D-9).

   → The *"Configuration Optokinetic Test"* dialog box will appear.

3. Open the *Calibration* tab and select the USE VISUALLAB field.

4. Click the [VisualLab Geometry] button.

   → The *"VisualLab Geometry"* dialog box will appear.



5. Now click this button.

   → On your projection screen/monitor, you will see a yellow frame.

6. Measure the height and width of this frame and then enter the values in the FRAME HEIGHT and FRAME WIDTH fields.

7. Measure the distance between the patients' eyes and the center of the projection screen. These values have to be entered in the DISTANCE EYE - SCREEN field.

8. In case, you are using the VGA output of the VisualLab, in the VIDEO MODE field select a refresh rate as high as possible.

   ☞ Consult the manual for your monitor or projector to find out which refresh rates are supported by the monitor.

9. Confirm your entries by using the [OK] button.

   → The VisualLab is now configurated.

4. VOG Masks

The VOG masks carry the video cameras that are used to record the eye images. They constitute a major component of your 2D VOG system. Therefore, please read this chapter carefully. It contains important notes that might have considerable affects on the quality of your measurements.

The video cameras of the VOG mask use infrared light (IR) which is not visible for people. Thus, using the VOG masks examinations can be performed in complete darkness. Although it seems to be dark, the eye images on your VOG computer are clear and visible due to the IR-technology.

VOG masks can be divided in two different types:

– Light occluding masks (LOM)
– Combi masks (COM)

☞ For a detailed description of these masks, refer to page B-22 and B-23.

Attention!

Before using the VOG masks, consider the following:

– The VOG masks contain sensible electronic and optical devices. Do not expose the masks to mechanic shocks.
– It is not allowed to use VOG masks in moist environments, as they are not specially protected against moisture (IPX o according to IEC 529).
– VOG masks are not to be used with explosive gases.

4.1 Connecting the VOG Masks to the VOG Computer

Attention!

Switch off your VOG computer, before connecting or disconnecting the VOG masks to the SMI SoftEyes board. Otherwise, your cameras and the SMI SoftEyes board might be damaged.

The VOG masks may be connected directly to the SMI SoftEyes board.

See chapter *Connecting the Mask Cameras* on page B-5 for a more detailed description.

Note

If you are using another VOG mask type or if you have connected two different VOG masks to your system, you will have to adapt the settings in the 2D VOG software.

4.2 Structure of VOG Masks

Your VOG mask has a modular structure. Depending upon the mask type and the configuration, one (monocular) or two (binocular) camera modules may be fixed at the front (LOM) or at one side (COM) of the mask. With monocular VOG masks, the open adapter is closed with a cover plate. The camera modules and the cover plate are fixed magnetically and can easily be removed.

4.3 Viewing the other Eye (Monocular Systems)

Due to the modular construction of the VOG masks, with monocular systems you can even view both eyes (though one after the other).

To do so, remove the camera module and the cover plate from their sockets and mount each of them at the opposite side.

4.4 Fixation Light and IR Illumination



Light emitting diodes (LEDs) are positioned at the top and at the side of the camera objective. The lateral LEDs are emitting infrared light (IR) which means they serve as illumination for the camera. The upper LED illuminates green and can be switched on and off using this button in the toolbar of the 2D VOG software. Furthermore, it can be used as fixation light to measure the visual nystagmus suppression (refer to the *Caloric Irrigation Test* chapter on page F-65 and to the *Buttons in the Toolbar* chapter on page D-10).

As the IR illumination is only slight, is doesn't damage the eye. In any case, the maximum total amount of radiation will be less than the limit of 1,9 mW for lasers, class 3A (IEC 825-1).

4.5 Fixing the VOG Masks

Attention!

Before the VOG mask can be placed, glasses and other optical aids like magnifying glasses have to be removed. Contact lenses are allowed, provided that they don't affect video recording of the eye by reflecting the IR illumination.

To fix the VOG mask, the elastic band can be opened. If the patient places the mask by himself, this won't be necessary in most cases. The VOG mask should sit comfortable for the patient but tight enough to ensure that it will not move even during a positioning test. When placing the VOG mask, make sure the patient's eyes and the camera objectives have the same height.

To provide optimal measurement of the eye movement, the pupil should be centered in the video image. Please check this with the eye images in the 2D VOG software. If this is not the case after fixing the mask, you have to readjust the mask.



Note

Please keep the mask in room temperature for approx. 30 minutes before starting the examination. Otherwise the mirrors (only COM masks) and the camera optics can be fogged during the tests and affect the measurements.

4.6 Cleaning the VOG Masks

The patient's face will be touched by the VOG masks. For hygienic reasons, they should be cleaned regularly.

Attention!

To clean the camera modules, they have to be switched off. When cleaning them, take care that the camera modules don't come into contact with liquids.

To clean the VOG masks, the face cloth, and the camera module, use a soft cloth lightly moistened or pads dampened with disinfectant fluids on isopropanal base. Do not use spiritus for cleaning.

The camera objectives and the infrared mirrors of the combi mask should carefully be cleaned using a soft cloth or special lens paper. Please avoid touching the lenses of the objectives with your hands.

4.7 The light-Occluding Mask (LOM)

The lightweight light-occluding mask (LOM) was designed for measuring eye movements in absence of any visual stimulation. It can be used for all tests where no visual stimulation is to be performed (e.g. caloric test or multi condition test).



Fig. B-7: Light-occluding mask (LOM2)



Fig. B-8: Light-occluding mask for children (LOM4)

4.7.1 Sizes

The light-occluding mask is available in two sizes:

- **LOM2**
  for adults
- **LOM4**
  for children (from 9 months).

There is no difference in handling these VOG masks.

4.7.2 Focusing Light-Occluding Mask

With the light-occluding mask, it is not necessary to focus the video image.

4.8 The Combi Mask (COM2)

In addition to examinations under light-occluding circumstances (e.g. multi condition test), the special design of the combi mask also allows you to perform examinations with visual stimulation (e.g. smooth pursuit test). Thus, it can be used for all VOG tests.



Fig. B-9: Closed COM2 mask for recording the eyes under light-occluding circumstances



Fig. B-10: COM2 mask as Free-Field-of-View mask. The front cover has been removed.

To make sure the patient can always look to the front, the front cover plate of the VOG mask that is magnetically fixed can easily be removed. If the mask is open, two mirrors can be seen that are diverting the eye image to the video cameras fixed to the side of the mask. The mirror coating reflects infrared light (IR) in the same way as it is required by the video cameras of the mask. However, visible light is hardly reflected so that the patient's looking to the front is not affected.

4.8.1 Vertical and Horizontal Adjustment of the Video Image

With the COM2 mask, horizontal and vertical pupil position in the video image can be adjusted by moving the camera within the camera cabinet. To do so, use the screws fixed to the back of the camera modules:



– The upper regulating screw with the markingt shown to your lef is adjusting the image in the vertical graph.



– The left regulating screw with the marking shown to your left is adjusting the image in the horizontal graph.

To adjust narrow eyes (e.g. with children), the mirrors can also be rotated. To do so, hold the mirror at its margin and turn it inwards by pressing gently.

4.8.2 Focusing the Combi Mask



The middle regulating screw at the back of the camera module is labeled "Focus". It allows you to focus the image. As the eye socket of each patient can be different, it may happen that a complete focussing of the image fails. However, observing the eye movement even works if the image is slightly fuzzy.

5. Foot Switch

Within the 2D VOG system, the foot switch replaces pressing the return key when you are performing a test. The advantage is that you can use both hands to help the patient or to handle other devices (e.g. caloric irrigators).

6. Servicing and Maintenance

The 2D VOG system doesn't require any special precaution for servicing or maintenance. You should consider the general service note for PCs.

☞ Please consider the description of how to clean VOG masks on page B-21.

C

Installing and Using the Software



1. Software Installation

In general, the 2D VOG software will be installed by your distributor or by the SensoMotoric Instruments staff.

1.1 Installing 2D VOG

To install the software yourself, follow the instructions below:

1. Insert the SMI installation CD into the CD-ROM drive of your computer.
2. From the Windows® start menu, select the command '*Run...*'.

   → The *Run* dialog box will appear.
3. In the OPEN field, type the letter of the drive containing the CD-ROM followed by **setup** (e.g. *d:\setup*). Press [OK] to confirm.

   → The setup program will be started.

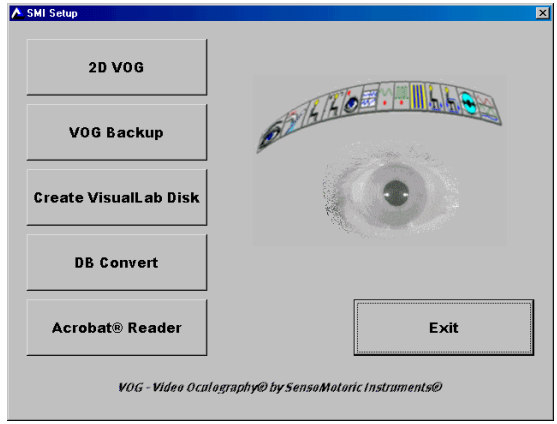


Fig. C-1: 2D VOG Setup Program



4. Click this button.

   → 2D VOG will be installed. Once installation has been successful and the software has been licensed, you can start working with 2D VOG.

1.2 Other 2D VOG Utilities

Apart from the proper 2D VOG program, the following additional modules will be delivered:

- VOG Backup
- Create VisualLab Disk
- DB Convert
- Acrobat® Reader

1.2.1 VOG Backup

This utility is used to backup and restore the VOG databases.

☛ See page C-25 for more detailed information.

1.2.2 Create VisualLab Disk

The VisualLab constitutes a separate computer that is used to produce visual stimulation. It is booted from a disk which is under normal circumstances always in the floppy drive of the computer. If you have lost the boot disk of if it has been damaged, you can easily create a new boot disk using the *Create VisualLab Disk* utility.

❍ Create a VisualLab Disk

To do so, proceed as follows:

1. Insert an formatted empty 3,5" disk (1,44 MB) into the floppy drive of your computer.

2. Insert the SMI installation CD into the CD-ROM drive of your computer.

3. From the Windows® start menu, select the command '*Run...*'.

   → The *Run* dialog box will appear.

4. In the OPEN field, enter the drive name of your CD-ROM drive followed by **setup** (e.g. *d:\setup*). Press [OK] to confirm.

   → The setup program will be started (see Fig. C-1).

Create VisualLab Disk

5. Click this button.

   → The VisualLab boot disk will be created.

1.2.3 DB Convert

This tool assists you to manage existing customer databases.

Using this tool, you can convert patient databases created with an older version of 2D VOG to the version 4.0 format.

With *DB Convert*, you can only convert demographical data. It is not possible to convert the recorded data.

❍ Installation



This module will be installed within the setup program, if you are using the button shown to your left. The rest of the procedure is identical with the installation routine described on page C-3.

→ When installation has finished, you can start the conversion program using the *DB Convert* command in the *2D VOG* folder of the Windows® start menu.

1.2.4 Acrobat® Reader

To view the online documentation, the Acrobat® Reader has to be installed on your computer.

❍ Installation



The Acrobat® Reader installation will be started, if you use the button shown to your left. The procedure is identical to the installation procedure described on page C-3.

After completing the installation, you can open the online documentation as described on page A-14.

1.3 Uninstalling 2D VOG

To delete 2D VOG from your computer, proceed as follows:

1. From the Windows® system control, select *Software*.

   → The Software Properties dialog box will be opened. In the lower part of the dialog box, all applications are listed that can be removed automatically from your system.

2. Now mark the *2D VOG* entry.

3. Click the [Add/Remove...] button. (This is only available, after you have marked an entry in the list).

   → The following security request will appear.

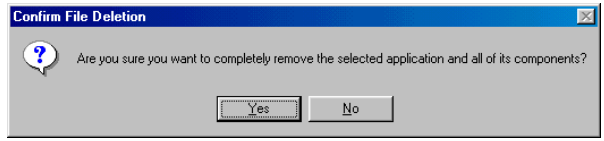


Fig. C-2: Security request when deleting 2D VOG

4. Confirm the request.

   → 2D VOG will be removed from your computer.

2. Using the Software

2.1 Starting 2D VOG

To start 2D VOG, proceed as follows:

1. In the *Start* menu, first select the *Programs* folder and then the *2D VOG* folder.

   → The contents of this folder will be displayed.

2. Click *2D VOG*.

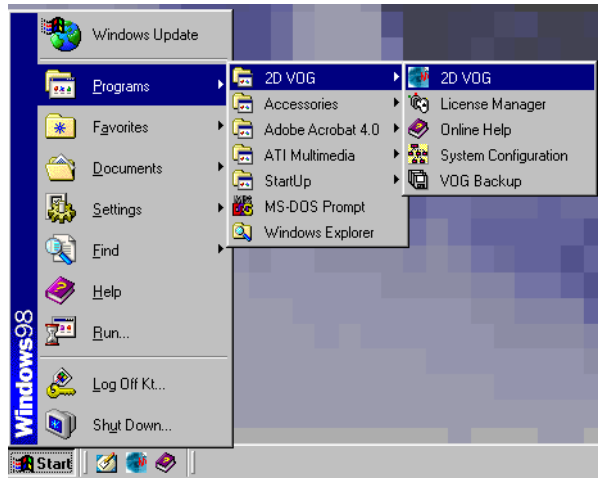


Fig. C-3: Prompting 2D VOG using the Start menu

→ 2D VOG will be started. The *User Identification* dialog box will appear. You will see the user name of the person last logged in.

Note

Instead of using the method described above, you may create a link on your desktop that allows you to start 2D VOG via double clicking the program icon.

☞ Please see the Microsoft® Windows® documentation for information on how to create links.

2.2 User Identification

After starting 2D VOG, the *User Identification* dialog box will appear where you can

– log in with your user name or
– create a new user name.

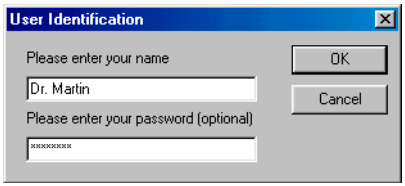


Fig. C-4: User identification

2.2.1 Login

For login proceed as follows:

1. Enter your name into the box and switch to the password box either by using the tab button or clicking into the box.

Note

If you are a new user unknown to the system, proceed as shown on page C-11.

2. If your user name is stored with a password, enter the password. For security reasons the entered password will be shown in encrypted form. If no password is used for your user name, please leave this field empty.

3. Confirm you entry by using the return key or clicking on the [OK] button.

   → The Patient Manager appears. You can now access the patient data base files last used in 2D VOG.

Note

The entered user name will be stored with each test and be shown on the monitor as well as on printouts. This allows you to identify the examiner in future.

2.2.2 Creating a new User

To create a new user name, proceed as follows:

1. In the name box, enter the new user name. You can also enter the optional password.

   → A note will appear saying that no user with this name has been registered.

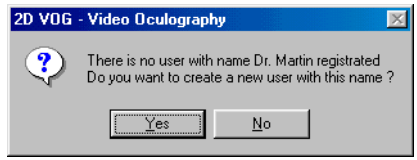


Fig. C-5: Creating a new user

2. Confirm this *request*.

   → For security reasons you will be prompted to confirm the password.

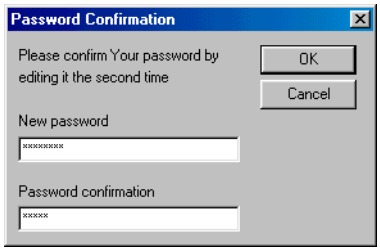


Fig. C-6: Confirming password

3. Enter the password again and confirm your entry.

   → The Patient Manager appears. You can now access the patient data base files last used in 2D VOG

Note

In case you have not created a data base yet, you should do it now. See chapter *Creating a New Database* on page D-67 for more detailed information.

You can now start using 2D VOG.

☞ Part **D (Working with 2D VOG)** of this manual contains a description of how to work with 2D VOG.

2.3 Exit 2D VOG

To exit 2D VOG, you have various possibilities

a) clicking on the Close button  in the upper right corner of the 2D VOG dialog box,

b) using the 'Exit' command from the 'Test' menu or

c) entering the shortcut <ALT> + <F4>.

3. License Manager

When designing modern systems, it is tried to replace complex hardware components to a maximum extent by sophisticated software used on standard computers. This is the way we have chosen when designing the 2D VOG – Video-Oculography® system.

Nevertheless, software is not free as its development especially for medical applications is complex and expensive. Thus, software producers have problems with simple and inexpensive copying of software. But this problem does not only affect the software development, it is also of great importance for you as you are using the software. Because a software producer who cannot cover the design and development costs by the sales profit, won't be able to maintain the software and develop new features.

This is the reason why we have decided to protect our software against misuse by using a license key. This key consists of a combination of letters and numbers that can be entered using the license manager.

License keys can have a time limit. Furthermore, depending upon the configuration of your 2D VOG system you deliberately may release the required software components.

If SensoMotoric Instruments has delivered a using license that has a special time limit or if you possibly wish to purchase additional licenses to be able to use more tests, you can ask for new license keys by using the form that can be printed in the License Manager.



Note

To ensure a continuous use of 2D VOG, we recommend to request for the new license **before** your existing license will expire.

3.1 Running the License Manager

There are two different possibilities to run the License Manager:

– direct start within the 2D VOG application
– using the Start menu

Starting the License Manager from 2D VOG

To run the License Manager from the program, from the 'Tools' menu select the 'Exit and Run License Manager' command.

→ 2D VOG will be closed. The License Manager will be started.

You now can

– print the form for requesting new license (see page C-18),
– enter/read the new license key (s. page C-19) or
– list the program features covered by your license (see page C-20).

Starting the License Manager from the Start menu

To run the License Manager from the Start menu, proceed as follows:

1. Close the 2D VOG application.
2. From the *Start m*enu, first choose the *Programs* folder and then select the *2D VOG* folder.
   → The contents of this folder will be displayed.
3. Click *License Manager*.
   → The License Manager will be started.

You now can

- print the form for requesting new license (see following description),
- enter/read the new license key (s. page C-19) or
- list the program features covered by your license (see page C-20)

3.2 Request Form

To prolong or extend your using license, the License Manager provides a request form especially designed for this purpose.

To print the form, proceed as follows:

1. Start the License Manager as already described.
2. Click the [Print License Key Request Form] button.

   → The Microsoft® Windows® *Printer Settings* dialog box will appear.
3. Enter the required settings and confirm your entries.

   → The form will be printed.

Please fill out the form completely and fax it to SensoMotoric Instruments. (The fax number can be found on the form.)



Note

SensoMotoric Instruments is legally obliged to acquire the customer addresses, since it is producing medical devices. Therefore, **please** enter your address on the form, so we can ensure that you will receive our service information.

3.3 Entering / Reading License Key

After receiving the new license key, in the License Manager you have to

– enter the key, if is has been sent to you in written form
– read the key, if it has been sent as file

To do so, start the License Manager (for more information, see page C-16).

Entering License Key

Enter the license key in the text box and confirm the entry.

→ The license key will be checked and – if it has been entered correctly – verified. You can now use all program features and components (like the VOG backup module) you have purchased a license for.

Reading License Key

To read the license key from a file, proceed as follows:

1. Click the [Read License Key from File] button.

   → The *"Read License Key from ..."* dialog box will appear.

2. Choose the folder and the file that contains the license key and click the [Open] button.

   → The license key will automatically be sent to the License Manager.

3. Confirm the License Manager using the [OK] button.

   → The license key will be checked and verified. You can

now use all program features and components (like the VOG backup module) you have purchased a license for.

3.4 Viewing Using Licences

If you want to know for which program features and components you still have a valid using license, you should proceed as follows:

1. Start the License Manager (for more information, see page C-16).

2. Click the [Details...] button.

   → A window will appear that contains all program features and components covered by a valid license.

3. For every program feature and component, you may display what kind of functions you are allowed to use (Creating, Saving, Performing a measurement etc.). By clicking on the plus symbol next the current program feature / test, a "path" will be opened showing the released program functions. To close a path, click the minus symbol.

4. System Configuration

If the hardware configuration of your computer has been modified, using this tool you can still change the system configuration of 2D VOG, e.g.

- Which masks have been installed?
- Where is the system data base stored?
- Which video standard is used within the system?
- How many eye-tracker boards have been installed in the computer?
- The VisualLab has been connected to which serial port?
- ...

❍ Starting the Program

When installation has finished, you can start configurating the system using the *System Configuration* command in the *2D VOG* folder of the Windows® start menu.

5. Backup

The backup operation is one of the most important tasks when working with data processing. For this reason, we want to emphasize how important it is to perform **regular backups**.

Accidental deleting of data, a hard disk crash or an operation system failure – every software user sooner or later experiences a loss of data. Usually, a backup is the only method to restore data. Even if a regular backup costs time, this labor is out of all proportion to manually reconstructing data – if this is still possible at all. For this reason, we recommend to perform a **weekly**, or better a **daily** backup, especially if several people are using the same data.

5.1 Backup Media

Depending upon number and size of the files to be saved (databases) the backup can be performed using different media, like for example

- streamer,
- disks or
- ZIP disks

Note

The size of the databases that require a backup mainly depends on to what extent 2D VOG is used. Usually, it can be assumed that the database size after one year is approx. 200 – 600 MB.

If you are not sure what is the apropiate backup medium, please contact your distributor who sold 2D VOG. They will be pleased to assist you .

5.2 How Often a Backup is Required?

The aim of every backup is to protect all existing files against undeliberate loss of data.

As certain files, especially the measured data produced with 2d VOG, are always changing, a daily backup should be preferred. For this case, we recommend you to perform an "either-way" backup. This is the changing saving of data from the hard disk to various media. This means you require at least two different backup media.

Example

1. day	Backup	data medium **1**
2. day	Backup	data medium **2**
3. day	Backup	data medium **1**
4. day	Backup	data medium **2**
5. day	etc.	

Of course, you can use even more backup data media (for example, five data media – from Monday to Friday)

Advantages of this backup method:

a) **Double Security**
In case the disk that contains the data backup last performed is damaged, you will always have a second version of your data. However, this version will be older (one day), but loosing data of one day is definitely better than loosing all data.

b) **Different Versions**
You can always access an older version of your data.

5.3 Backing up Using 2D VOG

For 2D VOG, the easy-to-use data rescue tool *VOG Backup* has been designed. Using this tool, you can backup all VOG patient databases and restore them if necessary.

Note

Please keep in mind that this tool is protected by a license key, too. Using the License Manager, you can check whether *VOG Backup* has been licensed for your system (refer to page C-20).

5.3.1 Installing *VOG Backup*

1. Insert the SMI installation CD into the CD-ROM drive of your computer.
2. From the Windows® start menu, select the command '*Run...*'.
   → The *Run* dialog box will appear.
3. In the OPEN field, type the letter of the drive containing the CD-ROM followed by **setup** (e.g. *d:\setup*). Press [OK] to confirm.
   → The setup program will be started (refer to Fig. C-1 on page C-3).

VOG Backup

4. Click this button.
   → The backup module will be installed. Once installation has been successful, you can perform the first backup.

5.3.2 Performing a Backup

To perform a backup, proceed as follows:

1. From the 'Tools' menu, select the 'Exit and Back up Database' command.

   → 2D VOG will be closed. *VOG Backup* will be started showing the main menu.

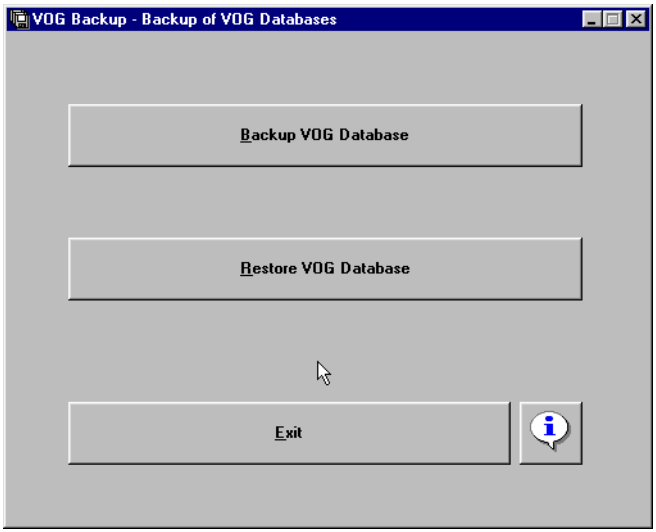


*Fig. C-7: VOG Backup m*ain menu

2. Click the [Backup VOG database] button.

   → The *Backup VOG database* dialog box will appear.

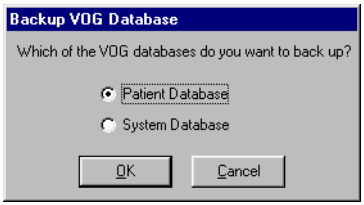


Fig. C-8: Selecting the Database

3. Now decide if you wish to back up the patient database or the system database.

Note

The patient database saves all patient and measured data acquired in the examinations and tests, whereas the system database contains data that is not specific for every patient, but files of common access for all patient databases (for example, text modules for letters, examination templates or a list of all program users).

→ The *Backup VOG database* dialog box will appear. The database used before starting the backup module will be shown in the upper section beginning with **1**.

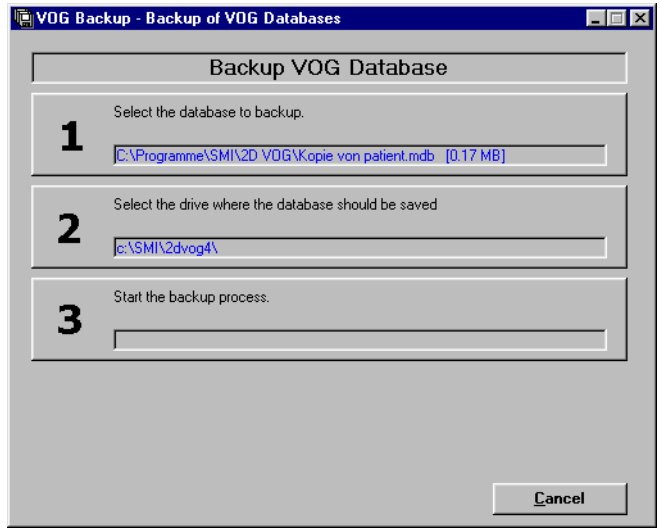


Fig. C-9: Backing up VOG database

4. To continue backing up this database, proceed with step **7.**

To perform backup for another database, using the mouse click section **1** titled "Select the database to backup".

→ The *Select Database* dialog box will be opened.

5. Select the directory, the folder and the database file to be opened.

Note

All databases that can be edited in 2D VOG have the extension **.mdb**.

6. Confirm you selection using the [Open] button.

→ The selected database will be shown.

7. Now, click section **2**.

→ The *Select Drive* dialog box will appear.

8. Select the drive and, if necessary, the destination folder.

Note

When running VOG Backup next time, in this line you will see the path last used.

9. Now click section **3**.

→ The database will be copied to the selected drive and to the given folder where the saved database will be compressed 2.5 times to save capacity.

Note

If you have selected an external medium for saving he database (e.g. disk), you will be prompted to insert a new disk when the disk capacity is exhausted.

→ If backup has been successful, you will receive the message: *"The database xxx has been backuped successfully."*

10. Confirm this message and then click the [Close] button.

    → The *VOG Backup* main menu will appear.

11. If necessary, you can now backup other databases. To do so, repeat step 2. to 10.

5.3.3 Restoring Backup Data

To restore a database, proceed as follows:

1. In the main menu (refer to Fig. C-7), click the [Restore VOG database] button.

    → The *Restore VOG database* dialog box will appear. In the upper section (Number 1) you will see the drive and the path that was last used for backing up a database.

2. If the saved database is found in the shown directory, proceed with step 3.

    Otherwise, click section **1**.

    → The *Select Drive* dialog box will appear. In the right window, all files generated by *VOG Backup* will be shown. These files usually have the extension .001.

    Note

    If a database has been saved on various tapes due to its size, these files are numbered successively (.002, .003 etc.). When restoring data, always choose the file that has the extension .001. The remaining files will automatically be restored.

3. Select the directory of the database and confirm using the [OK] button.

    → Section 1 now shows the selected path.

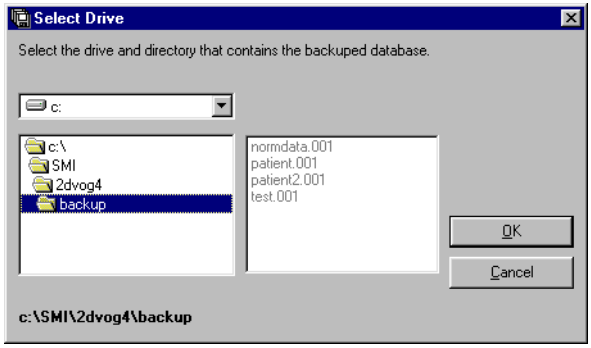


Fig. C-10: Displaying VOG Backup databases

4. Now click section **2**.

   → The *Select Backup File* dialog box will appear. All databases in the selected directory will be listed.

Note

For information purposes, the *Saved* column shows the source directory of this database. Thus, it can easily be seen where every single database was saved during the backup process.

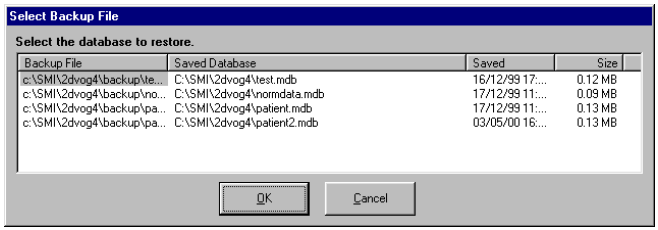


Fig. C-11: Displaying databases in the selected directory

5. Select the database to be restored and confirm using the [OK] button.

   → Section **2** will list the database.

6. Now click section **3**.

   → The *Restore Database* dialog box will appear.

7. Select the destination folder and enter the name of the database to be read. Confirm using the [Save] button.

   → Section **3** will show the destination folder and the file name.

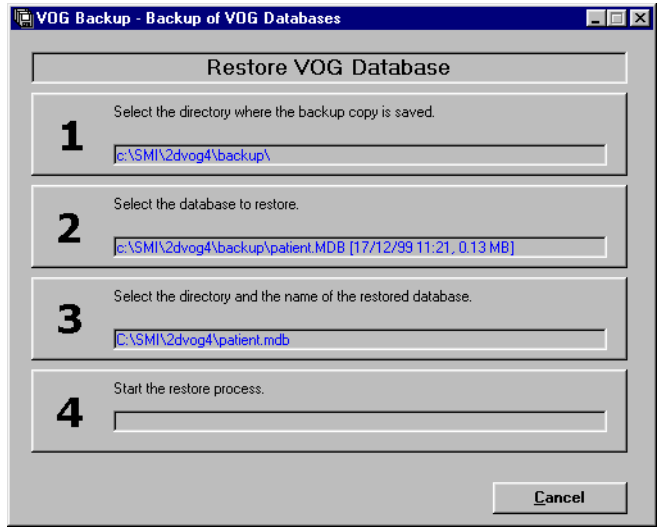


Fig. C-12: Restoring VOG database

8. Click section **4**.

   → The database will be restored to the selected folder.

Note

If a database has been saved on various media, VOG Backup will ask you to insert the next disk once data of the current disk has been copied.

→ When the restoring operation has been successful, the message *"The database xxx was restored suc-*

cessfully as yyy." will be shown.

9. Confirm this message and then click the [Close] button.

   → The *VOG Backup* main menu will appear.

5.3.4 Exit *VOG Backup*

To exit VOG Backup, in the main menu click the [Close] button.

→ The backup module will be closed.

You can now restart 2D VOG to continue working with the program.

D

Working with 2D VOG

SMI
SensoMotoric Instruments

1. The Patient Manager

The Patient Manager is used to manage your patient data. In this dialog box, all patients are stored with the following information:

- Surname, First name
- Date of birth, Sex
- ID number used within the clinic or the surgery
- Medical insurance number
- Address, Phone
- Optional note

Using the Patient Manager, you select the patient you want to perform an examination or test for or whose data you want to view.

Furthermore, with the Patient Manager you can view documents linked to a particular patient.

1.1 Starting the Patient Manager

The Patient Manager will automatically be started after prompting 2D VOG and logging in with your user name. It will be open until you have selected a patient or until you have clicked the [Cancel] button.

To open the Patient Manager again, for example if you want to select another patient,



- in the toolbar, click this button or
- select the 'Patient Manager...' command from the 'File' Menu.

1.2 Selecting a Patient

To select a patient, proceed as follows:

1. If necessary, open the Patient Manager (refer to page D-3).

   → All patients saved in the currently opened database will be listed in alphabetical order.

2. Select the desired patient by double clicking his name.

   Note

   You can also click the patient's name and then confirm, using the [OK] button.

   → The Patient Manager will be closed. The examination data for this patient will be shown in the Examination Tree.

The 2D VOG user surface appears, you can now start performing examinations and tests for this patient.

☞ See page D-9 for a detailed description of the user surface and the Examination Tree.

1.3 Creating a new Patient File

To perform an examination for a patient that has not been examined using 2D VOG up to now, you have to create a new file for this patient.

This may be done either

- via manual input in the Patient Manager or
- by **importing** data from your current patient management system, using a special interface (GDT)

Contact your distributor to receive more information on the "GDT" interface mentioned above.

❍ Manual Creating of a Patient File

To create a new patient file by hand, proceed as follows:

1. Open the Patient Manager (see page D-3) and click the [New] button.

   → The cursor will appear in the NAME box.

2. Enter the required patient information (Name, Address, etc.). An optional note can be saved for the patient in the lower area of the dialog box.

3. Confirm your entries using the [OK] button.

   → The Patient Manager will be closed. The new patient has been created and the Examination Tree for this patient will be displayed.

1.4 Editing Patient Files

To edit a patient file, proceed as follows:

1. Open the Patient Manager (refer to page D-3).
2. Click the name of the patient whose data you want to edit.
   → The patient information will be shown in the lower part of the dialog box.
3. After applying the changes, confirm with [OK].
   → The Patient Manager will be closed. The modified patient file has been saved and the Examination Tree for this patient will be displayed.

1.5 Documents

Using the Patient Manager, you can display documents linked with a patient.

1.5.1 Viewing and opening Documents

To view a linked document, proceed as follows:

1. In the Patient Manager, select the desired patient.
2. Click the [Documents] button.

   → The *documents* dialog box will be opened where all documents will be shown that are linked with the selected patient. Moreover, for you will get the following information for each document:

   – *Document*
   Folder and file name of the document

   – *Used Template*
   Nam of the letter template used to write the letter

   Note

   If you add a document that was not generated by the 2D VOG letter wizard, this column will be empty.

–*Status*
Date when the document was saved last

Note

If a document has been deleted or moved by the Windows Explorer, for example, the status *not found* will be indicated.

3. Select the desired document and then click the [Open Document] button.

4. The document will be opened, and simultaneously the application will be started that is linked with the type of the current file (See the Microsoft® Windows® documentation for more detailed information on linked objects).

1.5.2 Adding new Documents

You may always link a document/file to a patient when you think they might be important for the treatment or management of patients. These files don't have to be text files; you can even link files of any type with the patient.

To do so, proceed as follows:

1. In the Patient Manager, select the desired patient.

2. Click the [Documents] button.

   → The *Documents* dialog box will appear.



3. Click this button.

   → The *Open* dialog box will appear.

4. Select the file to be linked with the patient and confirm, using the [Open] button.

   → The desired file will be linked to the patient, being displayed in the *Documents* dialog box.

2. The User Interface

After selecting a patient in the Patient Manager dialog box, the proper 2D VOG user interface will appear.

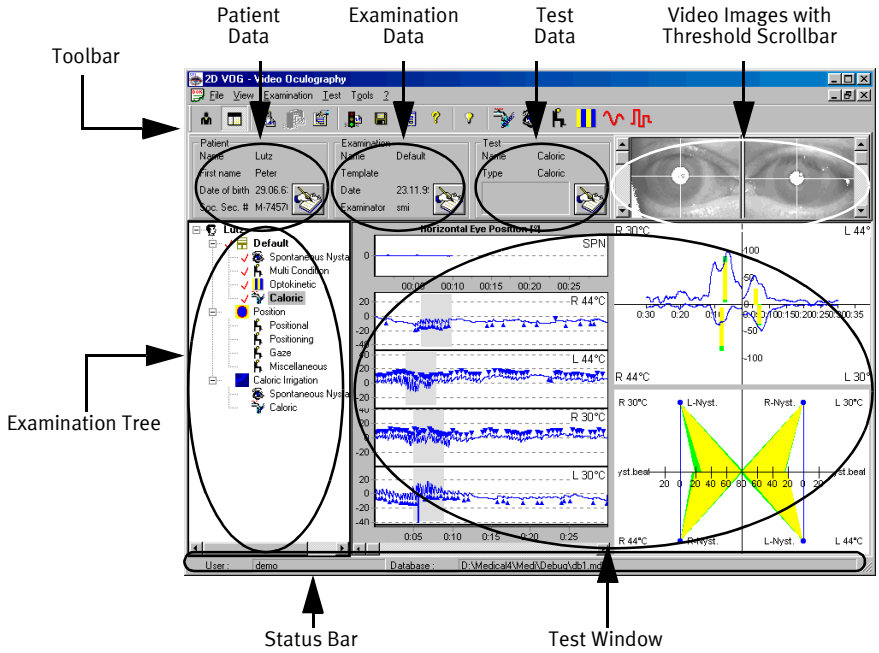


Fig. D-1: 2D VOG User Interface

2.1 Buttons in the Toolbar

Note

The toolbar can be turned on and off, using the 'Toolbar' command from the 'View' menu.

In the toolbar, you may use the following buttons:

General Buttons



Open the Patient Manager



Hide/Show Examination Tree

Examination Buttons



Create a new Examination



Start an Examination



Stop an Examination



Examination Properties



Print Examination

Test Buttons

 Start Measurement

 Stop Measurement

 Save Test Data

 Configure Test

 Print Test

 Turn on/off Fixation LED

Help

 Start Online Documentation

Creating Tests

 Spontaneous Nystagmus Test

 Gaze Test



Multi Condition Test



Saccade Test



Smooth Pursuit Test



Optokinetic Test



Caloric Test

2.2 "Patient", "Examination" and "Test" Areas

Note

Using the 'Test Information' command in the 'View' menu, you can turn on and off (show and hide) the "Patient" area, the "Examination" area, and the "Test" area. In this case, the eye images and the threshold scrollbars are also shown or hidden.

In the upper part of the dialog box, the following areas are shown:

- patient data,
- data for the selected examination and
- data for the selected test.

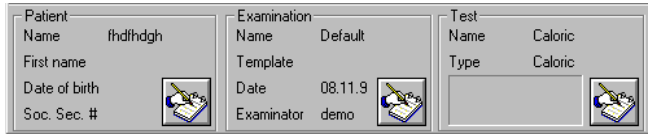


Fig. D-2: Showing Patient, Examination and Test Data

❍ Notes



If you click this button in one of the areas mentioned above, the *Notes* dialog box will appear.

With this dialog box, you can view notes already written or write/edit a note. To do so, just type the note into the text box and confirm with [OK]. The note will be stored automatically.

Note

For each record, a specific note will be saved which means that you can write a separate note

- for each patient,
- for each examination and
- for each test.

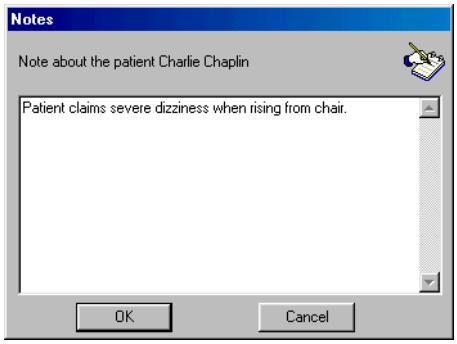


Fig. D-3: Showing notes saved for a patient

2.3 Video Images and Threshold Scrollbars

Note

Using the 'Test Information' command in the 'View' menu, you can turn on and off (show and hide) the eye images and the threshold scrollbars. In this case, the data area will also be shown or hidden.

❍ The Eye Image Area

To control your measurement, in this area the eye images recorded by the VOG mask cameras will be shown. Usually, this area always shows the current images of your VOG mask cameras. The video images are not stored with the data.

Note

According to the medical standards the right eye is displayed within the left video image while the left eye is displayed with the right video image.

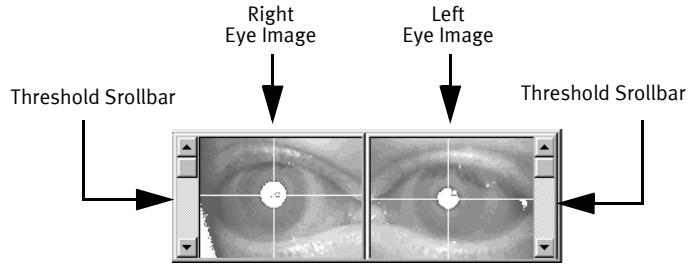


Fig. D-4: Eye Images and Threshold Scrollbar

❍ The Threshold Scrollbar

Your 2D VOG system is tracing eye movements by detecting the pupil center in every video image. The system automatically detects which part of the image represents the pupil by searching

for a round black area. However, the definition of "black" varies with brightness and contrast. The treshold when a point has to be defined as "black" has to be adjusted.

Usually, your VOG system automatically adjusts the treshold, so you don't have to care about this. Nevertheless, in some cases it can be necessary to correct adjustment of the treshold. To do so, use the treshold scrollbar that can be found to the left and to the right of the eye images (refer to Fig. D-4).

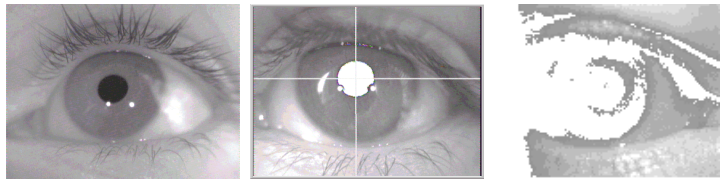
Using these srollbars, during and before performing the test you can adjust the treshold for pupil detection. This is done separately for the left and the right eye.

The treshold can be adjusted in two different ways:

a) Click the ▲ button to increase the treshold. Using the ▼ button, you will reduce the treshold.

b) Move the scrollbar upward with the mouse to increase the treshold and move it downward to reduce it, respectively.

The treshold will be adjusted correctly, if the pupil is colored completely white (refer to Fig. D-5).



Treshold adjustment to low | Correct treshold adjustment | Treshold adjustment to high

Fig. D-5: Changing Treshold

If threshold is set correctly, the crosshair will be in the center of the pupil and will follow the eye movements.

Note

Before starting an examination, check the treshold adjustment by asking the patient to look in succession to the left, to the right, upwards and downwards. If the crosshair follows the pupil until these eye positions are reached, threshold setting will be correct.

2.4 The Examination Tree

Note

The Examination Tree may either be turned on or off, using the 'Examination Tree' command from the 'View' menu.

In the Examination Tree, you can see all examinations and tests performed for this patient with the 2D VOG system. This allows you to review the complete history of your patient at a glance.

❍ Examination Tree Structure

The Examination Tree is similar to the one in Windows Explorer and can consist of up to three levels (refer to Fig. D-6).

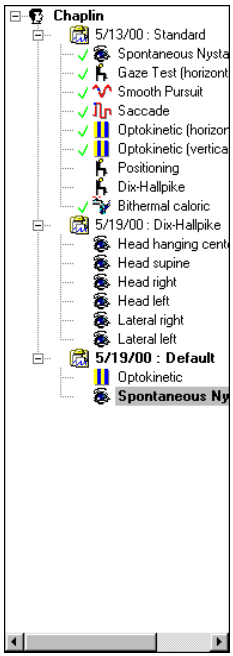


Fig. D-6: Examination Tree

First Level:	Patient name
Second Level:	Displays examinations
Third Level:	Displays planned and performed tests

Note

Tests that have been performed and saved are marked with a green tick (✓).

❍ Show and Hide Examinations / Tests

After selecting a patient from the Patient Manager, the Examination Tree will be shown where all existing levels still are open.

Hide Examinations / Tests:
You can hide single examinations or tests that belong to a specific examination, so the examination can be displayed in a more compact form.

Beside the level to be closed, click ⊟.

→ The examinations / tests that are beneath the corresponding level will be hidden.

Example

You click ⊟ beside the patient's name.

→ All examinations including the test that belong to this examination are hidden. The Examination Tree only shows the patient's name.

Beside the examination, you click ⊟.

→ All tests that belong to this examination are hidden.

Show Examinations / Tests:
To show the hidden examinations, click ⊞ beside the corresponding level.

→ The examinations / tests will be shown again.

❍ Creating a new Examination



If you create a new examination by clicking this button or selecting the 'Examination | New' command, this examination and including test will be inserted at the end of the Examination Tree.

❍ Creating a new Test

You can create a new test by clicking a test button (refer to page D-11) or using the command 'Test | New | ...'. When creating a test, you should consider the following:

– If you have selected the patient name or if you haven't made any selection before creating the new test, **at the end of the Examination Tree** a new examination entitled *Default* will be inserted that contains the corresponding test.
– If you have selected a special examination before adding the new test, this test will be inserted **at the end of the selected examination**.

❍ Opening a stored Examination

To open a stored examination, click the selected examination.

→ Tests that might be hidden are shown in the Examination Tree. The Examination Properties dialog will be opened where you can change the Properties.



Note

If the desired examination is not displayed, make sure the corresponding level is shown in the Examination Tree (refer to page D-19).

❍ Opening a stored Test

To open a stored test, click the selected test.

→ In the test window, the stored data of this test will be displayed. If the test has not been performed yet, the test window is empty.

Note

If the desired examination is not displayed, make sure the corresponding level is shown in the Examination Tree (refer to page D-19).

2.4.1 Context Menu

If you click an element of the Examination Tree using the right mouse button, the context menu will be opened. Depending on whether you have selected the patient's name, an examination or a test, the options that you have in this menu are different.

❍ Patient Context Menu

The patient context menu contains the following commands:

- New Examination...
  The "Create new Examination" dialog will appear where you can select a new examination which will be inserted at the end of the Examination Tree.
- Documents...
  The *"Documents"* dialog will be opened. It shows all letters that have been written and saved for this patient (refer to page D-7).
- Properties...
  The Patient Manager will be opened showing the patient data (refer to page D-3).

❍ Examination Context Menu

The Examination Context Menu includes the following commands:

- Add Test
  You can add a new test for an examination. After selecting the test it will be inserted at the end of the examination.
- Start
  The examination is started.
- Print...
  Examination data is printed (refer to page D-57).

– Delete
The examination including the related tests and recorded data is deleted.
– Properties...
The *"Examination Properties"* dialog box will appear. In this dialog box, you can edit the title of the examination and write a note for this examination.
– Letter Wizard
The Letter Wizard is started. You can now write a letter (refer to page D-61).

❍ Test Context Menu

Depending on whether a test is opened or not, using the context menu you can execute the following commands:

Test not opened:

– Open
The test is opened in the test window, the recorded data of the test is displayed.
– Delete
The test including all recorded data is deleted.

Open Test:

– Close
The test is closed, the test window is empty.
– Notes...
You can write and save a note for this test.
– Print...
The test data is printed (refer to page D-57).
– Save
The recorded data of a test is saved.
– Delete
The test including the recorded data is deleted.

– Export...
  The test data is exported, being saved as text file (*.txt*) for further processing.
– Properties...
  The Test Properties dialog box appears where you can configure the test (graphs, analysis settings, calibration etc.) (refer to part **F – Tests**).

2.4.2 Turn off the Examination Tree



Of course, you can hide the Examination Tree of a patient. To do so, click this button.



Note

To show the Examination Tree, click the button shown above.

2.5 The Test Window

The Test Window shows recorded data and analysis.

The type of diagrams and statistical data displayed depends on the test type. For detailed information, please refer to chapter F.

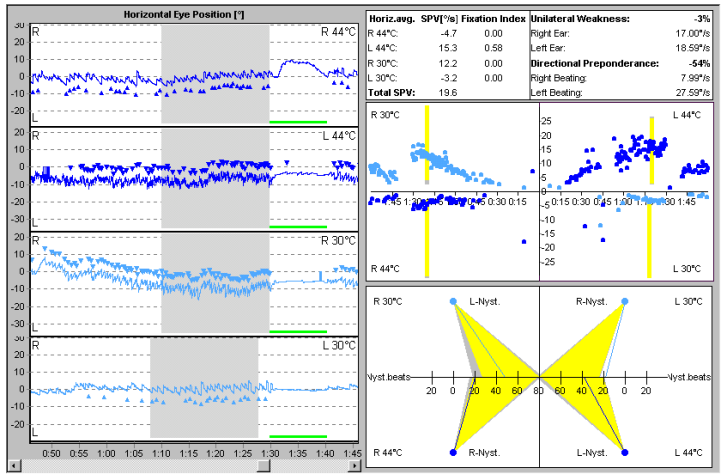


Fig. D-7: Displaying a caloric test in the Test window

2.6 The Status Bar

Note

The status bar can be turned on and off, using the command ’Status Bar’ from the ’View’ menu.

The status bar contains the following information:

– User logged in
– open Database



Fig. D-8: 2D VOG status bar

3. Test

To create a test and start recording the test data, please check the following:

- Have you selected the correct database (refer to page D-67),
- Has the VOG mask been set correctly on the patient's head (refer to page B-20),
- Has the treshold been adjusted properly (refer to page D-15)?

3.1 Creating a Test

To create a test, proceed as follows:

1. From the Patient Manager, select the desired patient (see page D-4).

2. You can create a new test by clicking a test button (refer to page D-11) or using the command 'Test | New | ...'. When creating a test, you should consider the following:

   - If you have selected the patient name or if you haven't made any selection within the Examination Tree before creating the new test, a new examination entitled *Default* that contains the corresponding test will be inserted **at the end of the Examination Tree**.

   - If you have selected a special examination before adding the new test, this test will be inserted **at the end of the selected examination**.

3.2 Configurating a Test

After creating a test, you can configurate it. To do so, you can apply various settings that might affect the measuring process of the test or the display of the recorded data.

In the following chapters, you will learn about

– how to open to configuration dialog box
– the elements of the configuration dialog box.

3.2.1 Opening Configuration Dialog Box

To open the configuration dialog box, proceed as follows:

1. Make sure the selected test is opened (refer to page D-39).
2. Open the "C*onfiguration XXX Test"* dialog box in one of the following ways:
   a) From the 'Test' menu, choose the 'Configuration...' command.
   b) Double click into the test window.
   c) In the Examination Tree, click the test using the right mouse button and select the 'Configuration...' command from the context menu.

   → The *"Configuration XXX Test"* dialog box will be opened.

3.2.2 Elements of the Configuration Dialog Box

The configuration dialog box is divided into various tabs ordered by topics.

The following description should help you to get an overview which configuration options you have in each tab.

Note

The tabs are similar for all test.

☞ All configuration features that are specific to one test are described in the **F – Tests** part in this manual.

"Test" Tab

In this tab, all settings that affect the measurement process are done. Select the mask that is used and indicate whether you wish to examine the left, the right or both eyes (as far as the system architecture allows this).

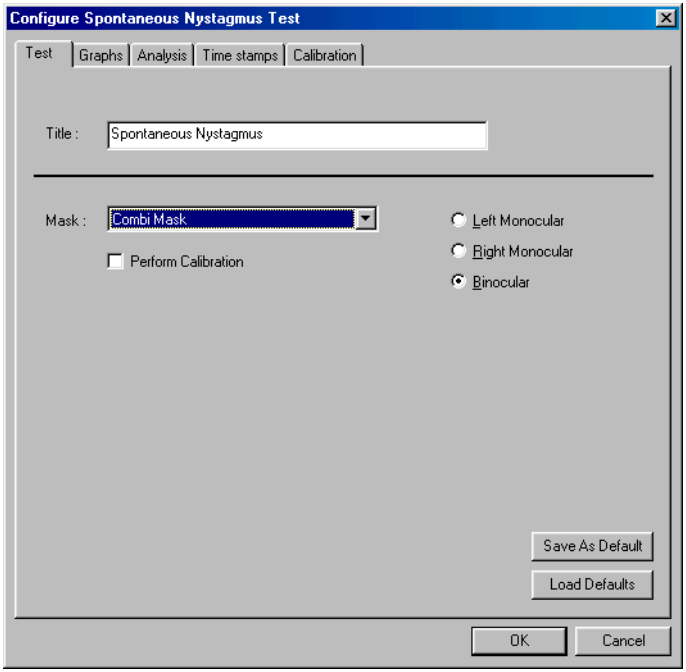


Fig. D-9: "Test" tab of the spontaneous nystagmus test

Additionally, you can enter an own title for the test that will be shown in the Examination Tree.

If you check the "Perform Calibration" option, a calibration will be performed before test recording is started.

"Graphs" Tab

Using this tab, you will define how your measurement results will be shown in the test window and in the "One Test Per Page" printout.

For all tests, you can

- turn on and off the display of each graph
- define scaling of the axes
- set grid lines
- mark the nystagmus beats detected by the 2D VOG system by triangles
- center the measure graphs.

Although these settings don't affect the measurement results in your database, they are saved to provide the same settings when you will open the test next time.

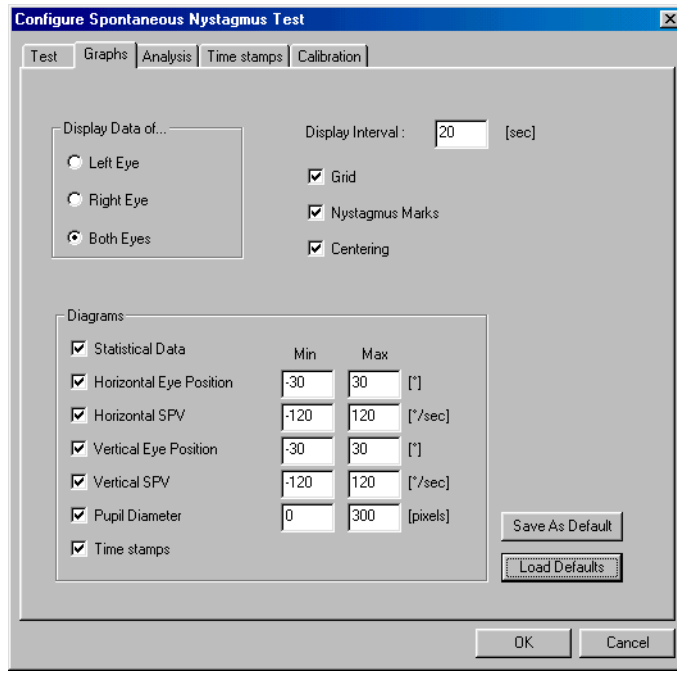


Fig. D-10: "Graphs" tab of the spontaneous nystagmus test

"Analysis" Tab

This tab allows you to optimize the automatic 2D VOG nystagmus analysis according to your individual requirements.

☞ For more information, refer to page D-69.

"Stimulus" Tab

This tab shows all control parameters for the stimulation devices. This is especially required for the VisualLab that is used with the optokinetic test. This tab is also available from the configuration dialog box of the caloric test.

"Time Stamp" Tab

This tab is only available in the configuration dialog box of the spontaneous nystagmus test.

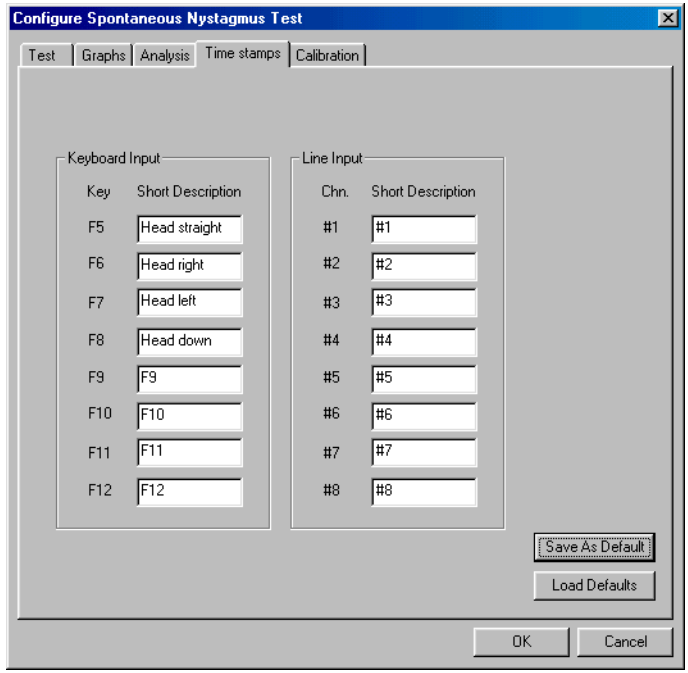


Fig. D-11: "Time stamps" tab in the spontaneous nystagmus test

Here you can define keys for "time stamps" that can be set when recording a Spontaneous Nystagmus Test by

– either using the keyboard
– or via the digital inputs (if you have purchased the *Digital In* option)

"Calibration" tab

Using this tab, you determine what stimulus type you want to use for calibration.

For more detailed information on calibration, refer to chapter *„Calibration“* on page D-49.

Note

The parameters that are defining the measurement procedure of a test, can not be changed after finishing measurement.

This especially includes the settings performed in the *Test*, *Stimulus* and *Calibration* tabs where you define, for instance, which mask is to be used and whether you deal with a derivation of the left or the right eye.

All other parameters can be changed after completing the measurement. This won't affect the stored measure data.

Advantage:
If you are using the procedure, it is ensured that the conditions under which the measurement is performed can be reproduced at any time.

3.2.3 Saving Configuration as Default

2D VOG allows you to save the settings as default that you have defined for a particular test. This is recommended if you perform this test type always using the same settings. For each of those tests that is created and performed the default settings will be used.

Example

In the *Graphs* tab of the configuration dialog box of a caloric test, you select the graphs

- Butterfly,
- Freyss,
- Claussen,
- Haid/Stoll,

saving them as default.

When creating and performing a caloric test next time, these default settings will be used.

Of course, you still can change the default settings for each test by configurating the test as described on page D-28.

Note

Tests that have already been created or performed are not affected by the new settings.

Even the tests created within the examination templates are not affected by the new default settings. However, when you create a new examination template, the new default settings will be used for those tests that will be created at that time.

To save a configuration as default, proceed as follows:

1. Open the configuration dialog box of the selected test (refer to page D-28).
2. Define the required settings by using the tabs.
3. 

Save these settings as default by clicking the [Save as Default] button.

   → The new default settings will be used for the current test and for all future tests that will be created for a patient.

3.2.4 Loading Default Settings

Changes that have been applied to the test settings can always be canceled by loading the default settings.

To do so, proceed as follows:

1. Open the configuration dialog box of the selected test (refer to page D-28).
2. 

Click the [Load Defaults] button.

   → The saved default settings will be loaded. The changes that have been applied during the configuration will be canceled.

3.3 Performing a Test

After creating a test, you can perform the complete measurement just by pressing the foot switch so you have both hands free to look after the patient or to operate other devices (e.g. caloric irrigators).

To perform the test, proceed as follows:



1. Start the test by pressing the foot switch, clicking this button or selecting the 'Start' command from the 'Test' menu.

   → Recording of the test data will start.

   Note

   Some tests may consist of various subtests. In this case, after starting the test a dialog box opens that shows all subtests.

   Now start recording by selecting the desired subtest and the confirm using the [OK] button or pressing the **foot switch**.

2. As some tests have a configurable time limit, they will stop automatically. However, you can stop them manually at any time.

   Other subtests **have** to be stopped manually.

   To stop recording, press again the foot switch, click the [OK] button or select this button from the toolbar.

3. Proceed as described above until the selected subtest have been performed. Of course, you may skip a single subtest. In this case, after finishing a subtest select the next subtest manually that is to be performed.

4. Now save the test by



a) clicking this button in the toolbar,

b) selecting the 'Save' command from the 'Test' menu

c) or closing the test and answering the request for saving data with [Yes].

→ Test data is saved. The test is completed.

Note

For each test, recorded data and graph settings are saved.

When opening a saved test, data and diagrams will be shown as configured when the test was saved.

☞ Detailed information on each test can be found in part **F – Tests** of this manual.

3.4 Opening Test / Displaying Recorded Data

To open a test (for instance, to display the data for this test or to print those data), proceed as follows:

1. In the Examination Tree, double click the desired test.

   Note

   If the desired test is not displayed, make sure the corresponding level is shown in the Examination Tree (refer to page D-19).

   → The test will be opened and test data will be displayed in the test window.

2. To close the test, open the context menu and select the 'Close' command.

3.5 Overwriting Recorded Data

To overwrite recorded data, proceed as follows:

1. In the Examination Tree, click the test whose data you wish to overwrite.

   → The saved data will be shown in the test window.



2. Start the test by pressing the foot switch, click this button or select the 'Start' command from the 'Test' menu.

   → You will be asked whether you are sure to overwrite the data.

3. Confirm this security request.

   → Recording of the test data will start.

4. Now proceed as described on page D-37, step 2.

3.6 Deleting Recorded Data

To delete the recorded data, proceed as follows:

To delete only the recorded data, not the test itself including its settings, proceed as follows:

1. In the Examination Tree, click the test whose recorded data you wish to delete.

   → The saved data will be shown in the test window.

2. From the 'Test' menu, select the 'Clear Data' command.

   → The recorded data will be deleted without deleting the test entry in the Examination Tree nor the test settings.

3.7 Deleting a Test

To delete a test including its entry in the Examination Tree, proceed as follows:

1. In the Examination Tree of the patient, select the desired test.

2. Now delete the test by using one the following ways:

   a) From the context menu, choose the 'Delete' command or

   b) Select the 'Delete' command from the 'Test' menu.

→ The test will be deleted including all recorded data and the selected test settings.

4. Examination

Several tests performed within a patient examination are summarized to build one "examination". Furthermore, 2D VOG allows you to define complete examination protocols as templates where each test can be configurated separately. After defining the examination templates, you only have to open your favorite template to create all tests of your examination protocol in the Examination Tree.

☞ On page E-3 you will find more information on how to create examination templates.

Once you have opened an examination template, 2D VOG will guide you through all tests of your protocol automatically. The only thing you have to do is pressing the foot switch. Of course, you can always interrupt the protocol procedure and create or start additional tests, using one of the test buttons in the toolbar.

Before creating an examination and performing the tests, please check the following:

– Have you selected the correct database (refer to page D-67)?
– Has the VOG mask been set correctly on the patient's head (refer to page B-20)?
– Is the treshold adjusted properly (refer to page D-15)?

4.1 Creating an Examination

To create an examination, proceed as follows:

1. From the Patient Manager, select the desired patient (see page D-4).



2. Click this button in the toolbar or select the 'New' command from the 'Examination' menu.

   → The Create new Examination dialog box will appear that shows all saved examination templates.

3. Select an examination and confirm, using the [OK] button.

   → The examination and all tests that belong to this examination will be inserted at the end of the Examination Tree.



Note

If you click the [Start] button after selecting the examination, the examination will be started immediately.

4.2 Performing an Examination

To perform an examination, proceed as follows:

1. If you haven't clicked on the [Start] button when selecting the examination template, you have to start the examination by hand.



To do so, mark the examination in the Examination Tree. Then click this button in the toolbar or choose the 'Start' command from the 'Examination' menu.

→ A window will appear showing all tests that have been defined in the Examination Tree. The first test not performed so far will be marked.

2. Click [OK] or press the foot switch.



Note

Of course, you may perform the test in any order by marking the desired test before you press the foot switch or click the [OK] button.

→ The selected test will be opened.

3. Press the foot switch again or click this button.



→ The test will be started.

4. To stop recording of the test data, press again the foot switch or click this button.



→ The test will be stopped.

5. Press again the foot switch or click this button.



→ The test data will be saved. The 'Perform xxx Examination' dialog box will reappear. The next test not performed so far will be marked.

6. Repeat steps 2 to 5 until you have performed all desired tests.

Note

You don't have to perform all tests of an examination. You may perform the other tests later.

To do so, start the examination as described in step 1. In the 'Perform xxx Examination', the first test not performed so far will automatically be marked.

4.3 Performing Additional Tests

If you want to perform further tests in addition to the test already given, you should proceed as follows:

1. In the toolbar, just click the button for the desired test.

   Note

   If the test shall become part of the current examination, make sure the examination it marked before clicking the test button in the toolbar.

   Otherwise, at the end of the Examination Tree will be inserted a new examination entitled *Default* that contains the new test.

2. Perform the additional tests as usually.

   For a detailed description of how to perform tests, refer to page D-37.

4.4 Changing Properties

You can always apply changes to the examination properties or add a note for the examination. To do so, proceed as follows:

1. In the Examination Tree of the patient, select the desired examination.

2. From the 'Examination' menu, choose the 'Properties' command.

   → The Examination Properties dialog box will appear.

3. You can now:
   - change the title of the examination
   - change the examination date
   - write a note
     This note can be shown at any time by clicking this button in the 'Examination' area.



4. Confirm the changes by using the [OK] button.

   → The changes will be saved.

5. Calibration

In contrast to Electro-nystagmograpic (ENG) devices using electrodes calibration is not absolutely necessary in a 2D VOG system to determine the amplitude of the eye movements in degrees. Your 2D VOG system has already been adapted to your 2D VOG mask ex factory. However, ex works calibration can never be exact, because with each patient you can have individual deviations (e.g. depth of the eye orbita). Therefore, 2D VOG allows you to perform an individual calibration for each patient.

Note

With tests where the position or velocity of a stimulus is not compared with the measured eye movements (e.g. spontaneous nystagmus test), calibration is not generally required. In these cases, the accuracy of ex works calibration is good enough.

Before performing a test where the eye movement is compared with a given stimulus of a known position or velocity (e.g. optokinetic test), you should calibrate the system.

The following table will help you to get an overview for which tests calibration is recommended:

Calibration not necessary	calibration required
– Spontaneous nystagmus test, – Caloric test, – Multi condition test	– Optokinetic test, – Smooth pursuit test, – Saccade test – Gaze test

During the calibration, the patient is fixating various given points on a wall or monitor in succession while the eye position is being measured with 2D VOG.

5.1 Configurating Calibration

The calibration parameters can be set for each test. To do so, proceed as follows:

1. Open the test for which you want to configure the calibration.
2. From the 'Test' menu, choose the 'Configuratioh...' command or double click on the Test window.

   → The configuration dialog box of the test opens.
3. Select the *Calibration* tab.

5.1.1 Selecting the Calibration Pattern

For the calibration, 2D VOG offers you one 2-point and two 5-point calibration patterns. The best results are achieved using the 5-point patterns.

5.1.2 VisualLab Calibration

You can use your VisualLab to project the calibration points either on a monitor or - using a projector - to the wall. In this case, dots are positioned automatically while calibration is performed.

Note

Make sure that the geometric conditions of the test arrangement have been set correctly in the *"Geometry of VisualLab Projection"* dialog box.

See page B-9, „*System Setup VisualLab*“ for more detailed information.

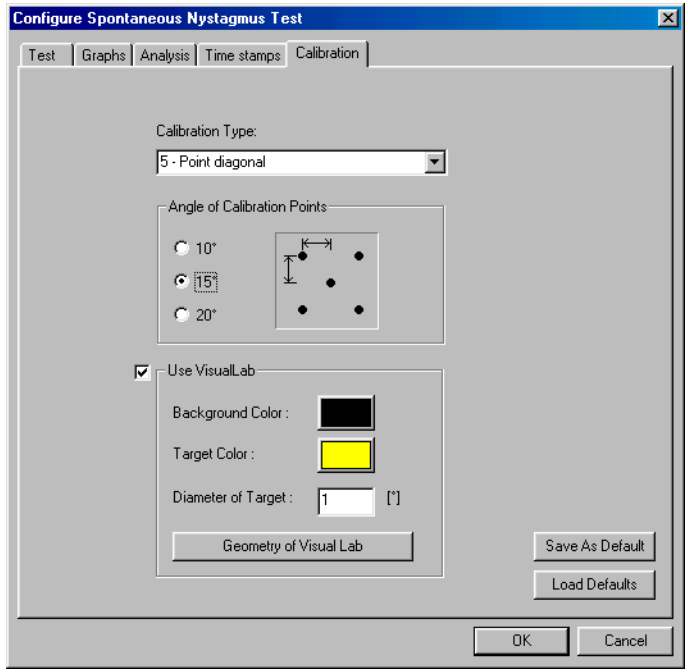


Fig. D-12: Configurating Calibration

To perform the calibration using the VisualLab, proceed as follows:

1. Select the USE VISUALLAB box.

2. In the BACKGROUND COLOR and TARGET COLOR fields, choose a color combination that will provide a good contrast on the wall.

3. In the DIAMETER OF TARGET box, choose a size as little as possible. But make sure that the point is still visible.

5.1.3 Calibration without VisualLab

To perform a calibration without the VisualLab module, the calibration points can be marked by using colored markings on a wall.

To do so, you have to calculate the position of these markings according to the environmental circumstances. shows an example of how the distances between the markings can be chosen if the distance between the patient and the wall is 1 m / 254".

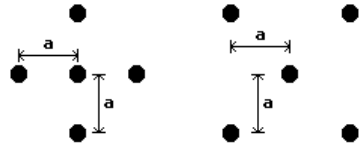
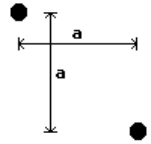
Calibration Type	Calibration Point Angles 10°	15°	20°
5-point linear / 5-point diagonal 	a = 7.1“	a = 10.6“	a = 14.2“
2-point 	a = 14.2“	a = 21.3“	a = 28.3“

Fig. D-13: Example for marking distances, if the distance between patient and wall is 1 m /254"

5.2 Performing Calibration

After creating a test, you can perform the calibration before the measurement is started.

Note

You can also define that each time a particular test is started calibration is to be performed.

For more information see chapter *„Calibration before Each Test“* on page D-55.

The perform the calibration, proceed as follows:



1. In the toolbar, click this button.

   → The *calibration* dialog box will open.

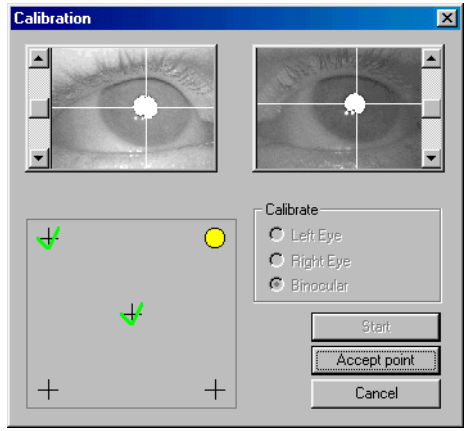


Fig. D-14: "Calibration" dialog box

The upper area of the dialog box shows the video images of the eyes that are being calibrated. If necessary, adjust the pupil threshold to achieve optimal eye detection (see page D-15).

2. Now ask the patient to look at the first calibration point and then select the desired calibration mode by clicking the [Automatic] or [Manual] button.

Automatic Calibration

If you are using this mode, the calibration algorithm will proceed automatically through the whole calibration sequence. Once a target point has been fixated well enough, the next target point will be displayed.

Manual Calibration
This mode should be used for patients that have difficulties in fixating the calibration points. For example, it may occur that the automatic fixation detection is not successful with patients showing excessive spontaneous nystagmus beating or with inattentive persons (e.g. children).

In this case, you can observe the video image of the calibrated eye on the screen. When you think the calibration point has been fixated exactly, click the [Accept Point] button.

→ The lower area of the dialog box shows the point pattern. Each point to be fixated is marked by a yellow circle. Points that have already been successfully measured, are marked by a green tick.

3. Ask the patient to fixate the currently active calibration point. (If you are using the VisualLab module, only the active calibration point will be displayed.)

→ When calibration has been successfully terminated, the dialog box will be closed automatically.

Note

You can always stop a current calibration by clicking the [Cancel] button.

❍ Calibration before Each Test

If you want to perform a calibration by default before starting a particular test, proceed as follows:

1. Create a new test of the desired type.

   → The test will be inserted in the Examination Tree, the test window is open.

2. In the 'Test' menu, select the 'Configuration...' command and double click in the test window.

   → The *"Configuration XXX Test"* dialog box will be opened.

3. Open the *Test* tab and check the PERFORM CALIBRATION box.

4. Click the [Save as Default] button.

→ The calibration will be included in the measurement procedure. Each time when a measurement is started the *"Calibration"* dialog box will be opened.

Note

Within a complete examination, calibration is necessary only once. If calibration has already been performed during a prior test, 2D VOG will load these values. In this case, the calibration dialog box won't be opened.



However, if the mask has been slipped down or removed since the last calibration has been performed, you have to repeat the calibration. In this case, start the calibration by hand by clicking this button in the toolbar.

6. Printing

You can choose whether you'd like to print

– a single test or
– particular / all tests of an examination.

6.1 Printing a Test

To print the measure data of a single test, proceed as follows:

1. Open the Patient Manager by selecting the desired patient (see page D-4).
2. In the Examination Tree of the patient, open the test to be printed (refer to page D-39).

   → The test data will be shown in the test window.
3. In the 'Test' menu, select the 'Print...' command.

   → The Microsoft® Windows® printer settings dialog box will appear where you can apply changes if necessary.
4. Confirm using the [OK] button.

   → The test will be printed.

Note

Data will be printed as "One Test per Page" (see page D-59).

6.2 Printing an Examination

If you want to print various or all tests of an examination, you should proceed as follows:

1. Open the Patient Manager by selecting the desired patient (see page D-4).

2. In the Examination Tree of the patient, select the examination.

3. In the 'Examination' menu, select the 'Print...' command.

   → The *"Print Examination"* dialog box will appear. In the Tests to be printed area, all tests are shown and selected that have been created for this examination.

4. **Printing particular Tests:**
   If you want to print only particular tests, select these by keeping pressed the CTRL key and then clicking the desired tests one after the other.

   Note
   If you want all tests to be included, click the [Select all] button.

5. In the lower area titled *Available Reports*, you have to select whether the tests are to be printed as
   – One Test per Page or
   – Four Tests per Page.

   (For more information, see the following descriptions).

6. Click the [Print] button.

   → The Microsoft® Windows® printer settings dialog box will appear where you can apply changes if necessary.

7. Confirm using the [OK] button.

   → The selected tests will be printed.

6.3 Four Tests per Page

The test data is printed in landscape orientation with a maximum of four tests on one page.

With this type of report, for each test only the master data will be printed.

6.4 One Test per Page

Using this option, you can define the printout yourself. To do so, proceed as follows:

1. Change the graph section until the desired information is displayed (refer to page F-3).

2. Save the test (refer to page D-37).

   → Data will be printed in the same form as it was shown in the test window when the test was saved last time.

7. Letter Wizard

For each examination that has been performed you usually need to write a letter to the referring doctor. To meet these purposes, a special letter wizard was included in the 2D VOG software that will assist you with this job.

You can create **letter templates** that can be defined according to your individual ideas. This means you can easily and nearly automatically create your letters using the letter wizard.

☞ For more detailed information on creating letter templates, see page E-11.

7.1 Creating a Letter

To create a letter, proceed as follows:

1. Open the Patient Manager by selecting the desired patient (see page D-4).
2. Select the examination and choose the 'Letter Wizard' command from the 'Examination' menu.

   → The letter wizard will open showing the following information:
   – Patient name
   – Examination name
   – Tests that have been performed
   – Letter template last used



3. If you require another letter template for your letter, click this button and select your template.
4. Confirm using the [Next] button.

5. Now you will be guided through the wizard step be step. For each field that has been defined in the letter template the wizard will open a dialog box where you can enter or select the required information.

Note

Most information will automatically be entered by 2D VOG according to your letter template. For further important information like "Referring Doctor", "Diagnoses" or "Proposed Treatment", 2D VOG provides an own database (refer to *„Inserting New Field Contents“*, page D-63). This means you only have to enter your information once, and when creating the next letter you can already use it.

However, as other information (e.g. "Case History") is related to one patient, it cannot be saved in a database for common use. Please enter this kind of information by hand in the required dialog box.

6. Browse through all windows of the letter wizard this way until you reach the last window.



This window will suggest you the folder and the file name of the letter to be stored. You may choose another folder or enter another file name. To do so, click this button and change the suggested options.

7. Decide whether the letter is to be
   - saved and edited
   - saved and printed
   - saved (only).

8. Confirm you entries by clicking the [Finalize] button.

   → According to your selection the letter will be saved and edited or printed. If you have chosen the Save and Edit option, the letter will be opened and the application will run that is linked to **.rtf** files (for more information, refer to the Microsoft® Windows® online documentation).

Note

The saved letter will automatically be linked to the related patient. All letters that are saved for the patient can be opened any time, by clicking the [Documents] button within the Patient Manager (see page D-7).

❍ Inserting New Field Contents

To extend the possible selections of the field contents that are available in the letter wizard, proceed as follows:



1. In the corresponding dialog box, click this button and enter the required information in the lower part of the window.

2. In most cases, this information is divided into two parts: First, you can enter a "title" in the list box. This title will be listed in the upper part of the window for later quick selection. The text entered in the lower text box will be inserted into your letter by the wizard.

   In the "Refering doctor" dialog, in the lower part of the window you can enter name, title and address of the doctor you want to address the letter to.



3. Then save your entries by clicking this button.

   → The new list box item can already be selected when you are creating the next letter.

In the long term, you will be very fast when you are writing your letters, as you only have to select the required data (except for very few items).

❍ Deleting List Box Items

To remove an item from the list box selection (e.g. if a particular doctor gave up his surgery), proceed as follows:

1. In the corresponding window, select the list box item.



2. Click this button.

   → The list box item will be removed.

7.2 Opening a Letter

A saved letter is automatically linked to the related patient. All letters that are saved for the patient can be opened at any time, using the Patient Manager (see page D-7)

☞ For more detailed information, see page D-7.

8. Databases

2D VOG saves all patient and test data that is acquired during an examination or a test in a database file. 2D VOG allows you to work with different databases. To do so, you can create as many databases as you like. This is useful if the system is shared by several doctors who wants to use separate databases.

Note

Different databases cannot be linked to build one database. Once you have created more than one database, it is not possible to copy data from one database to another. Please consider this fact before you decide to use several databases.

8.1 Creating a New Database

To create a new database, proceed as follows:

1. From the 'File' menu, select the 'New Database' command.

   → The *Save as* dialog box will open.

2. Select the folder and the file name of your new database file. Confirm using the [Save] button.

   → The new database will be created. The Patient Manager will appear where you can add a new patient.

8.2 Opening a Database

Usually, after starting 2D VOG you will use the database last opened.

To open another database, proceed as follows:

1. If necessary, close the Patient Manager.
2. Select the 'Open Database...' command from the 'File' menu.

   → The *Open* dialog box will appear.
3. Select the drive, the folder and the database file name to be opened.

   Note

   All database files that can be used in 2D VOG have the extension **.mdb**.
4. Confirm your selection using the [Open] button.

   → The database will open. The Patient Manager will appear showing all patients saved up to now. The status bar will show the file name of the opened database.

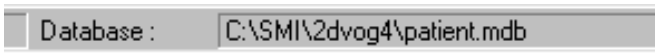


Fig. D-15: File name of opened database displayed in status bar

9. Nystagmus Analysis

The nystagmus analysis constitutes a major part of your 2D VOG system. It searches for nystagmus beats in the eye movement data and detects them automatically. In addition to detecting nystagmus beats it also calculates other parameters like the slow phase velocity or the frequency of nystagmus beating

2D VOG contains two different algorithms of the automatic nystagmus analysis:

– Continuous nystagmus analysis
– Discrete nystagmus analysis.

9.1 Continuous Nystagmus Analysis

With this nystagmus analysis, the current velocity of the eye movement is calculated for each data point acquired during the measurement which is

– 50 times per second for a PAL system,
– 60 times per second for a NTSC system.

As soon as the eye velocity exceeds a maximum velocity value, a quick phase start is recognized. The end of a quick phase is determined when the eye velocity falls below the maximum velocity value.

The treshold values used to determine the start and end points of a quick phase are adjusted automatically, and the detected points are shown graphically.

Velocity of the slow phase is determined of the temporal smoothing (low pass filtering) of the velocity calculated *outside from detected saccades and artefacts*. During the duration of the saccades or artefacts its velocity value is the average immediately

proceeding and following the saccade. Low pass filtering of the velocities which is constant in time is adapted to the corresponding requests of the tests (e.g. 2.0 s in the caloric test and 0.25 s in the smooth pursuit test) to avoid distortions of the test results.

❍ Advantage of the Continuous Nystagmus Analysis

The advantage of this technique of nystagmus analysis is that the eye velocity is determined for each data point which allows you to examine a continuous vestibular oculomotoric reaction like it is shown i.e. in the Optokinetic Test.

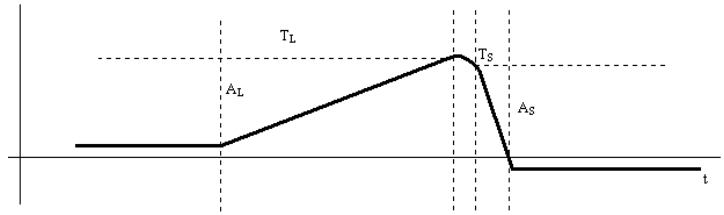
❍ Disadvantages of the Continuous Nystagmus Analysis

A disadvantage of this analysis technique is that with tests with low nystagmus activity like e.g. the spontaneous nystagmus test the slow phase velocity can be detected for any data point, but no separate measure value is generated for each nystagmus beat.

If the continuous analysis is used, with nystagmus beats appearing occasionally the real nystagmus velocity can not properly be detected. This was the reason for developing a second analysis technique for 2D VOG, the **Discrete nystagmus analysis** (see following description).

9.2 Discrete Nystagmus Analysis

This analysis technique looks for saw-tooth like nystagmoide eye movements that can be recognized by a quick eye movement which is preceded by a slow movement into the opposite direction.



$V_L = A_L / T_L$ (Velocity of the slow phase)
$V_S = A_S / T_S$ (Velocity of the quick phase)

Fig. D-16: Schematic view of a nystagmus beat

Possible nystagmus beats are identified, if

- the calculated velocity is higher than the given minimum velocity V_{Smin} set for a quick phase
- the amplitude of the quick phase is higher than the given minimum amplitude A_{Smin} set for a quick phase
- the time elapsed since the last quick phase is greater than the given minimum duration T_{Lmin} set for a slow phase.

If these conditions are met, further formal criteria will be checked to identify a nystagmus beat, as:

- Acceleration during the turning back phase (how is the turning back pointed?)
- Amplitude of the slow phase
- Velocity of the slow phase
- Duration of the slow phase
- Allowed deviation of data points from the calculated regression line (relative value).
  The higher this value is set, the more linear the eye movement has to be to be detected as slow phase.

The parameters for the analysis are set in a dialog box.

❍ Adjustable Parameters

Variable Identifier	Description	admissible Range	Default Values	Unit
B_{min}	min. acceleration in turning back phase between T_L and T_S	0 ... 10000	800	[°/s²]
V_{Smin}	min. velocity of quick phase	10 ... 400	50	[°/s]
A_{Smin}	min. amplitude of quick phase	0.5 ... 10.0	1.0	[°]
A_{Lmin}	min. amplitude of slow phase	0.5 ... 5.0	1.0	[°]
V_{Lmax}	max. velocity of slow phase	5 ... 150	70	[°/s]
T_{Lmin}	min. duration of slow phase	50 ... 1000	100	[ms]
T_{Lmax}	max. duration of slow phase	200 ... 3000	1500	[ms]
	max. dispersion of the measure values during the slow phases	0 ... 100	50	[%]

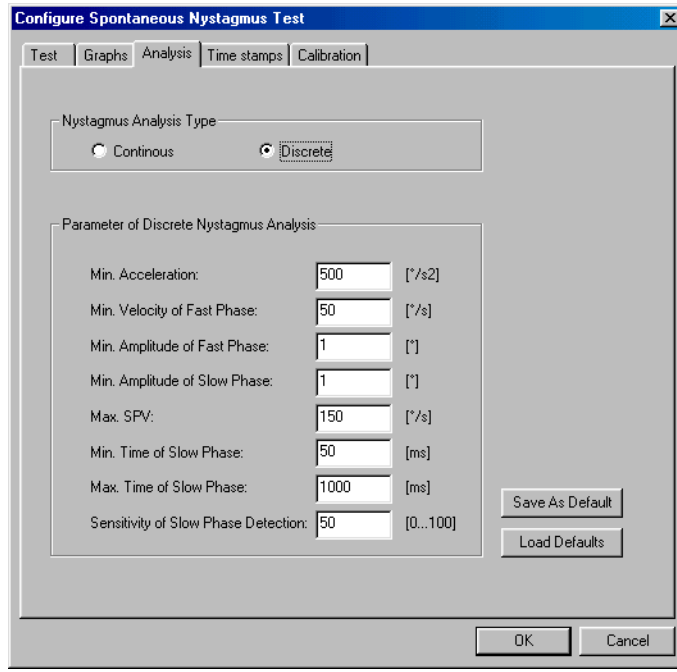


Fig. D-17: Configuration dialog box for spontaneous nystagmus test

These parameters allow you to detect the pattern of those nystagmus beats that are marked by a triangle.

❍ Advantages of the Discrete Nystagmus Analysis

This analysis technique should be chosen if you want to determine how often sporadic nystagmus beats appear, for example with spontaneous nystagmus examinations and the multi condition test, as each nystagmus beat is detected and then can be analyzed concerning frequency, amplitudes and velocity.

Even with caloric tests, this analysis may have good results, especially if there is only a low reaction to the irrigation.

This analysis technique is preferred with the following tests:

- Spontaneous nystagmus test
- Multi condition test
- Caloric test
- Gaze test.

E

Templates

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1. Examination Templates

2D VOG allows you to define complete examination protocols using examination templates. With these templates, you can create and configure those tests that are commonly used in the same way within a particular protocol.

Defining examination templates has the following advantages:

– You can create an infinite number of "default examinations" that are often performed for patients.

Example

Examination Template *"Position"* with:
– Position test
– Positioning test

"Caloric Irrigation" Examination Template
– Spontaneous nystagmus test
– Caloric test

"Default Examination" Examination Template
– Spontaneous nystagmus test
– Smooth pursuit test
– Optokinetic test

– If you are creating a new examination on the base of an existing examination template, all settings will automatically be applied that have been defined when creating the template for each test. This implies that you don't have to reconfigure each test before starting with an examination.
– The settings that have been defined for each test of an examination can only be applied to those examinations that base on this examination template. If a particular test is also used for another examination template, the settings might be completely different.

Example

The Spontaneous nystagmus test has been added to the *"Caloric irrigation"* examination template as well as to the *"Default Examination"* template. As the test configuration is different in each examination template, the measurement analysis and display will be different.

Mainly, in the examination template the following settings will be done:

- examination name
- choosing an icon
- selecting the default reporting format (refer to page D-58)
- selecting tests to be included to the protocol
- if necessary, changing of test sequence
- individual configuration of each test.

1.1 Defining an Examination Template

To define an examination template, proceed as follows:

1. From the 'Examination' menu, choose the 'Template Designer...' command.

   → The *Examination Template Designer* dialog box will appear. The upper part of the dialog box will show the existing examination templates.

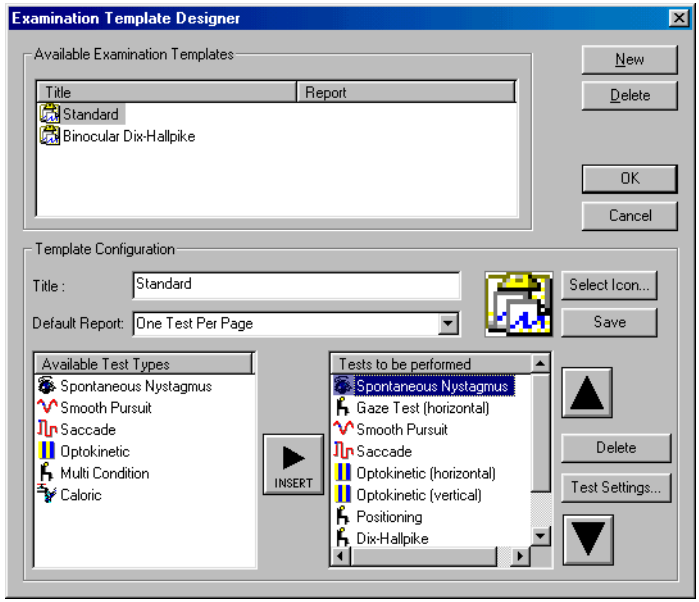


Fig. E-1: Examination Template Designer

2. Now click the [New] button.

3. Under "Template Configuration", enter a title for the new template. Then select the default report format.

4. Using the [Select Icon] button you can choose another icon that will be added to the examination name in the examination tree of the patient.

   You can easily create your own icons with the Windows built in *Paint* program. Just note that all icons must have a size of 16x16 pixel. Store your icons in bitmaps ('*.bmp') with 256 colors.

5. **Adding a new Test:**
   From the list of available tests, select the test to be added to the examination and then click this button.



   → The test will be shown in the *Tests to be performed* table.

   Now continue adding new tests until the right table will show all tests to are to be performed within this examination.



Note

You can always change the test order by selecting the test to be moved downward or upward and by clicking one of these buttons until the test has reached the correct position.



Deleting a Test:
To remove a test from the examination template, in the right table select the test and then click the [Delete] button.

6. **Test Settings:**
   Now you can change the settings for the tests added to the examination.

   To do so, select the desired test and click the [Test Settings] button.

   → A dialog box will appear where you can configure the selected test (for more detailed information on each

test and the test settings, see part **F – Tests** of this manual).

Note

These settings can only be applied for the test of an examination that uses the current examination template.

Tests beeing part of other examinations are not changed.

Repeat this step until configuration is finished.

7. Save the examination templates by clicking the [Save] button.

8. Close the *Examination Template Designer* by clicking the [OK] button.

   → Next time you open the *Create new examination* dialog box the examination template just created will be shown in the template list.

1.2 Editing an Examination Template

To edit an existing examination template, proceed as follows:

1. From the 'Examination' menu, choose the 'Template Designer...' command.

   → The *Examination Template Designer* dialog box will appear (refer to Fig. E-1). The upper part of the dialog box will show the existing examination templates.

2. From the list of available examination templates, select the template to be edited.

3. Apply the changes that are necessary, e.g.
   - Changing the title
   - Changing the default reporting format
   - Inserting or deleting single tests
   - Changing test settings
   - ...

4. Save the changes by clicking the [Save] button and close the *Examination Template Designer* by clicking the [OK] button.

   → The changed template will be saved.



Note

Changes applied to an examination template only refer to examinations created a new.
For existing examinations, the current properties won't be changed.

1.3 Deleting an Examination Template

To delete an examination template, proceed as follows:

1. From the 'Examination' menu, choose the 'Template Designer...' command.

   → The *Examination Template Designer* dialog box will appear (refer to Fig. E-1). The upper part of the dialog box will show the existing examination templates.

2. From the list of available examination templates, select the template to be deleted.

   → In the lower part of the dialog box, detailed information on this examination is displayed (Default reporting format, tests applied to the examination etc.).

3. Click the [Delete] button and answer the "Really Delete" security request with [Yes].

   → The examination template will be deleted.

2. Letter Templates

To use the 2D VOG option of writing letters using the letter wizard (refer to page D-61), you first have to define a letter template. Using the wizard, you then can open a predefined template and automatically copy any kind of patient information, test results and other 2D VOG data to the letter.

The letter templates can be created with any word-processing program (e.g. Microsoft® Word®) and are saved as RTF file. They consist of a mixture of fixed text modules and the so-called template fields that are used as wild cards. When creating the letter, these wildcards will automatically be filled with corresponding data. On page E-14 you will find a table that shows all available fields to define a letter template.

2.1 Defining a Letter Template

To define a letter template, we recommend the following procedure:

1. Start the word-processing program you are currently using.
2. Now start defining the letter template: First type the text of the letter in that way it is to be created by the letter wizard later.

   Note

   In this first step, we recommend to use existing patient data and your real contact. This will help you to find the correct phrases and the optimal template layout.

3. Now replace all "real" data by the corresponding template fields.

Note

Template fields are marked by angle brackets. So make sure that each template field begins with **<** and ends with **>**. If a particular template field doesn't have an angle bracket, the letter wizard will interpret them as normal text which won't be replaced by data.

☞ On the following pages, you will find a list of available template fields.

4. Apply your favorite layout to the text. You can also insert graphics (e.g. your logo).

5. Save your letter template as RTF file.

   → You can now use the letter template to write letters selecting the 'Letter Wizard' command from the menu.

2D VOG – Video-Oculography®

SMI SensoMotoric Instruments

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Dizziness Specialist
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<RefDoc_Title> <RefDoc_FirstName> <RefDoc_Name>
<RefDoc_Address>

Patient: <Patient_FirstName> <Patient_Name>, <Patient_Birthdate>

<Date>

Dear <RefDoc_Title> <RefDoc_Name>,

thank you very much for referring **<Patient_MS> <Patient_FirstName> <Patient_Name>**, born <Patient_Birthdate>. <Patient_MS> <Patient_Name> visited our praxis on <Examination_Date>.

Case history

<Case_History>

Examination

Due to the complains we performed the following Tests:

- <Performed_Tests>

Impressions

<Examination_Notes>

<Test_Notes>

Conclusion

<Diagnoses>

Treatment proposal

Due to our findings, we recommend the following:

<Proposed_Treatment>

We hope this results will prove useful to you.

Sincerely

Dr. Green

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Fig. E-2: Sample letter template with fixed text and template fields

2.2 Letter Template Fields

The following table shows all template fields that can be used to define letter templates.

Each template field that is inserted as wild card into the template begins with an open angle bracket (**<**) and end with a closed angle bracket (**>**). If an angle bracket is missing in the template field, the letter wizard will interpret the field as normal text which won't be replaced by data.

Name of Field	Description	Notes
<Date>	current date	format according to Windows country settings
<Patient_Name>	patient surname	
<Patient_Prename>	patient first name	
<Patient_MS>	form of address of patient	"Mr.", "Mrs.", "Ms."
<Patient_Birthdate>	patient birthdate	format according to Windows country settings
<Patient_Sex>	sex of patient	"male" / "female";
<Patient_ID>	patient ID	unrestricted definition of patient IDs by the user
<Patient_SocSecNo>	social insurance number of patient	
<Patient_Street>	street and house number of patient	
<Patient_City>	address city of patient	
<Patient_Zip>	address ZIP code of patient	

Name of Field	Description	Notes
<Patient_State>	patient state	only used within the United States
<Patient_Phone>	phone number of patient	
<Patient_Notes>	patient note	
<Examination_Notes>	examination notes	
<Examination_Date>	examination date (when file was **created**)	format according to Windows country settings
<RefDoc_Name>	surname of referring doctor	
<RefDoc_Prename>	first name of referring doctor	
<RefDoc_Title>	referring doctor title	
<RefDoc_MS>	form of address of referring doctor	
<RefDoc_Address>	address of referring doctor	
<User>	user name	entered during the login procedure
<Performed_Tests>	list that contains the names of all tests **performed** within an examination	– one test name per line – an examination may contain further tests, but in this case only those tests are listed that have been performed completely or partly. – the text corresponds to the first page of the letter wizard

Name of Field	Description	Notes
<Test_Notes>	list that contains name and notes of all tests **performed** within the examination	→ see <Performed_Tests> The structure is the following: *Test name1<new line>* *Note1 (can be more than one line)* *<space line>* *Test name2<new line>* *Note2 (can be more than one line)* *<space line>* and so on.
<Case_History>	case history	
<Diagnoses>	diagnoses	
<Proposed_Treatment>	proposed treatment	

2.3 Editing Letter Templates

To edit a letter template, proceed as follows:

1. Open the letter template by using your word-processing program.
2. Apply the changes.
3. Save again the letter template as RTF file.

   → The changes are applied for all future letters that will be created with the letter template.

F

Tests

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1. General

The following pages contain information on

- how measured data is displayed in the test windows,
- the display of measuring process diagrams and
- the color encoding.

1.1 Displaying Data in the Test Windows

To make the use of 2D VOG as ease as possible we intended to produce a similar design of the different test windows.

Graphs that display the measuring process over time are generally shown in the left part of the window. Those graphs that show the deviated parameters (e.g. the butterfly diagram in the caloric test) are shown in the right part.

In general, we tried to display all required information in one single test window. This allows you to analyze the complete test without having to switch between different diagrams.

1.2 Measuring Process Diagrams

1.2.1 Time Axis

Diagrams that are showing the measuring process over the time will have one time axis if they refer to the same measurement.

If the measurement is longer than the time section shown, you can move the displayed section using the scrollbar at the lower bottom of the test window.

1.2.2 Zooming

In each measurement process diagram, you can maximize and minimize the display of the measured data by using the zoom option.

Note

To change the graph axis scaling, enter the required settings in the *Graphs* tab in the configuration dialog box.

☛ For more information, refer to page D-30.

❍ Maximizing Display

To maximize the graph display, proceed as follows:

1. In the test window, click the area of the graph to be maximized using the left mouse button.
2. Pressing the mouse button, move the mouse in diagonal direction towards the lower right.

   → While you are moving the mouse, you will see a rectangular line that show the "selected" area.

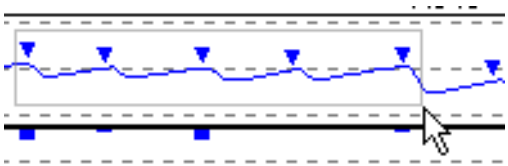


Fig. F-1: Maximizing display (zooming)

3. Stop pressing the mouse button if the rectangle covers exactly the area to be maximized.

   → Data will now be displayed in maximized form.

Note

You can repeat zooming as often as you like to continue maximizing the display.

❍ Minimizing Display

After maximizing the measure data display, of course, you can also switch back to the original size. To do so, proceed as follows:

1. In the test window, click the area of the graph to be displayed in its original size using the left mouse button. Place the mouse button at the lower right part of the graph.
2. Pressing the mouse button, now move the mouse in diagonal direction towards the upper left part of the graph.
   → While you are moving the mouse, you will see a grey rectangular line that shows the "selected" area.
3. Stop pressing the mouse button, if the rectangle covers an area of any size.
   → The data is now shown in its original size.

1.2.3 Marking of Detected Nystagmus Beats

Nystagmus Beats detected by the nystagmus analysis are marked by small triangles in the eye position graphs:

– triangles that point downwards show left and downward nystagmus beats
– triangles that point upwards show right and upward nystagmus beats.

☞ For more detailed information on the nystagmus analysis, refer to page D-69.

1.2.4 Color Encoding

To provide a clear display, the following colors are used for the graphs in 2D VOG:

red	measure data of right eye
blue	measure data of left eye
green	stimulus display
yellow	norm data range is 90% of the normal persons
grey	norm data range is 94% of the normal persons

2. Spontaneous Nystagmus Test

The spontaneous nystagmus test is suitable for recording non-evocative eye movements. To avoid distortions produced by visual stimulation a light-occluding mask should be used or the combi mask should be darkened with the front cover plate.

As it is very flexible, the spontaneous nystagmus test can be regarded as universal test. For instance, within an examination template you can create a succession of various spontaneous nystagmus tests to build a multi condition test. The time stamp allows you to exactly correlate your measurements with any stimulation.

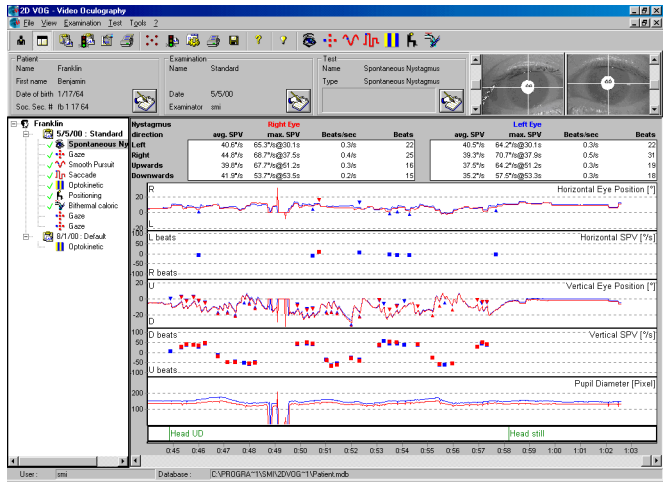


Fig. F-2: Spontaneous nystagmus test window

2.1 Spontaneous Nystagmus Test Diagrams

All spontaneous nystagmus test graphs are coupled by one time axis. In particular, the following measure value graphs are available:

- statistical data
- horizontal eye position
- horizontal velocity of the slow phase
- vertical eye position
- vertical velocity of the slow phase
- pupil diameter
- time stamp bar

2.1.1 Statistical Data

In the statistical data field, statistical data for the nystagmus beats of the left and right eye are shown.

Nystagmus	Right Eye				Left Eye			
	avg. SPV	max. SPV	Beats/sec	Beats	avg. SPV	max. SPV	Beats/sec	Beats
Left Beats	10.2°/s	16.2°/s@14.8s	0.2/s	5	9.3°/s	15.4°/s@17.6s	0.2/s	6
Right Beats	23.4°/s	30.4°/s@27.4s	0.1/s	3	20.6°/s	23.1°/s@28.2s	0.1/s	2
Up Beats	---	---	---	0	6.0°/s	6.0°/s@2.6s	0.0/s	1
Down Beats	---	---	---	0	9.1°/s	9.6°/s@2.2s	0.1/s	2

Fig. F-3: Statistical data field of spontaneous nystagmus test

For each direction of the nystagmus beat (right, left, up, down) the following parameters are calculated:

Abbreviation	Description	Unit
avg. SPV	average velocity of the slow phases of all detected nystagmus beats	[°/s]
max. SPV	maximum velocity of the slow phases of all detected nystagmus beats	[°/s]

Abbreviation	Description	Unit
beats/sec.	beats per second (number divided by recording time)	[Hz]
Beats	number of detected nystagmus beats	

Note

The statistical data field is only available, if you are using the discrete nystagmus analysis (for more information, refer to chapter *„Discrete Nystagmus Analysis“* on page D-70).

2.1.2 Horizontal and Vertical Eye Position

Horizontal and vertical eye positions are displayed by two separate graphs. Positive angles represent eye movements to the right and upwards, respectively, negative angles show eye movements to the left and downwards, respectively.

2.1.3 Horizontal and Vertical Velocity of the Slow Phase

Both graphs show the velocity of the slow phase (SPV) for horizontal/vertical eye movements in degrees per second. Positive values indicate the SPV for beats to the left and downwards, a negative value represents the SPV for beats to the right and upwards.

2.1.4 Pupil Diameter

This graph shows how the pupil diameter changes and can be used to perform the relative pupillometry. It is shown in pixels and refers to the pupil size shown on the CCD chip of the camera. It is not possible to indicate the absolute size (e.g. millimeter).

2.1.5 Time Stamp Bar

The time stamp bar allows you to exactly mark a point of time during measurement by setting time stamps (refer to page F-12).

The time stamps are displayed in form of vertical green lines and are marked by the key selected in the configuration dialog box in the *Time Stamp* tab.

For more information on this tab, refer to page D-33.

Note

Time stamps set shortly after another are shown on top of the other, so it is not possible to distinguish them in the graph. In this case, you should maximize the display using the zoom option (refer to page F-4).

2.2 Configurating the Spontaneous Nystagmus Test

To open the configuration dialog box,

a) double click into the test window or

b) select the 'Configuration' command from the 'Test' menu.

In the configuration dialog box, the test settings are shown in various tabs.

Note

The main settings have already been described in chapter *„Configurating a Test“* (page D-28), chapter *„Calibration“* (page D-49) and chapter *„Nystagmus Analysis“* (page D-69).

2.2.1 *Test* Tab

☞ Refer to page D-29.

2.2.2 Graphs Tab

☞ Refer to page D-30.

2.2.3 *Analysis* Tab

☞ Refer to page D-32.

2.2.4 *Time Stamps* Tab

Using this tab, you define the keys that are to be written to the time stamp bar when you are pressing a F5 to F12 key or when you are setting a TTL level.

2.2.5 *Calibration* Tab

☞ Refer to page D-34.

2.3 Setting a Time Stamp

To correlate the measured eye movements with other events 2D VOG offers to set time stamps during the measurement. This feature makes the spontaneous nystagmus test very versatile.

There are two ways to set a time stamp:

a) pressing the F5 to F12 key

b) setting a TTL level at one of the digital inputs (only possible with systems that use the *Digital In* option).

3. Multi Condition Test

The multi condition test has been designed to compare as many separate measurements as possible on one page. Therefore, it should preferably be used for positional or positioning tests. The statistical analysis is identical to the analysis of the spontaneous nystagmus test. As various tests can be displayed in a compact form on one page, it is not possible to display binocular measurements.

Note

To perform a binocular multi condition examinations

- open a gaze test and switch off the VisualLab stimulus (refer to *„Gaze Test“* on page F-19), or
- create an examination template and configure various successive spontaneous nystagmus tests that are titled according to the position or positioning of the patient.

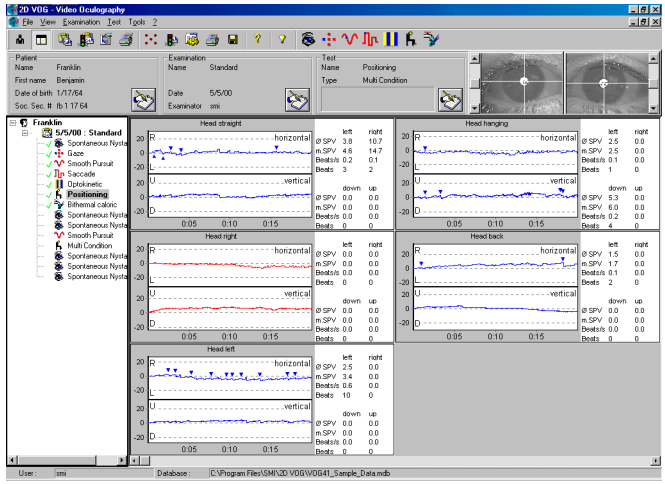


Fig. F-4: Multi condition test window

3.1 Multi Condition Test Diagrams

In the multi condition test, the following measure value diagrams are available:

- statistical data
- horizontal eye position
- vertical eye position

3.1.1 Statistical Data

In the statistical data field, for each measurement the essential parameters of the nystagmus analysis are shown.

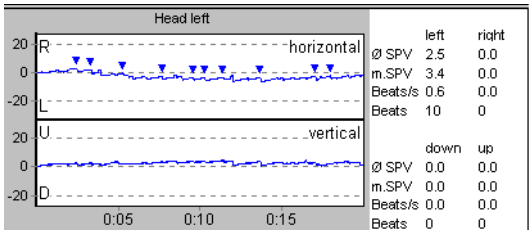


Fig. F-5: Statistical data field of multi condition test

The values are shown separately according to the direction of the nystagmus beat (left, right, down, up):

Note

The statistical data field is only shown if you are using the discrete nystagmus analysis (for more information, see chapter *„Discrete Nystagmus Analysis“* on page D-70).

Abbreviation	Description	Unit
avg. SPV	average velocity of the slow phases of all detected nystagmus beats	[°/s]
max. SPV	maximum velocity of the slow phases of all detected nystagmus beats	[°/s]
Beats/sec.	beats per second (number divided by recording time)	[Hz]
Beats	number of detected nystagmus beats	

3.1.2 Horizontal and Vertical Eye Position

Horizontal and vertical eye positions are displayed by two separate graphs. Positive angles represent eye movements to the right and upwards, respectively, negative angles show eye movements to the left and downwards, respectively.

3.2 Configurating the Multi Condition Test

To open the configuration dialog box,

a) double click into the test window or

b) select the 'Configuration' command from the 'Test' menu.

In the configuration dialog box, the test settings are shown in various tabs.



Note

The main settings have already been described in chapter *„Configurating a Test“* (page D-28), chapter *„Calibration“* (page D-49) and chapter *„Nystagmus Analysis“* (page D-69).

3.2.1 *Test* Tab

In the *Test* tab, you setup each subtest of your multi condition test.

To configure the test, proceed as follows:

1. Enter a title for the test (e.g. Dix-Hallpike manoeuvre).
2. In the MASK box, select the desired VOG mask.
3. If you always want to perform a calibration before starting the measurement, select the PERFORM CALIBRATION check box (refer to chapter *„Calibration“* on page D-49).
4. To analyze vertical eye movement, select the VERTICAL ANALYZE check box.

Note

If you don't need to analyze vertical eye movement, up to 10 subtests will be available. When you are performing the vertical analysis, due to the lack of space you can only define 6 subtests in the test window.

If you require more than 6 subtests, please create an examination template with various successive multi condition tests.

For more information, see chapter *„Examination Templates“* on page E-3.

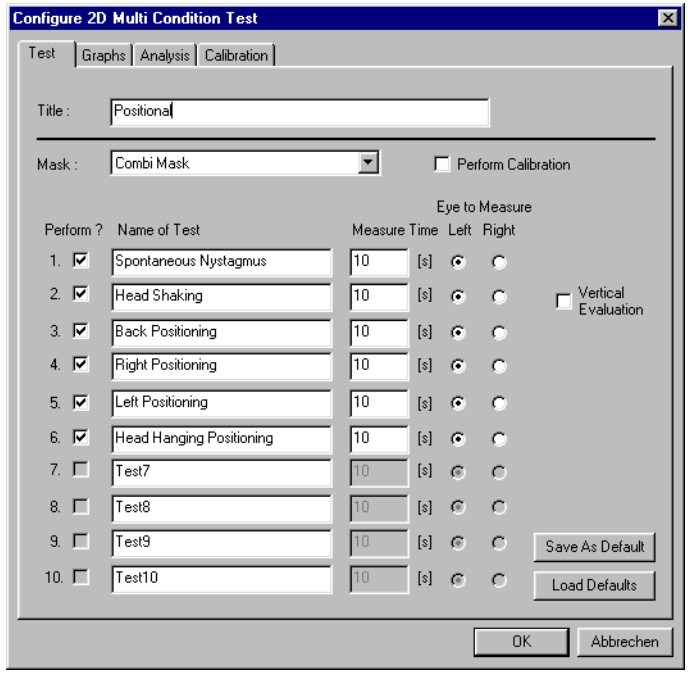


Fig. F-6: Multi condition test configuration

5. In the subtest lines, you have the following options for each subtests:
   - entering a title of your choice,
   - defining the recording time and
   - selecting the eye whose movement you want to record

6. In the PERFORM column, check each subtest to be performed.

3.2.2 *Graphs* Tab

☞ Refer to page D-30.

3.2.3 *Analysis* Tab

☞ Refer to page D-32.

3.2.4 *Calibration* Tab

☞ Refer to page D-34.

4. Gaze Test

During a gaze test eye movements are measured while the patient is fixating a target. In contrast to the saccade test where the movement towards the target (the saccade) is measured, the gaze test analyzes the nystagmus "beating" during the fixation period.

Typically in various successive measurements the fixation target is shown at different positions. To allow you to compare these measurements the gaze test can display up to six independent binocular recordings on a single page.

Using the VisualLab module a fixation target can be projected in front of the patient. However, it is also possible to perform the gaze test without using the VisualLab module (refer to *„Configurating the Gaze Test“* on page F-23).

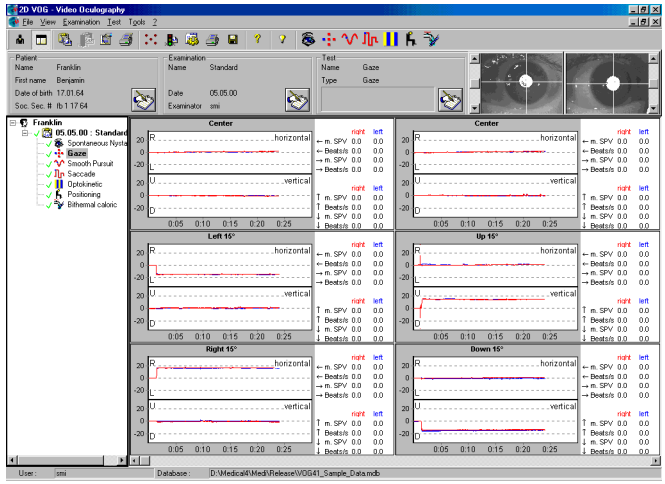


Fig. F-7: Gaze test test window

The statistical analysis of the gaze test is similar to the multi condition test. For this reason, the gaze test can be chosen to perform binocular positional and positioning tests.

Note

Please make sure the patient follows the target only with his eyes. The head must not be moved during the examination.

4.1 Gaze Test Procedure

If you perform the gaze test by using the VisualLab (see *„Configurating the Gaze Test“* on page F-23), the procedure is as follows:

1. Start:
   The target will be projected in front of the patient who is looking straight ahead (primary gaze position: horizontal = 0°, vertical = 0°). Recording has not started yet, so the patient will have enough time to fixate the target.

2. Two seconds after projection has started eye movement recording will begin (t = 0 in eye position diagrams). The target will still be shown in the primary gaze position.

3. After another second, the target will move to the given position (s. *„Configurating the Gaze Test“* on page F-23) and statistical analysis of the nystagmus beats will start.

In case you do not use the VisualLab module, eye movement recording and statistical analysis of the nystagmus beats will start as soon as the test is started.

4.2 Gaze Test Diagrams

For the gaze test, the following measure value diagrams are available:

– statistical data
– horizontal eye position
– vertical eye position

4.2.1 Statistical Data

In the statistical data field, for each measurement the essential parameters of the nystagmus analysis are shown.

	right	left
← m. SPV	0.0	0.0
← Beats/s	0.0	0.0
→ m. SPV	0.0	0.0
→ Beats/s	0.0	0.0

	right	left
↑ m. SPV	0.0	0.0
↑ Beats/s	0.0	0.0
↓ m. SPV	0.0	0.0
↓ Beats/s	0.0	0.0

Fig. F-8: Statistical data field of gaze test

The values are shown separately for the right and left eye and for each direction of nystagmus beating:

← left nystagmus beats
→ right nystagmus beats
↑ upward nystagmus beats
↓ downward nystagmus beats

Abbreviation	Description	Unit
Ø SPV	average velocity of the slow phases of detected nystagmus beats	[°/s]
m. SPV	maximum velocity of the slow phases of detected nystagmus beats	[°/s]
Nyst./s.	frequency of the nystagmus beats	[Nyst./s]

Use the *Graphs* tab in the gaze test configuration dialog box to select whether you wish to display the average or the maximum velocity of the slow phases of the detected nystagmus beats.

Note

The statistical data field is only available, if you are using the discrete nystagmus analysis (for more information, refer to chapter *„Discrete Nystagmus Analysis“* on page D-70).

If you are using the VisualLab module to project the targets, the analysis will start when the target moves to the pre-defined position (refer to *„Gaze Test Procedure“* on page F-20).

4.2.2 Horizontal and Vertical Eye Position

Horizontal and vertical eye positions are displayed by two separate graphs. A positive value represents eye movement to the right and upwards, respectively, a negative value shows eye movement to the left and downwards, respectively.

4.3 Configurating the Gaze Test

To open the configuration dialog box,

a) double click into the test window or

b) select the 'Configuration' command from the 'Test' menu.

In the 'Configuration' dialog box, the test settings are shown in various tabs.

Note

The main settings have already been described in chapter *„Configurating a Test“* (page D-28), chapter *„Calibration“* (page D-49) and chapter *„Nystagmus Analysis“* (page D-69).

Therefore, the following pages only contain configuration options that are specific for this test.

4.3.1 *Test* Tab

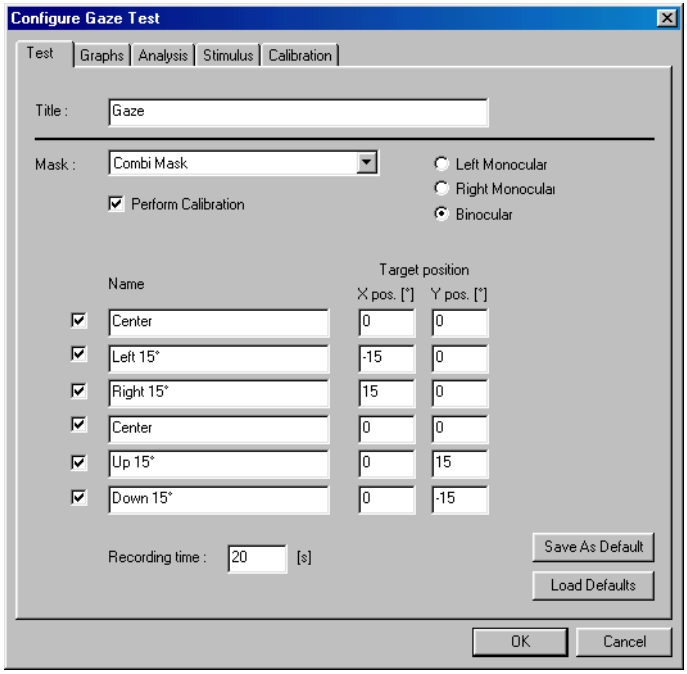


Fig. F-9: Gaze test configuration, "Test" tab

In the *Test* tab, you setup each subtest of your gaze test.

To configure the test, proceed as follows:

1. Enter a title for the test (e.g. Gaze 15°).
2. In the MASK box, select the desired VOG mask.
3. If you always want to perform a calibration before starting the measurement, select the PERFORM CALIBRATION check box (refer to chapter *„Calibration“* on page D-49).
4. Under RECORDING TIME, you define how long the eye movement is to be recorded and analyzed (in seconds).

Note

If you are using the VisualLab module, the recording time of a subtest will actually be different to the value given in the RECORDING TIME box, as it will require two more seconds (refer to *„Gaze Test Procedure“* on page F-20).

5. Within a gaze test, you can setup up to six subtests.

   For each subtest, you have the following options:

   – defining whether the subtest is to be performed
   – entering a title of your choice,
   – entering the horizontal and vertical coordinates of the target to be shown (X POS. and Y POS.) where a positive value represents a position to the right and upwards of the primary gaze position of the test person, a negative value shows a position to the left and downwards, respectively.

Note

This will only influence the test procedure, if the VisualLab module is installed with your system and if you have selected the "Use VisualLab" check box in the *Stimulus* tab of the gaze test.

If you require more than six subtests, we recommend to create an examination template containing several gaze tests.

☞ For more detailed information, refer to chapter *„Examination Templates“* on page E-3.

4.3.2 Graphs Tab

☞ Refer to page D-32.

4.3.3 *Stimulus* Tab

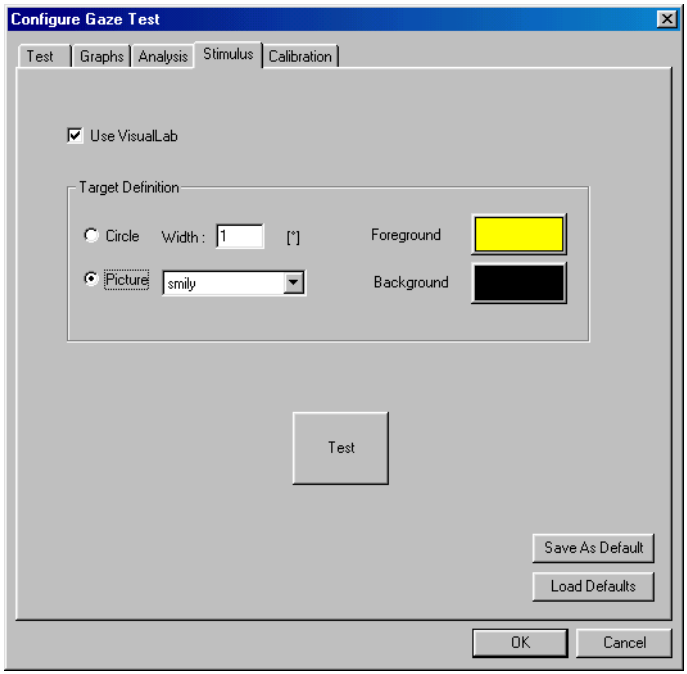


Fig. F-10: Gaze test configuration, "Stimulus" tab

Using the Stimulus tab, you can define the target parameters:

- **Circle / Picture**
  Selecting the target and its size.
- **Foreground color / Background color**
  Selecting target color and background color
- **Display Test**
  To control target displaying, click the [Test] button.

In addition, the stimulus tab allows you to disable the use of the VisualLab module during a gaze test. This option should be chosen, if

- your system is not equipped with a VisualLab module,
- the patient receives the targets to be fixated in a different way or
- the gaze test is used to perform positional or positioning tests.

4.3.4 *Analysis* Tab

☞ Refer to page D-32.

4.3.5 *Calibration* Tab

☞ Refer to page D-34.

5. Smooth Pursuit Test

In the smooth pursuit test, the patient's eyes have to follow with constant velocity a target that is moving across the projection screen. During the test, the target is swinging across the projection screen. To perform this test, you need the VisualLab module.

Note

Please make sure the patient follows the target only with his eyes. The head must not be moved during the test.

Before starting a smooth pursuit test you should perform a calibration (refer to page D-49).

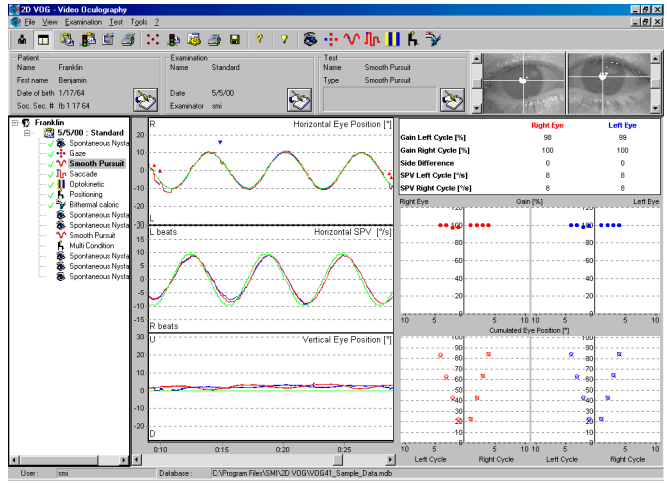


Fig. F-11: Smooth pursuit test window

5.1 Analysis

In general, each eye will be analyzed separately:

- **Red lines and points**
  in the graphs show the values for the **right eye**,
- **blue lines and points**
  indicate the values for the **left eye**.

Furthermore, the analysis distinguishes between eye movements to the right and to the left which in the following will be called "right cycles" and "left cycles".

Data for the "right eye" can easily be mixed up with data for "right cycles". Therefore we ask you to read carefully the following description.

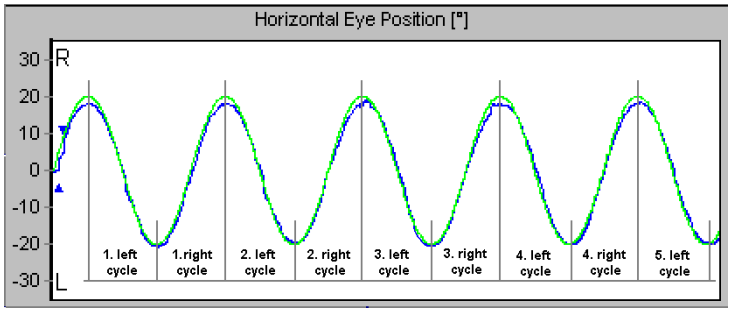


Fig. F-12: Smooth pursuit measurement cycles. The green line displays the stimulus position, the blue line shows the position of the left eye.

❍ Cycle

Half period of a target pendulum.

Right cycles contain a target movement from the left turning back to the right, and they end at the right turning back. Left cycles contain a target movement from the right turning back to the left, and they end at the left turning back (see Fig. F-12).

Gain

Percentage of constant smooth eye movement of the complete eye movement within a cycle:

$$G = \frac{Amplitude_{Smooth\ Eye\ Movement}}{Amplitude_{Smooth\ Eye\ Movement} + Amplitude_{Saccadic\ Eye\ Movement}} \times 100$$

The gain (G) is given in percent. If the patient's eye follow the target with a completely smooth eye movement, the gain value will be 100%. If jerky saccadic movements are measured apart from the smooth eye movement, the gain value will fall below 100%.

Side Difference

The side difference (S) constitutes a value that compares the gain (G) of eye movements to the right (right cycles) with movements to the let (left cycles). It is calculated according to the formula:

$$S = \frac{|G_{right}| - |G_{left}|}{|G_{right}| + |G_{left}|} \times 100$$

- Side Difference < 0
  There are more saccadic movements for eye movements to the left than for movements to the right.
- Side Difference > 0
  There are more saccadic movements for eye movements to the right than for movements to the left.

SPV

Like in the nystagmus analysis, the smooth eye movement velocity is called slow phase velocity (SPV). It should be the same as the velocity of the target movement.

The "maximum SPV of a cycle" is the fastest non-saccadic eye movement measured within a cycle. The values "SPV left cycles" and "SPV right cycles" are obtained by taking the arithmetic average of all maximum SPVs of left/right cycles. If the test person is healthy, these values should be the same as the given velocity of the target.

5.2 Smooth Pursuit Test Diagrams

5.2.1 Statistical Data

	Right Eye	Left Eye
Gain Left Cycle [%]	98	99
Gain Right Cycle [%]	100	100
Side Difference	0	0
SPV Left Cycle [°/s]	8	8
SPV Right Cycle [°/s]	8	8

Fig. F-13: Statistical data field of smooth pursuit test

Statistical data is calculated both for the left eye (right column) and for the right eye (left column). The following values are shown:

Abbreviation	Description	Unit
Gain left	Arithmetic average of the gains calculated for each left cycle.	%
Gain right	Arithmetic average of the gains calculated for each right cycle.	%
Side Difference	Value to determine the relation between gain left and gain right that is calculated from the arithmetic average of the measured gain	–

Abbreviation	Description	Unit
SPV left cycles	Arithmetic average of all maximum SPV (see above) measured for each left cycle.	°/s
SPV right cycles	Arithmetic average of all maximum SPV (see above) measured for each right cycle.	°/s

5.2.2 Horizontal and Vertical Eye Position

Horizontal and vertical eye positions are displayed by two separate graphs. Positive angles represent eye movements to the right and upwards, respectively, negative angles show eye movements to the left and downwards, respectively.

Additionally, the target position is indicated with green color for comparing the eye movement with the target movement. Sections that were detected as non-smooth eye movement during the analysis are identified by small triangles.

5.2.3 Horizontal SPV

This diagram shows the horizontal velocity of those parts of the eye movement that are smooth and constant. The value is given in degrees per second. For each section that is detected as saccadic movement the smooth eye movement velocity is being interpolated. A positive value identifies movements to the right, a negative value identifies movements to the left. The green line indicates the target velocity.

5.2.4 Gain

The "Gain" diagram shows the gains calculated for each cycle of the measurement. The values for right cycles (eye movements to the right) are shown on the right side of the diagram, left cycles (movements to the left) are shown on the left side of the diagram. To number the cycles, see Fig. F-12.

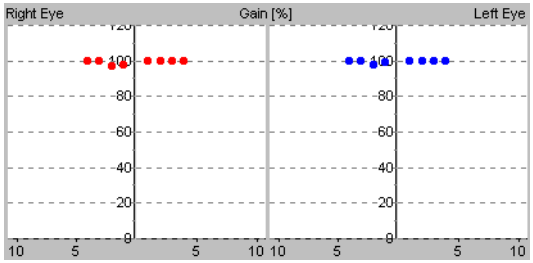


Fig. F-14: "Gain" diagram

5.2.5 Cumulated Eye Position

The "Cumulated Eye Position" diagram is the ideal form to analyze regular saccadic eye movements: Like in the "Gain" diagram, the abscissa contains the measurements sorted by cycle number and cycle direction. The ordinate shows the cumulated eye position in degree. This value results from adding all eye movements (A) of the previous cycles of the same direction.

For instance, the value at 3 (right cycle) is obtained as follows:

$$\textit{Cumulated Eye Position}_{3right} = A_{1right} + A_{2right} + A_{3right}$$

The cumulated eye position is calculated and displayed in two ways:

a) Each cross (X) indicates the eye position that results from adding up only the smooth eye movement during a cycle. This value is the numerator in the gain formula (see page F-31)

b) The circles (o) show the complete eye movement amplitude during a cycle: smooth and saccadic eye movements. This value is the denominator in the gain formula (see page F-31).

If the patient doesn't need any saccadic movements to follow the target, the diagram circles and crosses lie on top of each other. However, if regular saccadic movements are measured, with every cycle the cumulated eye position of the smooth eye movement will become smaller than the cumulated eye position for the complete eye movement amplitude. Therefore with increasing the number of cycles, the crosses and circles will separate from each other like a pair of scissors.

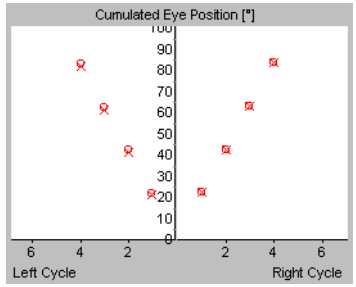


Fig. F-15: "Cumulated Eye Position" diagram of patient with smooth eye movements

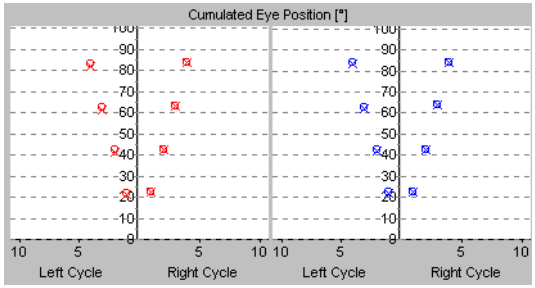


Fig. F-16: "Cumulated Eye Position" diagram of patient with regular saccadic movements

5.3 Configurating the Smooth Pursuit Test

To open the configuration dialog box,

a) double click into the test window or

b) select the 'Configuration' command from the 'Test' menu.

In the configuration dialog box, the test settings are shown in various tabs.

Note

The main settings have already been described in chapter *„Configurating a Test“* (page D-28), chapter *„Calibration“* (page D-49) and chapter *„Nystagmus Analysis“* (page D-69).

5.3.1 *Test* Tab

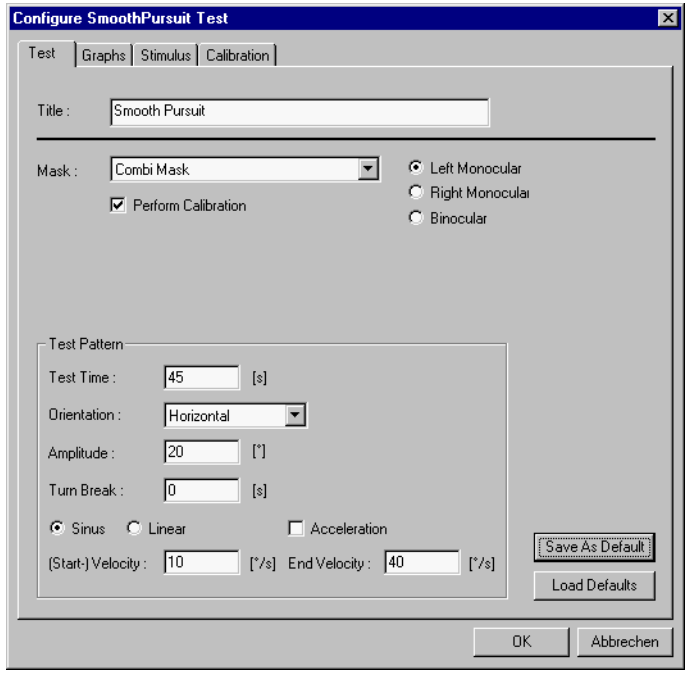


Fig. F-17: Smooth pursuit test configuration, "Test" tab

Using the *Test* tab, you set the complete smooth pursuit test procedure.

In the upper part of the tab, you have the following options:

– entering a title for the test,
– selecting the VOG mask used and
– defining whether the measurement is to performed for the right, for the left or for both eyes (binocular).

☞ For more information, refer to chapter „*Configurating a Test*“ on page D-28.

In the lower part of the tab called "Test Pattern", you define the test procedure. The following options have to be set:

1. Define the recording time.

2. In the ORIENTATION box, select the direction of the target movement:
   - *horizontal* or
   - *diagonal*

3. In the AMPLITUDE box, enter the target movement amplitude.

4. The TURN BREAK box should be used to define how long the target shall remain at the turning backs of its pendulum movement before the next cycle starts. This can be useful if the patient has difficulties in following the target or concentrating on the examination.

5. The option buttons SINUS and LINEAR allow you to choose between sinusoidal or linear, saw-tooth like pendulum movement of the target.

6. In the START VELOCITY box, you enter the target velocity.

7. If you want to obtain an increasing or decreasing target velocity during measurement, check the ACCELERATION box and enter a second velocity for the target in the END VELOCITY box.

Note
If you have checked the ACCELERATION box for the target velocity, the STATISTICAL DATA field won't show any statical data, as a statistical analysis for measure cycles of different target velocities does not make any sense.

5.3.2 *Graphs* Tab

Refer to page D-30.

5.3.3 Analysis

☞ Refer to page D-32.

5.3.4 *Stimulus* Tab

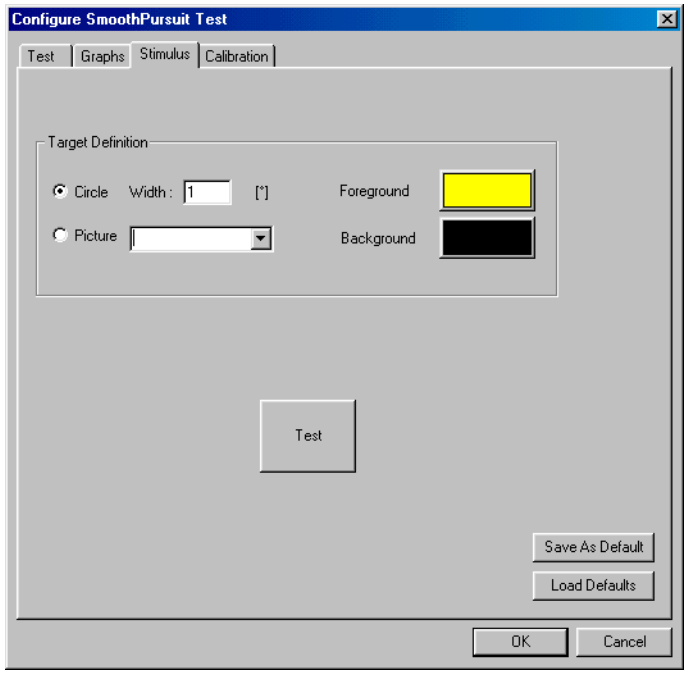


Fig. F-18: Smooth pursuit test configuration, "Stimulus" tab

Using the Stimulus tab, you can define the target parameters:

- **Circle / Picture**
  Selecting the target.
- **Foreground color / Background color**
  Selecting target color and background color
- **Display Test**
  To control target displaying, click the [Test] button.

5.3.5 *Calibration* Tab

☞ Refer to page D-34.

6. Saccade Test

If the saccade test is applied, the patient's eyes have to follow a target generated by the VisualLab module. However, the target is not moving constantly like in the smooth pursuit test, but it is jumping from one position to the next. The patient must try to fixate the target and also follow the different fixation points with fast eye movements (saccades). Ideally, the saccades should consist of one single fast movement that leads to an exact refixation of the target, to avoid that you have to correct the gaze position or that the gaze position is drifting away during the next fixation. To perform this test, the VisualLab module is required.

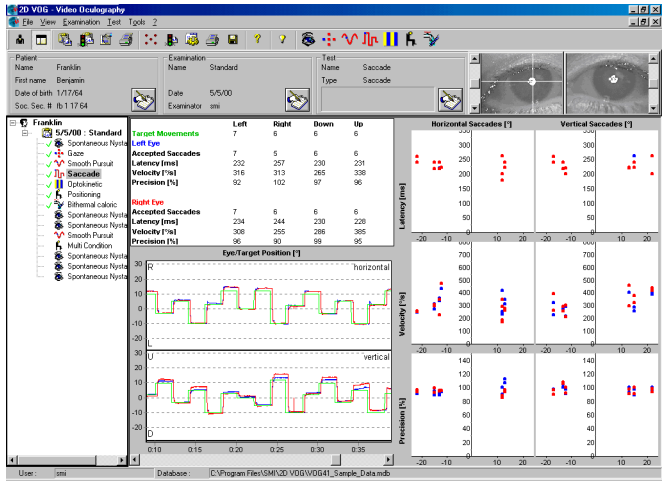


Fig. F-19: Saccade test window

Note

Please make sure the patient follows the target only with his eyes. The head must not be moved during the test.

Before starting a saccade test you should perform a calibration (refer to page D-49).

6.1 Analysis

The measured saccades are analyzed for latency, velocity and precision:

❍ Latency

Latency or reaction time is the time between the change of target position and the begin of the saccade (see Fig. F-20). To clearly distinguish between the saccade begin and fluctuations of eye position, the saccade begin is the point of time when the eye has passed a determinable percentage of the complete saccade amplitude. This percentage which is typically 10% can be set in the *Analysis* tab of the saccade test configuration dialog box (THRESHOLD FOR LATENCY box).

☞ For more information, see the description of the *Analysis* tab in chapter *„Configurating the Saccade Test“* (page F-51).

❍ Velocity

Velocity of eye movement during the saccade.

❍ Precision

The precision indicates how accurately a saccade is performed. It is equal to (Fig. F-20):

$$P = \frac{X2 - X0}{X3 - X0} \times 100$$

where:

- **a value near 100%**
  indicates that with this saccade the new position was well fixated
- **a value smaller than 100%**
  shows that the saccade amplitude was too small (undershoot)
- **a value larger than 100%**
  shows that the saccade amplitude was too high (overshoot).

If the value is too small or too large, the patient needs additional saccades in order to fixate the target at its new position.

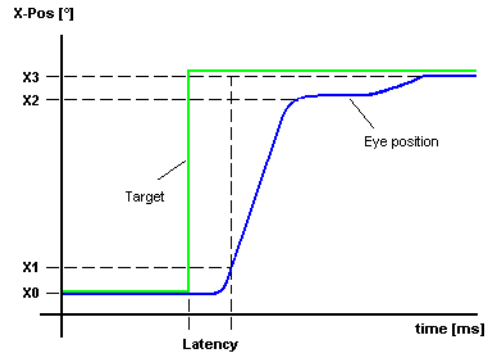


Fig. F-20: Important points of time within a saccade. This figure shows a saccade where the amplitude is too small. To focus the target, correcting is required.

❍ Temporal Resolution of Saccade Measurement

Saccades are very fast eye movements. To exactly measure them, systems that have a minimum temporal resolution of 200 Hz should be used. As your 2D VOG system is equipped with stan-

dard video cameras, the spatial resolution will only be 50 Hz with PAL and 60 Hz with NTSC systems.

Therefore, the 2D VOG system is not the optimal tool for saccade research. For this purpose, SensoMotoric Instruments offers the EyeLink system, a special system that can record with a spatial resolution of 250 Hz.

For the clinical analyses of the saccade test, the spatial resolution of your 2D VOG system is certainly sufficient (for instance, refer to: J. T. Enright (1998): Estimating peak velocity of rapid eye movements from video recordings. Behavior Research Methods, Instruments, & Computers 30(2), 349-353). However, you should meet the following conditions to reduce measurement errors:

– The saccade amplitude should be at least 10°.
– The saccade should be scanned with at least two, better three measure values.

Both options can be set in the *configuration* dialog box of the saccade test (see page F-47). Consequently, saccade measurements that don't meet these conditions won't be included to the statistical data.

6.2 Saccade Test Diagrams

6.2.1 Statistical Data

	Left	Right	Down	Up
Target Movements	7	6	6	6
Left Eye				
Accepted Saccades	7	5	6	6
Latency [ms]	232	257	230	231
Velocity [°/s]	316	313	265	338
Precision [%]	92	102	97	96
Right Eye				
Accepted Saccades	7	6	6	6
Latency [ms]	234	244	230	228
Velocity [°/s]	308	255	286	385
Precision [%]	96	90	99	95

Fig. F-21: Statistical data field for saccade test

In this test, the following measure data will be calculated for both the left and the right eye:

– saccades to the right,
– saccades to the left,
– upward saccades and
– downward saccades.

Abbreviation	Description	Unit
target movements (right, left, up, down)	Number of target movements to the given direction	–
accepted saccades	Number of saccades that have been included to the analysis (for more information, see chapter *„Analysis“* on page F-42 and chapter *„Configurating the Saccade Test“* on page F-47).	

Abbreviation	Description	Unit
latency	Reaction time measured between target position change and exceed of threshold X1 (refer to chapter *„Analysis“* on page F-42 and Fig. F-20)	ms
velocity	Average saccade velocity	°/s
precision	Precision of the saccade Defines how precisely the saccade hits the new target position (refer to *„Analysis“* on page F-42).	%

6.2.2 Horizontal and Vertical Eye and Target Position

Horizontal and vertical eye positions are displayed by two separate graphs. Positive angles represent eye movements to the right and upwards, respectively, negative angles show eye movements to the left and downwards, respectively. Additionally, the target position is indicated with green color for comparing the eye movement with the target movement.

6.2.3 Latency, Velocity and Precision

The three essential parameters (latency, velocity and precision) are displayed in separate graphs for horizontal and vertical saccades. Each saccade is represented by a single dot.

Saccades to the left and downwards can be seen on the left side of the diagram, saccades to the right and upwards are on the right side of the diagram.

6.3 Configurating the Saccade Test

To open the configuration dialog box,

a) double click into the test window or

b) select the 'Configuration' command from the 'Test' menu.

In the configuration dialog box, the test settings are shown in various tabs.

Note

The main settings have already been described in chapter *„Configurating a Test“* (page D-28), chapter *„Calibration“* (page D-49) and chapter *„Nystagmus Analysis“* (page D-69).

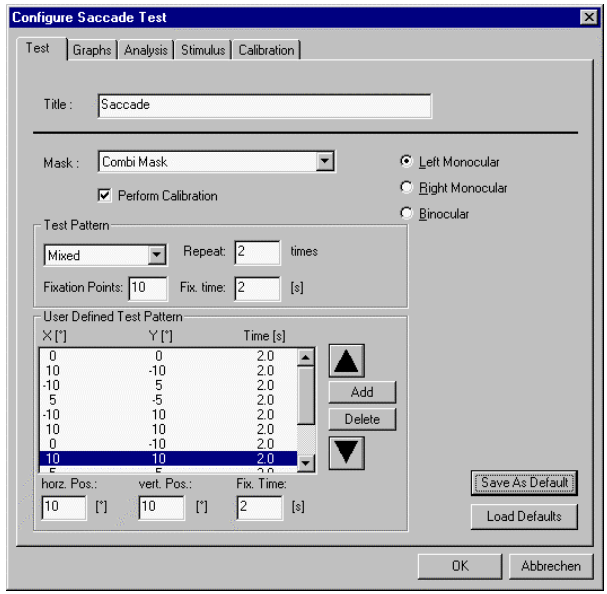


Fig. F-22: Saccade test configuration, "Test" tab

6.3.1 *Test* Tab

Using the *Test* tab, you set the saccade test procedure.

In the upper part of the tab, you have the following options:

- entering a title for the test,
- selecting the VOG mask used and
- defining whether the measurement is to be performed for the right, for the left or for both eyes (binocular).

☞ For more information, refer to chapter „*Configurating a Test*“ on page D-28.

Under TARGET POSITIONS (in the middle of the tab) you define how the target will be positioned.

The target position sequence can be set automatically, but you can also define a list of positions yourself.

❍ Automatic Defining of Target Positions

If you want the target positions to be defined automatically, proceed as follows:

1. In the "Test Pattern" area, select one of the following target positions:
   - *horizontal*
     The target will only move in horizontal direction.
   - *vertical*
     The target will move in the vertical direction.
   - *mixed*
     The target will move in horizontal and vertical direction.
2. In the TARGET POSITIONS box, enter the desired number of target positions.

3. Use the REPEAT box to determine how often the pattern is to be repeated.

4. In the FIX. TIME box, determine after how many seconds the target changes to the next position.

   → Target positions will now be set automatically by the 2D VOG software.

❍ User Defined Target Positions

To define user defined target positions, proceed as follows:

1. In the "Test Pattern" area, select the pattern:
   – *User Defined*
   The target will move to the positions determined by the user in the USER DEFINED TEST POSITIONS box.

2. In the "User Defined Test Pattern" area, now enter each target position. To do so, proceed as follows:

   2a) Click the [New] button to add a new position to the list.

   2b) Under X-POS and Y-POS, enter the desired target position.

   2c) In the FIX. TIME box determine after how many seconds the target will move to the next position.

   2d) Repeat steps 2a) to 2c) until you have entered all positions.

Note

Positions already entered can be corrected by first selecting the position in the list and then reentering the values under X-POS, Y-POS and FIX. TIME.



The position order can be changed by clicking one of these buttons.

6.3.2 *Graphs* Tab

☞ Refer to page D-30.

6.3.3 *Analysis* Tab

This tab is used to define the conditions for selecting the saccades that are to be analyzed.

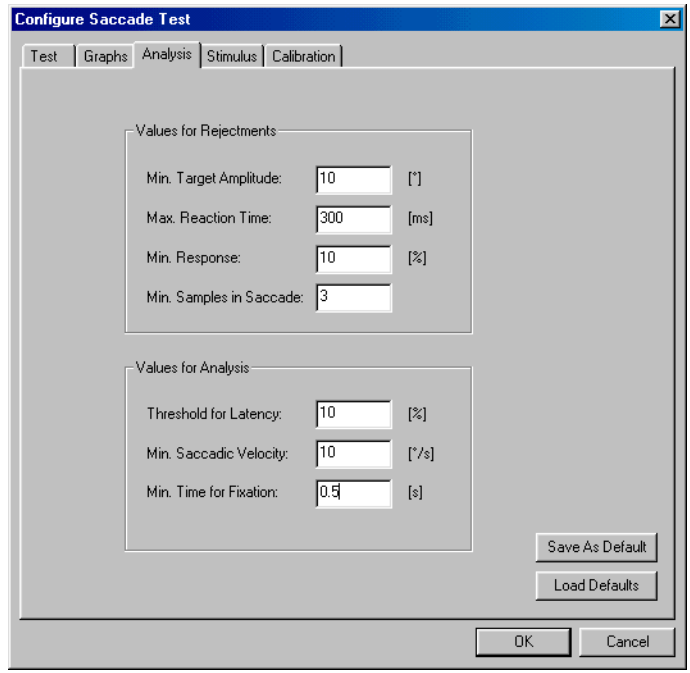


Fig. F-23: Saccade test configuration, "Analysis" tab

❍ *"Values for Rejectments"*

- **Min. Target Amplitude**
  Minimum amplitude value for a saccade (target jump). The duration of small saccades is not long enough to obtain an exact measurement with 2D VOG (see *„Analysis“* on page F-42). To avoid distortions in the statistical analysis, saccades that have an amplitude value smaller than 10° should be excluded.

- **Max. Latency**
  If the patient reacts very late, you can assume that he might have not concentrated enough on the stimulus. Therefore, saccades performed after a large latency should not influence the analysis. The recommended value is 300 ms.

- **Min. Response**
  Minimum saccade amplitudes measured as percentage of the target jump amplitude. Saccades of a smaller amplitude are not analyzed as a response to the target position change, so will be excluded from the analysis. The recommended value is 10%.

- **Min. Number of Data Points**
  Define the minimum number of data points required to scan a saccade so they can be used for the analysis. At least two data points are recommended, three points are better.

❍ *"Values for Analysis"*

- **Latency for Threshold**
  The amplitude that has to be passed by the eye to regard the saccade as "started" (see *„Latency“* on page F-42). The treshold is given in relation to the total amplitude of the saccade. Recommended value: 10%.

– **Min. Velocity**
  Minimum eye movement velocity to be detected as saccade

– **Min. Fixation Time**
  Minimum duration required by the patient to fixate the target after the saccade.

6.3.4 *Stimulus* Tab

Using this tab, you can define the target parameters:

– **Circle / Picture**
  Selecting the target.
– **Foreground color / Background color**
  Selecting target and background color
– **Display Test**
  To control target displaying, click the [Test] button.

6.3.5 *Calibration* Tab

☞ Refer to page D-34.

7. Optokinetic Test

With the optokinetic test you can examine the movement of the eyes when they are looking at a large moving pattern. Using the VisualLab module, you can generate various patterns (see *„Stimulus Tab“* on page F-63) that are moving across the projection screen in horizontal or vertical direction in front of the patient.

Large movements in the environment cause nystagmus beats that are used to avoid blur due to movement on the retina. Ideally, during the slow phase of the nystagmus the eyes have to move with the velocity of the projection pattern, but to the opposite direction to finally jump back with a fast saccade to the original position.

In the optokinetic test, the patients ability is tested in up to six subtests to follow moving stimulus patterns of different direction and speed.

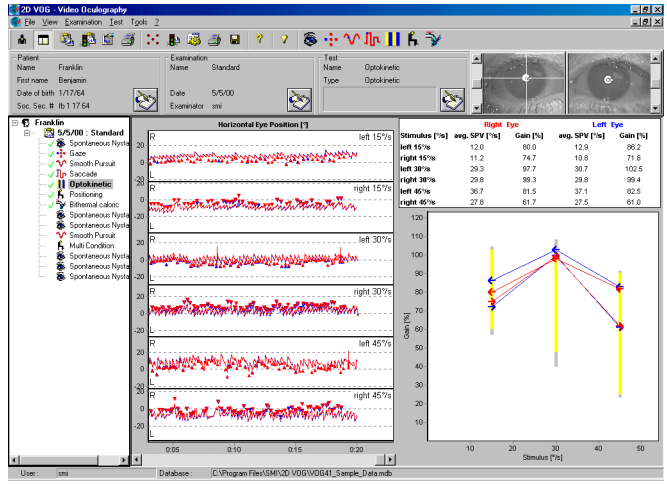


Fig. F-24: Optokinetic test window

Note

Please make sure the patient follows the target only with his eyes. The head must not be moved during the test.

Before starting an optokinetic test, you should perform a calibration (see page D-49).

7.1 Analysis

The analysis of the optokinetic test calculates the velocity of the movement during the slow phase (SPV) and compares this velocity with the projection pattern velocity. The relation of both velocity values is called gain (G):

$$G = \left| \frac{SPV}{Stimulus\ Velocity} \right|$$

Note

Data acquired in the first three seconds of a measurement are not analyzed.

7.2 Diagrams

7.2.1 Statistical Data

	Right Eye		Left Eye	
Stimulus [°/s]	**avg. SPV [°/s]**	**Gain [%]**	**avg. SPV [°/s]**	**Gain [%]**
left 15°/s	12.0	80.0	12.9	86.2
right 15°/s	11.2	74.7	10.8	71.8
left 30°/s	29.3	97.7	30.7	102.5
right 30°/s	29.8	99.3	29.8	99.4
left 45°/s	36.7	81.5	37.1	82.5
right 45°/s	27.8	61.7	27.5	61.0

Fig. F-25: Statistical data field for Optokinetic test

In the statistical data field, the average slow phase velocity and the average gain are given for each subtest and each eye recorded:

Abbreviation	Description	Unit
avg.SPV	Average slow phases velocity of detected nystagmus beats during one subtest.	[°/s]
gain	Relation between speed of eye movement and speed of stimulus movement averaged for each subtest.	[%]

7.2.2 Horizontal and Vertical Eye Position

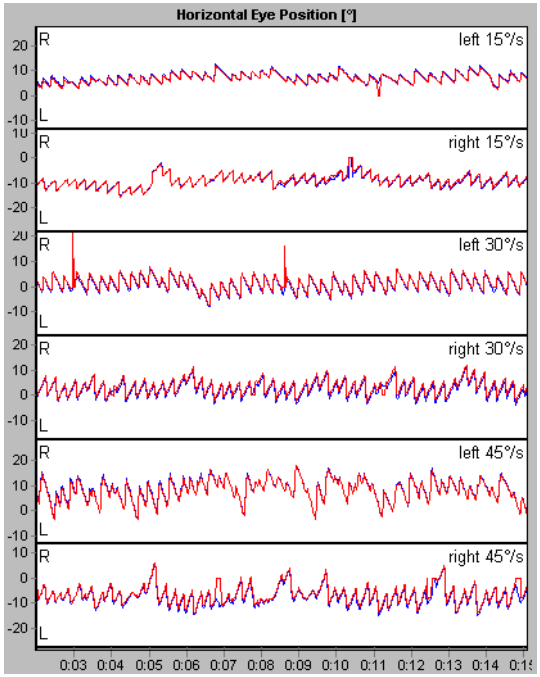


Fig. F-26: Horizontal and Vertical Eye Position

For each subtest the eye position data is shown. The direction of eye movements displayed and analyzed depends on the configured orientation of stimulus movement (see *Stimulus* tab of Optokinetic test configuration on page F-63):

– If horizontal stimulus movement is selected, the horizontal eye movement is recorded and analyzed.
– If vertical stimulus movement is selected, the vertical eye movement is recorded and analyzed.

Direction and speed of stimulus movement are given in the upper right corner of each eye position diagram.

7.2.3 Gain Diagram

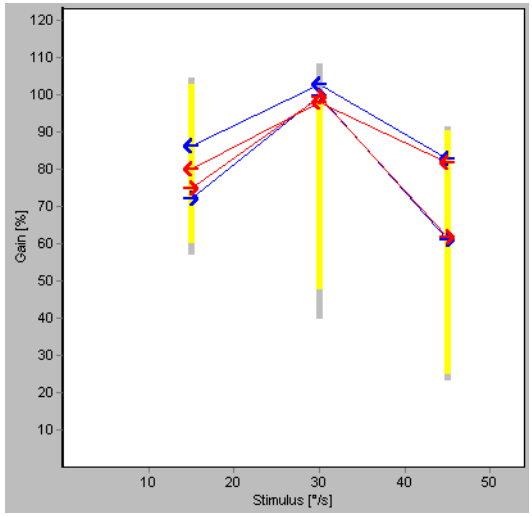


Fig. F-27: Gain Diagram Optokinetic Test

In the gain diagram the average gain during each subtest is plotted over speed of stimulus movement and compared to normative data. The direction of stimulus movement during each subtest the data is plotted for is indicated by an arrow:

← Stimulus moving from right to left

→ Stimulus moving from left to right

↑ Stimulus moving upward

↓ Stimulus moving downward

The normative data range is shown as yellow and grey bars:

– The yellow bars indicate the range the data of 90% of healthy subjects would fall in.
– The grey bars indicate the range the data of 94% of healthy subjects would fall in.

7.3 Configurating the Optokinetic Test

To open the configuration dialog box, double click into the test window or select the 'Configuration' command from the 'Test' menu.

In the configuration dialog box, the test settings are shown in various tabs.

Note

The main settings have already been described in chapter *„Configurating a Test“* (page D-28), chapter *„Calibration“* (page D-49) and chapter *„Nystagmus Analysis“* (page D-69).

7.3.1 *Test* Tab

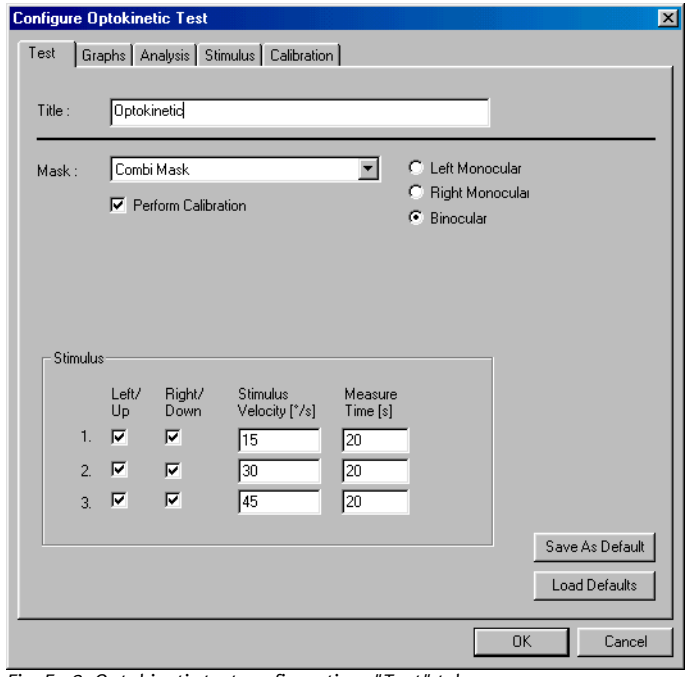


Fig. F-28: Optokinetic test configuration, "Test" tab

Using the *Test* tab, you setup the optokinetic test procedure.

In the upper part of the tab, you have the following options:

– entering a title for the test,
– selecting the VOG mask used and
– defining whether the measurement is to performed for the right, for the left or for both eyes (binocular).

In the lower part of the *Stimulus* tab, you define the test procedure. The following options have to be set:

- Define up to three different velocities of stimulus movement.
- For each velocity defined, select the direction of the stimulus movement that shall be tested for. Depending of the orientation of stimulus movement selected in the *Stimulus* tab you choose for right and left or upward and downward stimulus movement.
- Define the time after which measurement shall stop automatically.

☞ For more information on the *Test* tab, refer to page D-29.

7.3.2 *Graphs* Tab

☞ Refer to page D-30.

7.3.3 *Analysis* Tab

☞ Refer to page D-32.

7.3.4 *Stimulus* Tab

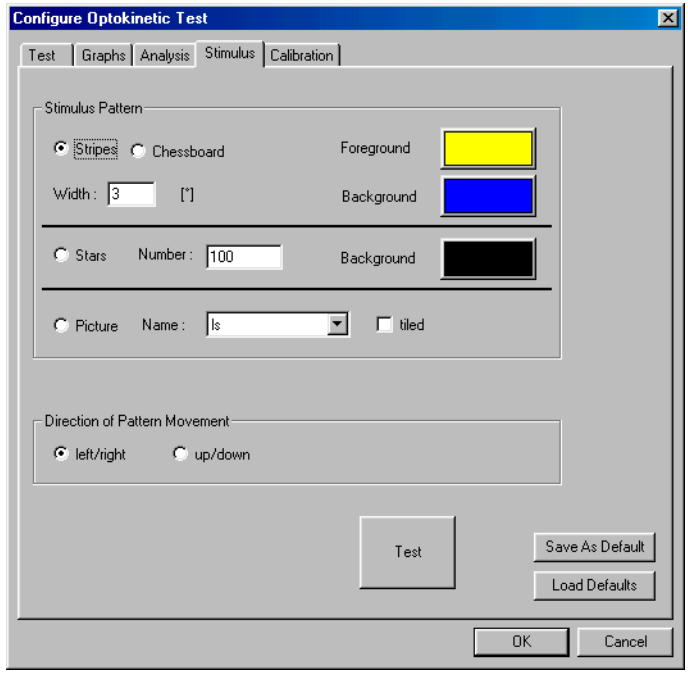


Fig. F-29: Optokinetic test configuration, "Stimulus" tab

Using the *Stimulus* tab, you can define the stimulus pattern:

– **Stripes / Chessboard**
Select the kind of pattern you wish to use and select the foreground and background colors for stripe / chessboard.

– **Stars**
Choose the number of stars displayed in the stars pattern and select the color for star patterns.

– **Picture**
Select a picture.

– **Direction of Pattern Movement**
Choose between horizontal and vertical movement of the stimulus pattern.

To control the stimulus pattern defined, press the [Test] button.

7.3.5 *Calibration* Tab

☞ Refer to page D-34.

8. Caloric Irrigation Test

The caloric irrigation test acquires eye movement data during four caloric irrigations, like they are typically performed during a bithermal caloric irrigation test. Recorded data are analysed for slow phase velocity and frequency of nystagmus beating and displayed in various standard diagrams. In addition standard statistical parameters like unilateral weakness and directional preponderance are calculated (see below).

Preceding to the caloric irrigations spontaneous nystagmus (SPN) beating can be recorded. This is recommended for patients showing a sever nystagmus beating even when not irrigated. The average slow phase velocity determined during the SPN test is taken into account for calculation of the directional preponderance (see Analysis).

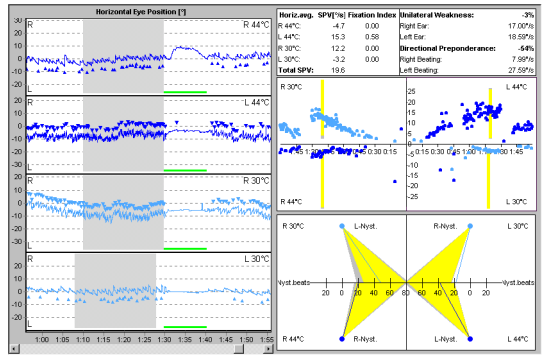


Fig. F-30: Caloric Irrigation test window

8.1 Analysis

Attention!

While in the eye position diagrams either horizontal or vertical eye movement data can be plotted, data shown in the statistical field and diagrams on the right hand side of the caloric irrigation test window are always based on horizontal eye movement data.

8.1.1 Culmination Phase

The culmination phase refers to the phase of the fastest nystagmus beating during an irrigation. If you configure your 2D VOG system to detect the culmination phase automatically, it will search for the period of the highest average slow phase velocity.

8.1.2 Average Slow Phase Velocity (ASPV)

The average slow phase velocity (ASPV) given in the statistical data field for each irrigation is calculated on the base of the detected nystagmus beats during the culmination phase:

$ASPV_{RightWarm}$	average SPV during culmination phase of **warm** irrigation of the **right** ear
$ASPV_{RightCold}$	average SPV during culmination phase of **cold** irrigation of the **right** ear
$ASPV_{LeftWarm}$	average SPV during culmination phase of **warm** irrigation of the **left** ear
$ASPV_{LeftCold}$	average SPV during culmination phase of **cold** irrigation of the **left** ear
$ASPV_{SPN}$	average SPV during the spontaneous nystagmus subtest

8.1.3 Total Slow Phase Velocity (TSPV)

The total slow phase velocity (TSPV) is the sum of the averaged SPV calculated separately for each caloric irrigation:

$$TSPV = ASPV_{RightWarm} + ASPV_{RightCold} + ASPV_{LeftWarm} + ASPV_{LeftCold}$$

On a healthy subject, the total SPV should sum up to zero.

TSPV << o	indicates a directional preponderance for right nystagmus beating
TSPV >> o	indicates a directional preponderance for left nystagmus beating

8.1.4 Unilateral Weakness

The Unilateral Weakness constitutes a value that compares the nystagmus beating induced by irrigation of the left ear with the one induced by irrigation of the right ear. It is calculated according to the following formula:

$$Unilateral\ Weakness = \frac{ASPV_{RightEar} - ASPV_{LeftEar}}{ASPV_{RightEar} + ASPV_{LeftEar}} \times 100$$

where $ASPV_{RightEar}$ and $ASPV_{LeftEar}$ refer to the summed amounts of the average slow phase velocities in the culmination phase recorded during irrigations of the right and left ear, respectively:

$$ASPV_{RightEar} = |ASPV_{RightCold}| + |ASPV_{RightWarm}|$$

The unilateral weakness is given in percent [%] and, on a healthy subject, should be near zero. Values that deviate from zero can be interpreted as follows:

Unilateral Weakness << o	The vestibular system shows **less sensitivity** to caloric irrigations of the **right** ear than of the left ear.
Unilateral Weakness >> o	The vestibular system shows **less sensitivity** to caloric irrigations of the **left** ear than of the right ear.

8.1.5 Directional Preponderance

The directional preponderance constitutes a value that compares the patients ability to perform right beating nystagmus with his ability to do left beating nystagmus.

Caloric irrigations which should induce right nystagmus beating (right warm and left cold) are set in relation with the ones that should result in left nystagmus beating (left warm and left cold).

The formula is as follows:

$$Directional\ Preponderance = \frac{ASPV_{RightBeating} - ASPV_{LeftBeating}}{ASPV_{RightBeating} + ASPV_{LeftBeating}} \times 100$$

where $ASPV_{RightBeating}$ and $ASPV_{LeftBeating}$ refer to the summed amounts of the average slow phase velocities during the culmination phase recorded during irrigations which should result in right and left nystagmus beating, respectively:

$$ASPV_{RightBeating} = |ASPV_{RightWarm}| + |ASPV_{LeftCold}|$$

$$ASPV_{LeftBeating} = |ASPV_{LeftWarm}| + |ASPV_{RightCold}|$$

If a spontaneous nystagmus (SPN) recording has been performed as part of the caloric irrigation test (refer to chapter *„Configurating the Caloric Irrigation Test“* (*Test* tab) on page F-81), the average slow phase velocity during the SPN test ($ASPN_{SPN}$) is subtracted:

$$ASPV_{RightBeating} = |ASPV_{RightWarm} - ASPV_{SPN}| + |ASVP_{LeftCold} - ASVP_{SPN}|$$

$$ASPV_{LeftBeating} = |ASVP_{LeftWarm} - ASVP_{SPN}| + |ASVP_{RightCold} - ASVP_{SPN}|$$

The directional preponderance is given in percent [%] and, on a healthy subject, should be near zero. Values that deviate from zero can be interpreted as follows:

Directional Preponderance << o	Nystagmus beating to the left is faster than to the right.
Directional Preponderance >> o	Nystagmus beating to the right is faster than to the left.

Attention!

As the average slow phase velocity during the spontaneous nystagmus subtest is subtracted irrespectively of the frequency of the nystagmus beating, the spontaneous nystagmus subtest should only been included in the caloric irrigation test protocol, if the patient shows a substantial spontaneous nystagmus beating. Otherwise a few nystagmus beats, having no physiological meaning, can lead to distortion of the directional preponderance value.

8.1.6 Fixation Index

The fixation suppression index quantifies the ability of a patient to willingly suppress the nystagmus beating by fixating a given target. It is usually determined just after the culmination phase by switching on the fixation light of the VOG mask (refer to chapter *„Fixation Light and IR Illumination“* on page B-19).

The fixation light can either be switched on manually (using the buttons in the toolbar, refer to chapter *„Buttons in the Toolbar“* on page D-10) or automatically (refer to chapter *„Configurating the Caloric Irrigation Test“* (*Test* tab) on page F-81).

To calculate the fixation index, the average SPV while fixation light is switched on ($ASPV_{Fixation}$) is divided by the average SPV during an equal length of time, just before the fixation light has been switched on ($ASPV_{PraeFixation}$):

$$Fixation\ index = \frac{ASPV_{Fixation}}{ASPV_{PraeFixation}}$$

Healthy subjects can totally suppress nystagmus beating when a fixation target is given. In this case $ASPV_{Fixation}$ will be zero which results in a fixation index value of zero. For patients that are not able to suppress the nystagmus beating the fixation index will be greater than zero. A fixation index of one indicates that the patient is not able to suppress nystagmus beating at all.

8.2 Diagrams

8.2.1 Color Encoding

In order to distinguish between data recorded during cold and warm irrigations two additional colors are used in the caloric irrigation test diagrams:

Light blue		Data recorded during **cold** irrigations from the **left** eye
Blue		Data recorded during **warm** irrigations from the **left** eye
Purple		Data recorded during **cold** irrigations from the **right** eye
Red		Data recorded during **warm** irrigations from the **right** eye

☞ To get more information on the other colors used, refer to chapter *„Color Encoding“* on page F-6.

8.2.2 Horizontal and Vertical Eye Position

Depending on the settings (see *„Configurating the Caloric Irrigation Test“* (*Graphs* tab) on page F-83), horizontal or vertical eye position data is shown.

Eye position data is plotted separately for each irrigation and for an optional preceding spontaneous nystagmus recording as subtest diagrams. Direction and temperature of the respective irrigation are given in the upper right corner of the each diagram.

When recording has stopped, the culmination phase which is the phase of fastest eye movement (see page F-66) is indicated by a grey background (see Fig. F-31 on page F-72).

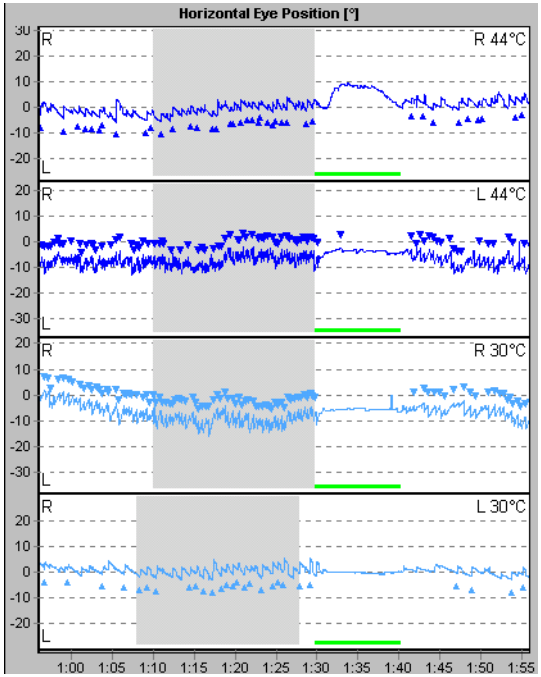


Fig. F-31: Eye position diagram – The grey shaded area indicates the period of fastest nystagmus beating (culmination phase).

The green bar marks the period when the fixation light within the VOG mask has been switched on.

8.2.3 Statistical Data

Horiz.avg.	SPV[°/s]	Fixation Index	Unilateral Weakness:	-6%
R 44°C:	-4.8	0.00	Right Ear:	16.59°/s
L 44°C:	15.2	0.60	Left Ear:	18.97°/s
R 30°C:	11.8	0.00	**Directional Preponderance:**	**-51%**
L 30°C:	-3.8	0.00	Right Beating:	8.56°/s
Total SPV:	18.4		Left Beating:	27.00°/s

Fig. F-32: Statistical data field for caloric test

Statistical data is calculated for horizontal eye movements only. The following values are shown:

Abbreviation	Description	Unit
avg. SPV	Average slow phase velocity during culmination phase	[°/s]
Fixation index	Fixation suppression index	
Total SPV	Sum of averaged slow phase velocities	[°/s]
Unilateral weakness	Unilateral weakness	[%]
Right Ear	Sum of average slow phase velocities during irrigations of the right ear ($ASPV_{RightEar}$)	[°/s]
Left Ear	Sum of average slow phase velocities during irrigations of the left ear ($ASPV_{LeftEar}$)	[°/s]
Directional preponderance	Directional preponderance	[%]
Right Beating	Sum of average slow velocities during warm irrigation of the right ear and cold irrigation of the left ear ($ASPV_{RightBeating}$)	[°/s]
Left Beating	Sum of average slow velocities during warm irrigation of the left ear and cold irrigation of the right ear ($ASPV_{LeftBeating}$)	[°/s]

8.2.4 Butterfly Diagram

In the butterfly diagram, the slow phase velocity (SPV) of horizontal nystagmus beats are plotted over time.

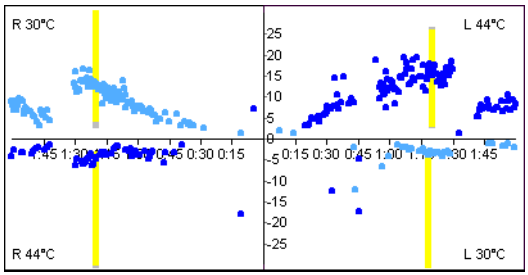


Fig. F-33: Butterfly Diagram

The diagram is divided into four quadrants each representing data from one caloric irrigation:

- The **upper** two quadrants show data from irrigations which should result in **left nystagmus beating** (right cold and left warm).
- The **lower** two quadrants show data from irrigations which should result in **right nystagmus beating** (right warm and left cold).
- The **two quadrants** on the **left** hand side show data from irrigation of the **right ear** (right cold and right warm).
- The **two quadrants** on the **right** hand side show data from irrigation of the **left ear** (left warm and left cold).

Note

In the quadrants on the left hand, time axis runs from right to left.

In healthy subjects, this specific arrangement of data plots results in a shape similar to the outlines of butterfly wings.

Like in the Optokinetic test the yellow and grey bars represent normative data ranges (refer to chapter *„Color Encoding“* on page F-6) for the culmination phase:

– The yellow bars indicate the range the data line of 90% of healthy subjects would cross.
– The grey bars indicate the range the data line of 94% of healthy subjects would cross.

8.2.5 Freyss Diagram

The Freyss Diagram graphically represents the average slow phase velocities (ASPV) during the culmination phase of each irrigation. These values are also listed in the statistical data field by numbers.

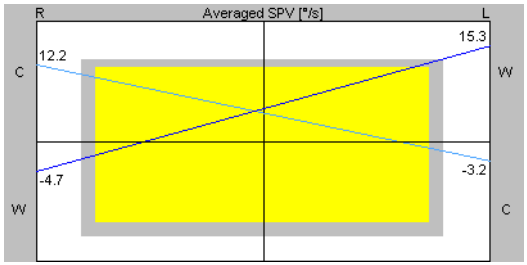


Fig. F-34: Freyss Diagram

On the left vertical axis, the ASPV calculated for the irrigations of the right ear are plotted; data points on the right vertical axis show the ASPV values for left ear irrigations.

The diagram is completed by two lines: one connecting the data points obtained for the two warm irrigations (blue or red), the other one connecting the data points representing the two cold irrigations (light blue or purple).

The normative data range in the Freyss Diagram has the shape of a box and refers to the junction of the two lines described above which should be close to the centre of the graph:

– For 90% of healthy subjects the junction will be within the yellow marked area.
– For 94% of healthy subjects the junction will be within the grey bounded area.

8.2.6 Claussen Diagram

The Claussen diagram gives information about the frequency of nystagmus beating during the culmination phases of each irrigation. Here, nystagmus beats in the non-expected direction (i.e. right beats in the right cold irrigation) are subtracted from the number of beats in the expected direction (i.e. left beats in the right cold irrigation).

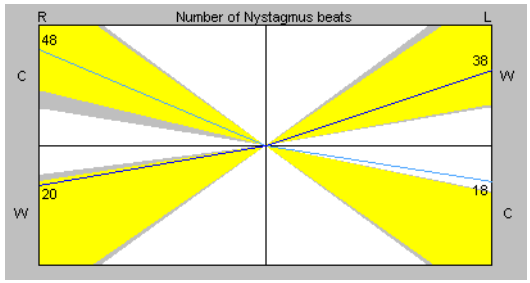


Fig. F-35: Claussen Diagram

Like in the Butterfly diagram, the diagram is divided into four quadrants. Each quadrant represents data from one caloric irrigation (refer to chapter *„Butterfly Diagram“* on page F-74). On the outer vertical axis of each quadrant, the number of nystagmus beats is plotted. For better graphical representation, a line is drawn from each data point to the center of the diagram.

The normative data range shown refers to the lines:

– For 90% of healthy subjects the lines should be within the yellow marked area.
– For 94% of healthy subjects the lines should be within the grey bounded area.

8.2.7 Scherer Diagram

The Scherer Diagram consists of two parts: directional preponderance and unilateral weakness.

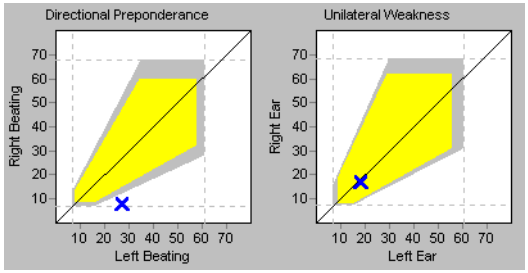


Fig. F-36: Scherer Diagram

In the *directional preponderance* graph, the average slow phase velocity right beating ($ASPV_{RightBeating}$) value is plotted versus the average slow velocity left beating ($ASPV_{LeftBeating}$) value (please refer to chapter *„Directional Preponderance“* on page F-68).

In the *unilateral weakness* graph, the average slow phase velocity right ear ($ASPV_{RightEar}$) value is plotted versus the average slow velocity left ear ($ASPV_{LeftEar}$) value (please refer to chapter *„Unilateral Weakness“* on page F-67).

All values shown in the Scherer diagram are also listed in the statistical data field (refer to chapter *„Statistical Data“* on page F-72).

In healthy subjects the data crosses shown should be close to the diagonal plotted in the graphs. The normative data range refers to the position of the data cross:

- For 90% of healthy subjects the crosses will be within the yellow marked areas.
- For 94% of healthy subjects the crosses will be within the grey bounded areas.

8.2.8 Haid / Stoll Diagram

Like the Claussen diagram, the Haid/Stoll diagram represents the frequency of nystagmus beating.

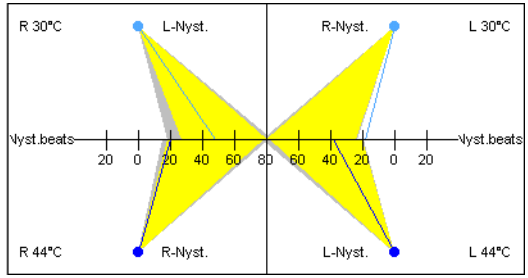


Fig. F-37: Scherer Diagram

The four quadrants of the diagram represent the four caloric irrigations. Like in the Claussen diagram, the number of nystagmus beats into the non-expected direction (i.e. right in the right cold irrigation) is subtracted from the number of beats into the expected direction (i.e. left in the right cold irrigation).

For each irrigation, the number of nystagmus beats is plotted on the X-Axis by drawing a line from the outside of each quadrant to the center of the diagram when the sum of beats into the expected direction minus the ones into the non-expected directions is positive, and to the outside in case there are more beats into the non-expected direction than into the expected direction.

So on a healthy subject it is expected that all four lines are going from the outside of the quadrants to the inside of the diagram.

The normative data range shown refers to the lines:

– For 90% of healthy subjects the lines will be within the yellow marked area.
– For 94% of healthy subjects the lines will be within the grey bounded area.

8.3 Configurating the Caloric Irrigation Test

To open the configuration dialog box,

a) double click into the test window or

b) select the 'Configuration' command from the 'Test' menu.

In the configuration dialog box, the test settings are shown in various tabs.

Note

The main settings have already been described in chapter *„Configurating a Test“* (page D-28), chapter *„Calibration“* (page D-49) and chapter *„Nystagmus Analysis“* (page D-69).

8.3.1 *Test* Tab

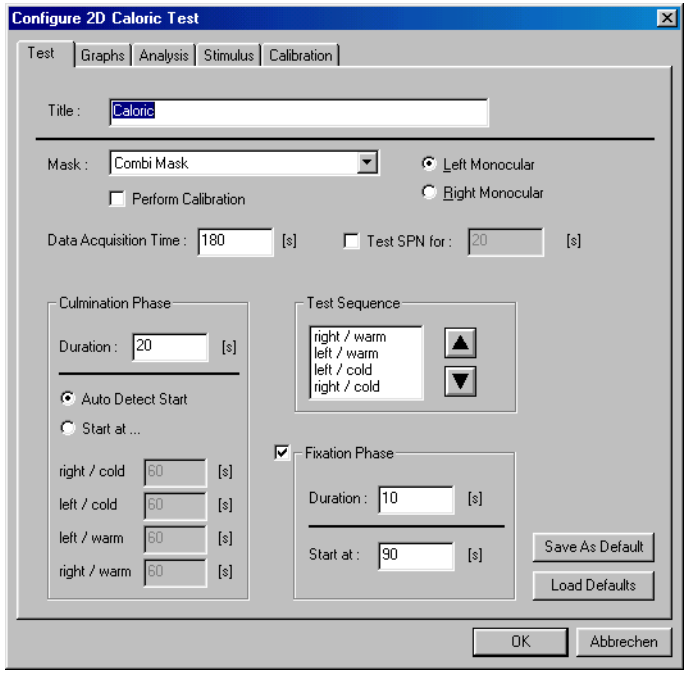


Fig. F-38: Caloric test configuration, "Test" tab

In the *Test* tab, you setup each subtest of your caloric irrigation test. To configure the test, proceed as follows:

1. Enter a title for the test (e.g. Bithermal caloric irrigation).
2. In the MASK box, select the desired VOG mask.
3. If you always want to perform a calibration before starting the measurement, select the PERFORM CALIBRATION check box (refer to chapter *„Calibration“* on page D-49).
4. Select whether you want to record from the right or the left eye.

Note

The Caloric irrigation test allows for monocular recording only.

5. Enter the duration of data acquisition for the irrigations and - if wanted - select the TEST SPN check box to perform a spontaneous nystagmus test before the irrigations.

Attention!

The results of the spontaneous nystagmus (SPN) test are taken into account to calculate the directional preponderance value shown in the statistical data field. You may want to skip the SPN test when the patient does not show a profound spontaneous nystagmus beating (refer to chapter *„Analysis“* on page F-66).

6. Under CULMINATION PHASE, enter the duration of the culmination phase and select whether 2D VOG shall automatically search for the most active phase (in terms of maximal average slow phase velocity) or enter a starting point of the culmination phase for each irrigation.

7. The TEST SEQUENCE box allows you to change the sequence of irrigations by selecting one irrigation and change its position within the sequence by clicking on the arrows.

8. Select the FIXATION INDEX check box to let 2D VOG switch on the fixation light within your VOG mask (refer to chapter *„Fixation Light and IR Illumination“* on page B-19) for a given duration at a given starting time in relation to the beginning of the recording.

8.3.2 *Graphs* Tab

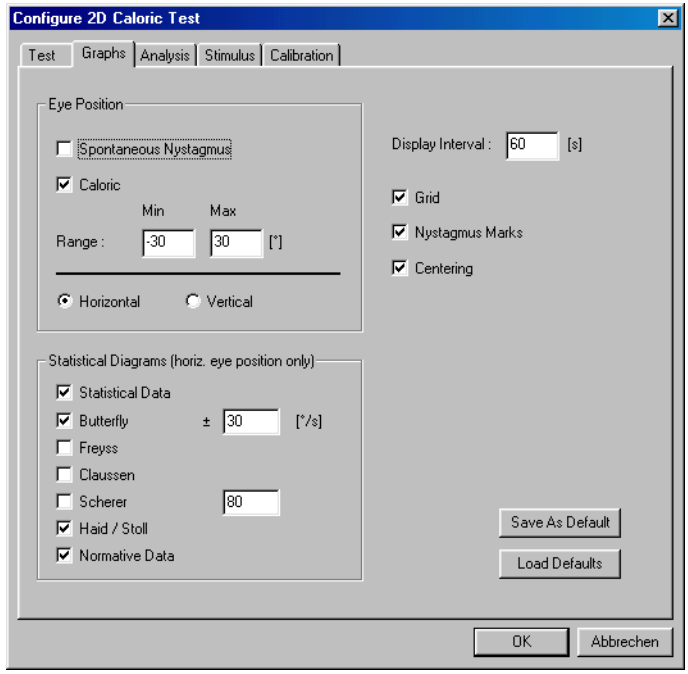


Fig. F-39: Caloric test configuration, "Graphs" tab

Like in the other tests, the *Graphs* tab allows you to configure the layout of the diagrams in the test window (refer to page D-30).

Note

If you choose the vertical eye position, the diagrams will show the vertical eye movement. All other diagrams (Statistical data, butterfly diagram, Freyss diagram, Claussen diagram, Scherer diagram, Haid/Stoll diagram) are not influenced by this settings. They will always refer to the horizontal eye movements.

8.3.3 *Analysis* Tab

☞ Refer to page D-32.

8.3.4 *Stimulus* Tab

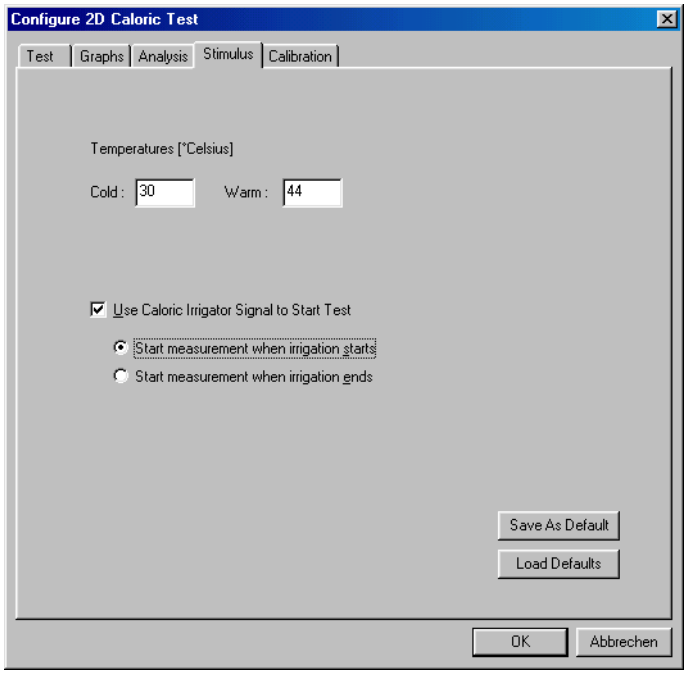


Fig. F-40: Caloric test configuration, "Stimulus" tab

Using the *Stimulus* tab, you can define the following:

– **Temperature of irrigations**
Enter temperature of cold and warm irrigation. The entry does not affect the analysis of data but is used for graphical representation of the used temperatures in the diagrams of the caloric irrigation test.

– **Starting the test by a trigger signal from caloric irrigator**
The 2D VOG allows to start the caloric irrigation test by means of a trigger signal from the caloric irrigator (Digital In option required).

For more information please refer to your dealer or to SensoMotoric Instruments.

8.3.5 *Calibration* Tab

☞ Refer to page D-34.

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Facsimile Form

To improve our products we also need your help. Therefore we would like to know your opinion about our product and the documentation.

On the following pages, you will find a facsimile form that can be used to

- describe problems or errors arised when using the product
- write down your opinion or suggestions concerning the documentation.

Please fill out the form and send or fax it to the following address:

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Germany

Fax: +49 (3328) 39 55 99

Facsimile Form

To: **SensoMotoric Instrumtents GmbH**
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Instrument: **2D VOG - Video-Oculography® Version 4**
Manual version: **E1.0**

Please describe your problem / the error:

This Manual

1. The manual ...

... is easy to understand ☐

... has a logical structure ☐

... is well layouted ☐

(Please enter numbers from 1 to 6 where 1 is the best.)

2. How are the topics described?

... too detailed ❍

... not detailed enough ❍

... adequate description ❍

3. All information can easily be found ...

... correct ❍

... partially ❍

... not correct ❍

4. I miss the following information/description:

5. The following information should be described in a more detailed form:

6. The following information is not correct: